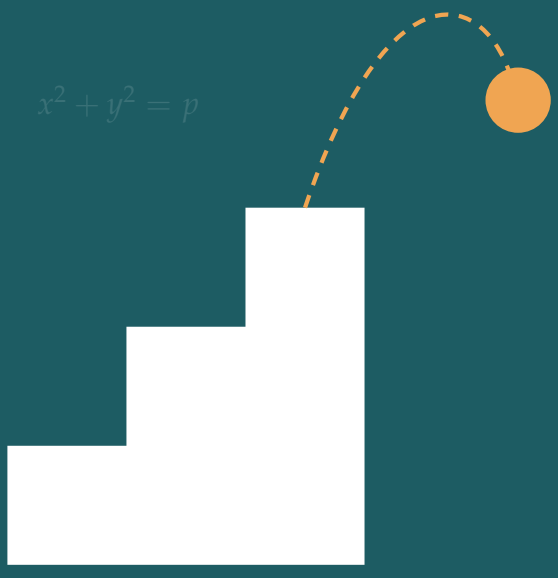
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1993 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1993 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1993 Worked Solutions

Worked solutions to all 48 questions of the 1993 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1993, Question 1

Pure • Counting Dice Totals

STEP I 1993, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q1

This question has a full guided treatment in *Beyond STEP*, Section 9.6, *Strange Dice*: https://beyondstep.keithmcnulty.org/ch09/strange-dice.html

The key idea

Every face is a different colour, so a "way" of throwing a total is a choice of one face from each die. If one die has a_k faces numbered k and the other has b_k, the number of ways of throwing a total n is

$$N(n) = \sum_{i+j=n} a_i\, b_j.$$

In the second part we read the table from the smallest totals upwards. Each new total brings in only one or two unknowns we haven't met yet, so the face counts are forced one at a time, and we only have to rule out the few branches that appear along the way.

Solution

The total of 6

On the first die the counts are $a_1 = 1$, $a_2 = 2$, $a_3 = 2$, $a_6 = 1$. On the second they are $b_1 = 1$, $b_3 = 2$, $b_4 = 1$, $b_5 = 1$, $b_6 = 1$. A total of 6 needs the first die to show 1, 2 or 3; a 6 would need a 0 on the second die. So

$$N(6) = a_1b_5 + a_2b_4 + a_3b_3 = 1\cdot 1 + 2\cdot 1 + 2\cdot 2 = 7,$$

which is the seven ways required.

Finding the new numbering

Call the dice P and Q. Let P have a_k faces numbered k, and Q have b_k, for $k = 1, \ldots, 6$. These are non-negative integers with

$$a_1 + \cdots + a_6 = b_1 + \cdots + b_6 = 6.$$

(As a sanity check, the entries in the table add up to $36 = 6 \times 6$.)

Totals 2 and 3. $N(2) = a_1b_1 = 0$, so the two dice can't both have a 1. But $N(3) = a_1b_2 + a_2b_1 = 2$, so some die has a 1. Call that die P. Then $a_1 \geq 1$ and $b_1 = 0$, and

$$N(3) = a_1b_2 = 2, \qquad \text{so} \qquad (a_1, b_2) = (2,1) \text{ or } (1,2).$$

The case $a_1 = 2$, $b_2 = 1$ is impossible. Since Q has no 1, $N(7) \geq a_1b_6 = 2b_6$, and $N(7) = 3$ gives $b_6 \leq 1$. In this case Q has one face numbered 2, and we'll check that it has no 3 or 4:

- $N(4) = a_1b_3 + a_2b_2 = 2b_3 + a_2 = 1$, so $2b_3 \leq 1$, which forces $b_3 = 0$ and then $a_2 = 1$.
- $N(5) = a_1b_4 + a_2b_3 + a_3b_2 = 2b_4 + 0 + a_3 = 1$, so $b_4 = 0$ and $a_3 = 1$.

So Q has one 2, no 1, 3 or 4, and its other five faces are 5s and 6s: $b_5 + b_6 = 5$. Since $b_6 \leq 1$, this gives $b_5 \geq 4$, and then

$$N(6) \geq a_1b_5 = 2b_5 \geq 8,$$

which contradicts $N(6) = 4$. So this case can't happen.

So $a_1 = 1$ and $b_2 = 2$. Now we work up the table, remembering $b_1 = 0$.

- $N(4) = a_1b_3 + a_2b_2 = b_3 + 2a_2 = 1$. Since $2a_2$ is even, $a_2 = 0$ and $b_3 = 1$.
- $N(5) = a_1b_4 + a_2b_3 + a_3b_2 = b_4 + 0 + 2a_3 = 1$. In the same way, $a_3 = 0$ and $b_4 = 1$.
- $N(6) = a_1b_5 + a_2b_4 + a_3b_3 + a_4b_2 = b_5 + 2a_4 = 4$.

So far Q has $b_1 + b_2 + b_3 + b_4 = 0 + 2 + 1 + 1 = 4$ faces accounted for, and so $b_5 + b_6 = 2$. From $b_5 = 4 - 2a_4$, b_5 is even, so $b_5 = 0$ or $b_5 = 2$.

? **Ask yourself** How do we decide between $b_5 = 0$ and $b_5 = 2$?

We use the next total. P has no 2 or 3 and Q has no 1, so

$$N(7) = a_1b_6 + a_4b_3 + a_5b_2 = b_6 + a_4 + 2a_5 = 3.$$

- If $b_5 = 0$, then $a_4 = 2$ and $b_6 = 2$. Now $N(7) \geq b_6 + a_4 = 4 > 3$, which is impossible.
- If $b_5 = 2$, then $a_4 = 1$ and $b_6 = 0$. Now $N(7) = 0 + 1 + 2a_5 = 3$ gives $a_5 = 1$. P has $a_1 + a_4 + a_5 = 3$ faces accounted for, so $a_6 = 3$.

The table has forced everything:

$$P: \ 1, 4, 5, 6, 6, 6 \qquad\qquad Q: \ 2, 2, 3, 4, 5, 5.$$

Checking the rest of the table. We used only the totals 2 to 7, so we still have to confirm that this numbering gives the remaining entries. With $a_1 = a_4 = a_5 = 1$, $a_6 = 3$ and $b_2 = 2$, $b_3 = b_4 = 1$, $b_5 = 2$:

$$\begin{aligned} N(8) &= a_4b_4 + a_5b_3 + a_6b_2 = 1 + 1 + 6 = 8, \\ N(9) &= a_4b_5 + a_5b_4 + a_6b_3 = 2 + 1 + 3 = 6, \\ N(10) &= a_5b_5 + a_6b_4 = 2 + 3 = 5, \\ N(11) &= a_6b_5 = 6, \\ N(12) &= a_6b_6 = 0. \end{aligned}$$

All of these agree with the table. So the dice must be numbered $1,4,5,6,6,6$ and $2,2,3,4,5,5$, and this numbering really does give the table. It is the only possible numbering, apart from swapping which die we call which.

Checks

- Draw the 6×6 grid of sums for $1,4,5,6,6,6$ against $2,2,3,4,5,5$ and count each total. You should get $0,2,1,1,4,3,8,6,5,6,0$.
- A computer search over every way of numbering two dice with 1 to 6 (there are 462 multisets per die) finds exactly this pair and nothing else, which confirms that the solution is unique.

Common slips

- Guessing a numbering that fits the table and stopping. The question asks for reasoning, so you need to show that the table forces the numbering.
- Forgetting the case $a_1 = 2$, $b_2 = 1$, or dismissing it without a reason.
- Fixing the face counts from the small totals and never checking the totals 8 to 12.

Beyond STEP

The neat way to package $N(n)$ is with a generating function. If $p(x) = \sum a_k x^k$ and $q(x) = \sum b_k x^k$, then the coefficient of x^n in $p(x)q(x)$ is exactly $N(n)$, so the question asks you to factorise a given polynomial into two pieces of a particular shape. The same idea gives the *Sicherman dice*, $1,2,2,3,3,4$ and $1,3,4,5,6,8$, which have the same distribution of totals as two ordinary dice. Herbert Wilf's *generatingfunctionology* is a lively introduction to the method.

WORKED SOLUTION

STEP I 1993, Question 2

Pure • Geometric Series

STEP I 1993, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q2

The key idea

Each series is a geometric progression with some terms removed. The exponents in $S_n(r)$ are the numbers that are *not* multiples of 3. The exponents in $T_n(r)$ are the numbers that are even *or* multiples of 3. So we write each sum as a combination of complete geometric progressions: we take a full progression and subtract the unwanted terms, or add two progressions and subtract the terms we have counted twice.

Solution

The sum of even powers

The left-hand side is a geometric progression with first term 1, common ratio r^2 and $n+1$ terms. Call it E. Then

$$E - r^2E = (1 + r^2 + \cdots + r^{2n}) - (r^2 + r^4 + \cdots + r^{2n+2}) = 1 - r^{2n+2},$$

since everything else cancels. If $|r| \neq 1$ then $r^2 \neq 1$, so we may divide by $1 - r^2$:

$$1 + r^2 + r^4 + \cdots + r^{2n} = \frac{1 - r^{2n+2}}{1 - r^2},$$

as required. (For real r, the condition $r^2 \neq 1$ is exactly $|r| \neq 1$.)

The series $S_n(r)$

The exponents $1, 2, 4, 5, 7, 8, 10, \ldots, 3n-1$ are the integers from 1 to $3n-1$ that are not multiples of 3. They come in pairs $3k+1,\ 3k+2$ for $k = 0, 1, \ldots, n-1$. So

$$S_n(r) = \sum_{k=0}^{n-1} \left(r^{3k+1} + r^{3k+2}\right) = (r + r^2) \sum_{k=0}^{n-1} r^{3k}.$$

The last sum is a geometric progression with ratio r^3 and n terms. For real r, $r \neq 1$ gives $r^3 \neq 1$, so

$$S_n(r) = \frac{r(1+r)(1-r^{3n})}{1-r^3}.$$

Equivalently, we can take all powers $r, r^2, \ldots, r^{3n}$ and remove the multiples of 3:

$$S_n(r) = \frac{r(1-r^{3n})}{1-r} - \frac{r^3(1-r^{3n})}{1-r^3}.$$

This second form is the one we need for the limit.

The limit. If $|r| < 1$ then $|r^{3n}| = |r|^{3n} \to 0$ as $n \to \infty$. Hence

$$S_n(r) \to \frac{r}{1-r} - \frac{r^3}{1-r^3}.$$

Now $\dfrac{r}{1-r} = \dfrac{1}{1-r} - 1$ and $\dfrac{r^3}{1-r^3} = \dfrac{1}{1-r^3} - 1$. The two -1s cancel, which leaves

$$S_n(r) \to \frac{1}{1-r} - \frac{1}{1-r^3},$$

as required.

The series $T_n(r)$

The exponents $0, 2, 3, 4, 6, 8, 9, 10, 12, \ldots, 6n$ are the integers from 0 to $6n$ that are even or divisible by 3 (or both).

> **Ask yourself** If we add the even powers to the powers that are multiples of 3, which terms have we counted twice?

An exponent is both even and a multiple of 3 exactly when it is a multiple of 6. So we use inclusion–exclusion: add the even powers, add the multiples of 3, and subtract the multiples of 6. Up to $6n$ there are

- even powers $r^0, r^2, \ldots, r^{6n}$, which form a GP with ratio r^2 and $3n+1$ terms;
- multiples of 3, $r^0, r^3, \ldots, r^{6n}$, which form a GP with ratio r^3 and $2n+1$ terms;
- multiples of 6, $r^0, r^6, \ldots, r^{6n}$, which form a GP with ratio r^6 and $n+1$ terms.

By the first result with n replaced by $3n$, together with the same argument for ratios r^3 and r^6, we get

$$T_n(r) = \frac{1-r^{6n+2}}{1-r^2} + \frac{1-r^{6n+3}}{1-r^3} - \frac{1-r^{6n+6}}{1-r^6}.$$

Every denominator is nonzero because $|r| \neq 1$. This is why the question needs $|r| \neq 1$ here and only $r \neq 1$ for S_n: $r = -1$ makes $1-r^2$ and $1-r^6$ vanish.

The limit. If $|r| < 1$, then r^{6n+2}, r^{6n+3} and r^{6n+6} all tend to 0. So

$$T_n(r) \to \frac{1}{1-r^2} + \frac{1}{1-r^3} - \frac{1}{1-r^6}.$$

This tidies up over the common denominator $1-r^6$, using $1-r^6=(1-r^2)(1+r^2+r^4)=(1-r^3)(1+r^3)$:

$$\lim_{n\to\infty}\mathrm{T}_n(r)=\frac{(1+r^2+r^4)+(1+r^3)-1}{1-r^6}=\frac{1+r^2+r^3+r^4}{1-r^6}.$$

The cases $r>1$, $r=1$, $r=-1$

We count the terms first. In each block of six exponents $6k, 6k+1, \ldots, 6k+5$, the ones that are even or divisible by 3 are $6k, 6k+2, 6k+3, 6k+4$, which is four of them. The blocks $k=0,\ldots,n-1$ cover the exponents up to $6n-1$, and the final term is r^{6n}. So $\mathrm{T}_n(r)$ has $4n+1$ terms.

- $r>1$**:** every term r^e satisfies $r^e\geq 1$, so $\mathrm{T}_n(r)\geq 4n+1$, and therefore $\mathrm{T}_n(r)\to\infty$.
- $r=1$**:** every term equals 1, so $\mathrm{T}_n(1)=4n+1\to\infty$. The formula breaks down here because its denominators are zero.
- $r=-1$**:** $(-1)^e$ is $+1$ for even e and -1 for odd e. In each block the exponents $6k, 6k+2, 6k+4$ are even and $6k+3$ is odd, so each block contributes $1+1-1+1=2$. Hence
$$\mathrm{T}_n(-1)=2n+1\to\infty.$$

In all three cases $\mathrm{T}_n(r)$ diverges to $+\infty$. Note that $r=-1$ also diverges: the odd multiples of 3 do not cancel enough of the even terms. Contrast this with a simple alternating series such as $1-1+1-\cdots$, which merely oscillates.

Checks

- Put $n=1$ in the S_n formula: $\frac{r(1+r)(1-r^3)}{1-r^3}=r+r^2$, which is correct.
- Put $r=\frac{1}{2}$ in the limit of T_n. The formula gives $\frac{1+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}}{1-\frac{1}{64}}=\frac{92}{63}\approx 1.4603$, and adding several hundred terms of $\mathrm{T}_n(\frac{1}{2})$ directly gives the same value.
- As $r\to 1^-$, the limit of T_n behaves like $\frac{4}{6(1-r)}=\frac{2}{3(1-r)}$. This is consistent with four of every six powers being present, about two-thirds of the full series $\frac{1}{1-r}$.

Common slips

- Forgetting to subtract the multiples of 6, which counts $1, r^6, r^{12}, \ldots$ twice.
- Getting the number of terms wrong in each GP. There are $3n+1$ even exponents from 0 to $6n$, not $3n$.
- Assuming $\mathrm{T}_n(-1)$ oscillates like an alternating series, without actually adding up a block of terms.

Beyond STEP

Counting the terms in a union by adding the two sets and subtracting their intersection is the *inclusion–exclusion principle*. Combined with generating functions, it is a standard tool in combinatorics and number theory: the limit we found is the generating function of the non-negative integers that share a factor with 6. The same sieve idea, applied with every prime, gives Euler's product for the Riemann zeta function. Undergraduate analysis makes the step "$r^n \to 0$ when $|r| < 1$" rigorous and asks when infinite series may be rearranged in this way. Abbott's *Understanding Analysis* covers this well.

WORKED SOLUTION

STEP I 1993, Question 3

Pure • Diophantine Equations

STEP I 1993, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q3

The key idea

Because $p \le q \le r$, the reciprocal $\frac{1}{p}$ is the *largest* of the three terms. So the sum is at most $\frac{3}{p}$, and that pins p down to a couple of values. Once p is fixed, the same trick on the two remaining terms bounds q, and then r is forced. The idea is to use the ordering to turn an equation (or an inequality) in three unknowns into a short finite check.

Solution

Part (i)

Since $1 \le p \le q \le r$, we have $\frac{1}{r} \le \frac{1}{q} \le \frac{1}{p}$. So

$$1 = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} \le \frac{3}{p},$$

which gives $p \le 3$. Also $p \neq 1$, because then $\frac{1}{q} + \frac{1}{r} = 0$, which is impossible for positive q, r. So $p = 2$ or $p = 3$.

Case $p = 2$**.** Now $\frac{1}{q} + \frac{1}{r} = \frac{1}{2}$. Since $\frac{1}{r} \le \frac{1}{q}$,

$$\frac{1}{2} = \frac{1}{q} + \frac{1}{r} \le \frac{2}{q} \quad \Longrightarrow \quad q \le 4,$$

and since $\frac{1}{r} > 0$ we need $\frac{1}{q} < \frac{1}{2}$, so $q \ge 3$.

- $q = 3$: $\frac{1}{r} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, so $r = 6$.
- $q = 4$: $\frac{1}{r} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$, so $r = 4$.

Case $p = 3$**.** Now $\frac{1}{q} + \frac{1}{r} = \frac{2}{3} \le \frac{2}{q}$, so $q \le 3$. With $q \ge p = 3$ this forces $q = 3$, and then $\frac{1}{r} = \frac{1}{3}$, so $r = 3$.

Every case has been covered, so the only solutions are

$$(p,q,r) = (2,3,6), \quad (2,4,4), \quad (3,3,3),$$

and there are no others.

Part (ii)

If $p = 1$ the sum is $1 + \frac{1}{q} + \frac{1}{r} > 1$ for every q and r. These are the solutions we're told about. (The ordering $p \le q \le r$ means these are $q = n$, $r = m$ with $1 \le n \le m$. The inequality the question writes between m and n is the wrong way round, but it doesn't matter: $p = 1$ works for any q and r.) So we look for solutions with $p \ge 2$.

As before, $\frac{1}{p}$ is the largest term, so

$$1 < \frac{1}{p} + \frac{1}{q} + \frac{1}{r} \le \frac{3}{p} \quad \Longrightarrow \quad p < 3.$$

With $p \ge 2$ this gives $p = 2$. We now need

$$\frac{1}{q} + \frac{1}{r} > \frac{1}{2}.$$

Ask yourself Why is the inequality now strict in the bound on q, and does it matter?

Since $\frac{1}{r} \le \frac{1}{q}$, we have $\frac{1}{2} < \frac{1}{q} + \frac{1}{r} \le \frac{2}{q}$, so $q < 4$. The strict inequality is what rules out $q = 4$. (In part (i), $q = 4$ was allowed, and gave the equality case $(2,4,4)$.) With $q \ge p = 2$ we get $q = 2$ or $q = 3$.

- $q = 2$: the sum is $\frac{1}{2} + \frac{1}{2} + \frac{1}{r} = 1 + \frac{1}{r} > 1$ for *every* r. So $(2,2,r)$ is a solution for every integer $r \ge 2$.
- $q = 3$: we need $\frac{1}{r} > \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, so $r < 6$. With $r \ge q = 3$ this gives $r = 3,4,5$.

So, apart from the $p = 1$ family, the solutions are

$$(p,q,r) = (2,2,r) \text{ for any integer } r \ge 2, \qquad (2,3,3), \quad (2,3,4), \quad (2,3,5).$$

Each step above was forced, so there are no others.

Checks

Substitute directly. For example, $\frac{1}{2} + \frac{1}{3} + \frac{1}{5} = \frac{31}{30} > 1$, but $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$ exactly. That shows why $(2,3,5)$ is the last of the $q = 3$ family, with $(2,3,6)$ the boundary case from part (i). Likewise $(2,4,4)$ gives exactly 1, so nothing with $p = 2$, $q = 4$ makes it into part (ii). A computer search over all $1 \le p \le q \le r < 400$ confirms both lists exactly.

Common slips

- Forgetting the infinite family $(2, 2, r)$ in part (ii). Once $p = q = 2$, the first two terms already add up to 1.
- Writing $p \leq 3$ instead of $p < 3$ in part (ii) and then wasting time on $p = 3$ (which gives a sum of at most 1).
- Stating the list of solutions without showing that every case has been checked. The "no others" is half the marks.

Beyond STEP

The solutions of $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$ are exactly the triples for which a triangle with angles $\frac{\pi}{p}, \frac{\pi}{q}, \frac{\pi}{r}$ lives on a sphere. They match the finite symmetry groups in three dimensions: dihedral $(2, 2, r)$, tetrahedral $(2, 3, 3)$, octahedral $(2, 3, 4)$ and icosahedral $(2, 3, 5)$. Part (i) gives the three regular tilings of the flat plane. Equations like this, solved in integers, are Diophantine equations. Davenport's *The Higher Arithmetic* is a good introduction, and Allenby's *Rings, Fields and Groups* covers the symmetry groups.

WORKED SOLUTION

STEP I 1993, Question 4

Pure • Integration by Substitution

STEP I 1993, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q4

The key idea

The substitution $t = \pi - x$ reflects the interval $[0, \pi]$ onto itself, and it leaves $\sin x$ unchanged, because $\sin(\pi - t) = \sin t$. So the awkward factor x turns into $\pi - t$, and the integral we want appears again on the right-hand side with a minus sign. That lets us solve for it. For the integrals that follow, the main job is to write $1 + \cos^2 x$ in terms of $\sin x$. Over $[\pi, 2\pi]$, a shift by π brings us back to $[0, \pi]$.

Solution

The general result

Let $I = \displaystyle\int_0^\pi x\, \mathrm{f}(\sin x)\, \mathrm{d}x$ and put $t = \pi - x$. Then $\mathrm{d}x = -\mathrm{d}t$. When $x = 0$, $t = \pi$, and when $x = \pi$, $t = 0$. Also $\sin x = \sin(\pi - t) = \sin t$. Hence

$$\begin{aligned} I &= \int_\pi^0 (\pi - t)\, \mathrm{f}(\sin t)\, (-\mathrm{d}t) = \int_0^\pi (\pi - t)\, \mathrm{f}(\sin t)\, \mathrm{d}t \\ &= \pi \int_0^\pi \mathrm{f}(\sin t)\, \mathrm{d}t - \int_0^\pi t\, \mathrm{f}(\sin t)\, \mathrm{d}t. \end{aligned}$$

The last integral is I again, since the name of the variable of integration doesn't matter. So $2I = \pi \int_0^\pi \mathrm{f}(\sin t)\, \mathrm{d}t$, which gives

$$\int_0^\pi x\, \mathrm{f}(\sin x)\, \mathrm{d}x = \frac{\pi}{2} \int_0^\pi \mathrm{f}(\sin x)\, \mathrm{d}x,$$

as required. (We need the integrals to exist, for example with f continuous on $[-1, 1]$.)

The integral over $[0, \pi]$

Ask yourself The integrand involves $\cos^2 x$, not just $\sin x$. Is it really of the form $x\,\mathrm{f}(\sin x)$?

Yes. Since $\cos^2 x = 1 - \sin^2 x$, we have $1 + \cos^2 x = 2 - \sin^2 x$, so

$$\frac{x \sin x}{1 + \cos^2 x} = x\,\mathrm{f}(\sin x) \qquad \text{with} \qquad \mathrm{f}(s) = \frac{s}{2 - s^2},$$

and f is continuous on $[-1, 1]$. By the general result,

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\,\mathrm{d}x.$$

The remaining integral is easy because $\sin x$ is (minus) the derivative of $\cos x$. Substituting $u = \cos x$, with $\mathrm{d}u = -\sin x\,\mathrm{d}x$,

$$\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x}\,\mathrm{d}x = \int_1^{-1} \frac{-\mathrm{d}u}{1 + u^2} = \int_{-1}^{1} \frac{\mathrm{d}u}{1 + u^2} = \Big[\arctan u\Big]_{-1}^{1} = \frac{\pi}{4} - \Big(-\frac{\pi}{4}\Big) = \frac{\pi}{2}.$$

Therefore

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x = \frac{\pi}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{4}.$$

The integral over $[0, 2\pi]$

The general result only covers $[0, \pi]$, so we split the range:

$$\int_0^{2\pi} = \int_0^{\pi} + \int_{\pi}^{2\pi}.$$

Ask yourself How can we bring $\int_{\pi}^{2\pi}$ back to an integral over $[0, \pi]$?

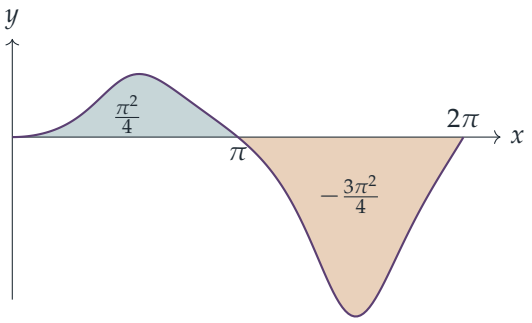
Shift by π: put $x = u + \pi$, so u runs from 0 to π. Then $\sin x = -\sin u$ and $\cos x = -\cos u$, so $\cos^2 x = \cos^2 u$. Hence

$$\begin{aligned}
\int_{\pi}^{2\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x &= \int_0^{\pi} \frac{(u + \pi)(-\sin u)}{1 + \cos^2 u}\,\mathrm{d}u \\
&= -\int_0^{\pi} \frac{u \sin u}{1 + \cos^2 u}\,\mathrm{d}u - \pi \int_0^{\pi} \frac{\sin u}{1 + \cos^2 u}\,\mathrm{d}u \\
&= -\frac{\pi^2}{4} - \pi \cdot \frac{\pi}{2} = -\frac{3\pi^2}{4},
\end{aligned}$$

using both results from the previous part. Adding the two pieces,

$$\int_0^{2\pi} \frac{x \sin x}{1 + \cos^2 x}\,\mathrm{d}x = \frac{\pi^2}{4} - \frac{3\pi^2}{4} = -\frac{\pi^2}{2}.$$

The graph shows why the answer is negative. On $[\pi, 2\pi]$ the factor $\sin x$ is negative, and there the factor x is larger than it is on $[0, \pi]$. So the area below the axis outweighs the area above it.



Checks

- Take $f \equiv 1$: the general result says $\int_0^{\pi} x\,dx = \frac{\pi}{2} \cdot \pi = \frac{\pi^2}{2}$, which is correct.
- The integrand is positive on $(0, \pi)$, so the first answer must be positive, and $\pi^2/4 \approx 2.467$ is. Simpson's rule gives 2.4674011 for the first integral and -4.9348022 for the second, matching $\pi^2/4$ and $-\pi^2/2$.
- The general identity also checks numerically for several random choices of f.

Common slips

- Applying the general result directly on $[0, 2\pi]$. It relies on the reflection $x \mapsto \pi - x$ mapping the interval to itself, and on $[0, 2\pi]$ it gives the wrong answer $\frac{\pi}{2} \cdot 0 = 0$ (the correct reflection there is $x \mapsto 2\pi - x$, under which $\sin x$ changes sign).
- Losing the minus sign from $dx = -dt$, or forgetting to swap the limits back.
- Writing $\arctan(-1) = 3\pi/4$: the arctangent takes values in $(-\pi/2, \pi/2)$, so $\arctan(-1) = -\pi/4$.

Beyond STEP

The trick here is to exploit a symmetry of the domain that the integrand nearly respects: replacing x by $\pi - x$ and averaging the two versions of the integral. The same idea gives $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx = \frac{\pi}{4}$ without any integration. At university, symmetry arguments like this turn up throughout analysis and in Fourier series, where the parity of a function decides which terms survive. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with substitutions of this kind, and Abbott, *Understanding Analysis*, makes the change-of-variables rule rigorous.

WORKED SOLUTION

STEP I 1993, Question 5

Pure • Modulus and Triangle Inequality

STEP I 1993, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q5

The key idea

Everything rests on the triangle inequality: the length of a sum is at most the sum of the lengths. Applied term by term, it bounds the polynomial by a geometric series in $|z|$, and with $|z| \leq \frac{1}{2}$ that series stays strictly below 1. A root must make the polynomial part equal to -1, which has modulus exactly 1, so a root can't lie in the disc $|z| \leq \frac{1}{2}$.

Solution

The modulus and the product rule

For $z = x + \mathrm{i}y$ with x, y real, the modulus is the distance from the origin to the point (x, y):

$$|z| = \sqrt{x^2 + y^2} \qquad \text{(the non-negative square root)}.$$

Let $z_1 = a + \mathrm{i}b$ and $z_2 = c + \mathrm{i}d$ with a, b, c, d real. Then $z_1 z_2 = (ac - bd) + \mathrm{i}(ad + bc)$, so

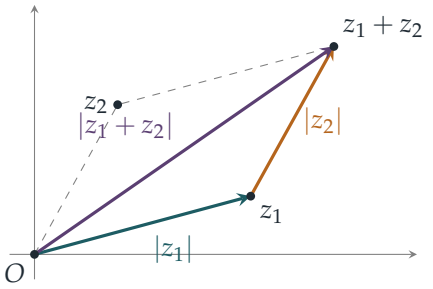
$$\begin{aligned} |z_1 z_2|^2 &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2 \\ &= a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 \\ &= (a^2 + b^2)(c^2 + d^2) = |z_1|^2 |z_2|^2. \end{aligned}$$

Both $|z_1 z_2|$ and $|z_1||z_2|$ are non-negative, so taking square roots gives $|z_1 z_2| = |z_1||z_2|$, as required.

The triangle inequality

In the Argand diagram, let O be the origin, A the point z_1 and C the point $z_1 + z_2$. Then $\overrightarrow{AC}$ represents z_2 (it is the same vector as O to z_2), so OAC is a triangle with side lengths

$$OA = |z_1|, \qquad AC = |z_2|, \qquad OC = |z_1 + z_2|.$$



A straight line is the shortest route between two points: going from O to C directly is no longer than going via A. So

$$|z_1 + z_2| \leq |z_1| + |z_2|,$$

with equality exactly when O, A, C are collinear with A between them, that is, when z_1 and z_2 point in the same direction (or one of them is 0).

If you prefer algebra: writing $\bar{z}$ for the conjugate,

$$\begin{aligned} |z_1 + z_2|^2 &= |z_1|^2 + |z_2|^2 + 2\,\mathrm{Re}(z_1\bar{z}_2) \\ &\leq |z_1|^2 + |z_2|^2 + 2|z_1||z_2| = (|z_1| + |z_2|)^2, \end{aligned}$$

since the real part of a complex number is at most its modulus, and $|z_1\bar{z}_2| = |z_1||z_2|$ by the product rule.

Applying the inequality repeatedly (an easy induction on the number of terms) gives

$$|w_1 + w_2 + \cdots + w_n| \leq |w_1| + |w_2| + \cdots + |w_n|$$

for any complex numbers $w_1, \ldots, w_n$.

The bound on the polynomial

Suppose $|a_j| \leq 1$ for each j and $|z| \leq \frac{1}{2}$. Then

$$\begin{aligned} \left|a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z\right| &\leq |a_n||z|^n + |a_{n-1}||z|^{n-1} + \cdots + |a_1||z| \\ &\leq |z|^n + |z|^{n-1} + \cdots + |z| \\ &\leq \left(\tfrac{1}{2}\right)^n + \left(\tfrac{1}{2}\right)^{n-1} + \cdots + \tfrac{1}{2} \\ &= 1 - \left(\tfrac{1}{2}\right)^n \\ &< 1. \end{aligned}$$

The first line is the n-term triangle inequality together with the product rule (used repeatedly, so that $|a_k z^k| = |a_k||z|^k$). The second uses $|a_j| \leq 1$, the third uses $0 \leq |z| \leq \frac{1}{2}$, and the fourth sums the finite geometric series. This is the required strict inequality.

Ask yourself Where does the *strict* inequality come from? Each step above is only "$\leq$".

It comes from the finite geometric series: $\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}$ falls short of 1 by exactly $\frac{1}{2^n}>0$. That gap is what the deduction needs.

The roots

(The paper's final line says $|x|>\frac{1}{2}$; this is evidently a misprint for $|w|>\frac{1}{2}$, since w is the root being discussed.)

Let w be a root, so

$$a_nw^n+a_{n-1}w^{n-1}+\cdots+a_1w=-1, \qquad \text{and hence} \qquad |a_nw^n+\cdots+a_1w|=|-1|=1.$$

Suppose, for contradiction, that $|w|\leq\frac{1}{2}$. Then the previous part, with $z=w$, gives $|a_nw^n+\cdots+a_1w|<1$, which contradicts the modulus being exactly 1. Therefore every root satisfies $|w|>\frac{1}{2}$.

Checks

- The bound is attained in the limit: with every $a_j=1$ and $z=\frac{1}{2}$ the left-hand side is exactly $1-2^{-n}$.
- The root result is sharp too. Taking every $a_j=-1$ gives $1-z-z^2-\cdots-z^n=0$, whose positive root is 1 for $n=1$, 0.618 for $n=2$ (the golden ratio conjugate), 0.5087 for $n=5$ and 0.50025 for $n=10$: always above $\frac{1}{2}$, but tending to it.
- Numerically, we tested 50 000 random choices of a_j in the unit disc and z in the disc of radius $\frac{1}{2}$ (largest value found 0.79), and found all roots of 2000 random such polynomials: the smallest root modulus was 0.562.

Common slips

- Writing $|a_1z+\cdots+a_nz^n|=|a_1||z|+\cdots+|a_n||z|^n$. The triangle inequality only gives "$\leq$".
- Using the infinite sum $\frac{1}{2}+\frac{1}{4}+\cdots=1$, which gives only ≤ 1 and spoils the deduction. Use the finite sum.
- Forgetting to justify $|z^k|=|z|^k$: it follows from the product rule you have just proved.

Beyond STEP

This is a baby version of the bounds on polynomial roots that underpin numerical root-finding: Cauchy's bound puts every root of a polynomial in a disc whose radius

depends only on the sizes of the coefficients, and the same "dominant term wins" argument drives Rouché's theorem in complex analysis. The triangle inequality itself is the defining property of a *metric*, the abstract notion of distance at the heart of first-year analysis. For a beautifully geometric treatment of all this, read Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP I 1993, Question 6

Pure • Curve Sketching

STEP I 1993, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q6

The key idea

Everything turns on how x^N behaves when $N = 10^{100}$. If $|x| < 1$, even a little below 1, then x^N is unimaginably small; if $|x| > 1$, even a little above 1, it is unimaginably large; and at $x = \pm 1$ it is exactly 1 (because N is even). So the fraction $x^N/(1+x^N)$ behaves like a switch: 0 inside $(-1, 1)$ and 1 outside. For h the idea is the opposite one: $10^{-N}x$ is so tiny that sin is indistinguishable from its argument.

Solution

Throughout, write

$$s(x) = \frac{x^N}{1+x^N} = \frac{1}{1+x^{-N}} \quad (x \neq 0), \qquad s(0) = 0,$$

so that $\mathrm{f}(x) = 2 + s(x)$ and $\mathrm{g}(x) = x\,s(x)$.

The features of f

- *Symmetry and range.* N is even, so $x^N = (-x)^N \geq 0$. Hence f is even, and $0 \leq s(x) < 1$ gives $2 \leq \mathrm{f}(x) < 3$.
- *The level $y = 2$ for $|x| < 1$.* Here $x^N \to 0$ with staggering speed: for example $0.99^N = 10^{N \log_{10} 0.99} \approx 10^{-4.4 \times 10^{97}}$. So $s(x) \approx x^N$ is far below anything a graph could show, and $\mathrm{f}(x) = 2$ to any visible accuracy. Exactly, $\mathrm{f}(0) = 2$.
- *The level $y = 3$ for $|x| > 1$.* Now x^{-N} is just as tiny, so $s(x) = 1/(1+x^{-N})$ is 1 to any visible accuracy and $\mathrm{f}(x) = 3$, approached from below. (The line $y = 3$ is in fact an asymptote as $|x| \to \infty$.)
- *The jumps at $x = \pm 1$.* At $x = \pm 1$, $x^N = 1$, so $\mathrm{f}(\pm 1) = \frac{5}{2}$, halfway up. The function is continuous and differentiable, so the vertical segments in the sketch are really an

extremely steep climb. Differentiating,

$$\mathrm{f}'(x) = \frac{Nx^{N-1}}{(1+x^N)^2}, \qquad \mathrm{f}'(1) = \frac{N}{4} = 2.5 \times 10^{99}.$$

? **Ask yourself** How narrow is the "vertical" part of the graph?

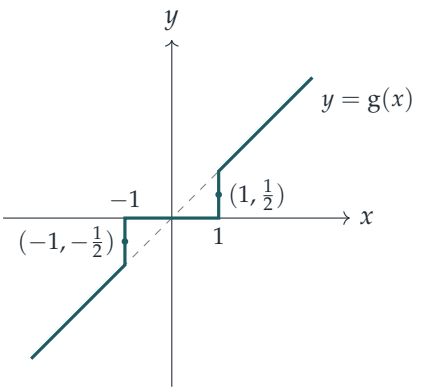
Put $x = 1+\delta$ with δ small, so $x^N = \mathrm{e}^{N\ln(1+\delta)} \approx \mathrm{e}^{N\delta}$ and $s(x) \approx 1/(1+\mathrm{e}^{-N\delta})$. This climbs from 0.01 to 0.99 as $N\delta$ goes from $-\ln 99$ to $\ln 99$, that is, over an interval of width about 9×10^{-100}. On any drawing, that is a vertical line. The same happens at $x = -1$ by symmetry. (The sign of f' also tells us that f increases for $x > 0$ and decreases for $x < 0$, since x^{N-1} has the sign of x.)

The graph of g

Since $\mathrm{g}(x) = x\,s(x)$, the switch s now multiplies the line $y = x$:

- for $|x| < 1$, $s(x)$ is negligible, so $\mathrm{g}(x) = 0$ to any visible accuracy (exactly 0 at $x = 0$);
- for $|x| > 1$, $s(x)$ is 1 to any visible accuracy, so $\mathrm{g}(x) = x$; precisely, $x - \mathrm{g}(x) = \dfrac{x}{1+x^N}$, which is negligible;
- at $x = \pm 1$, $\mathrm{g}(\pm 1) = \pm\frac{1}{2}$, halfway up each jump.

g is odd, because s is even and x is odd. It is also increasing: $\mathrm{g}'(x) = s(x) + x\,s'(x)$, and both terms are ≥ 0 because x and $s'(x)$ have the same sign. At $x = 1$ the gradient is $\mathrm{g}'(1) = \frac{1}{2} + \frac{N}{4}$, so again we see an apparently vertical jump, this time from 0 up to 1 at $x = 1$ and from -1 up to 0 at $x = -1$.



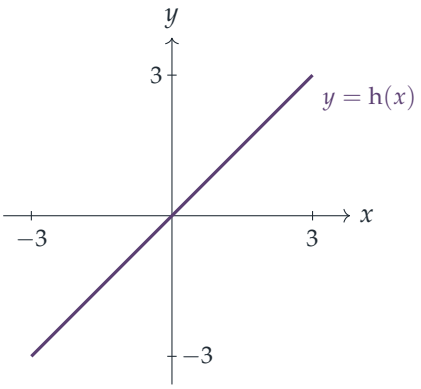
The sketch of g on $[-3, 3]$: zero on $(-1, 1)$, the line $y = x$ (dashed) outside, joined by near-vertical climbs through $(\pm 1, \pm\frac{1}{2})$.

The graph of h

Let $u = 10^{-N}x$. For $|x| \le 3$ we have $|u| \le 3 \times 10^{-N}$, which is extraordinarily small. For small u, $\sin u = u - \frac{u^3}{6} + \cdots$, and for $u > 0$ we have the bounds $u - \frac{u^3}{6} \le \sin u \le u$. Multiplying by 10^N, for $0 \le x \le 3$,

$$0 \le x - \mathrm{h}(x) \le \frac{x^3}{6}\,10^{-2N} \le \frac{27}{6} \times 10^{-2N},$$

and h is odd, so the same holds for $-3 \le x \le 0$ with the signs reversed. So on $[-3,3]$ the graph of h is the straight line $y = x$ through the origin to any conceivable accuracy, lying a whisker below it for $x > 0$ and above it for $x < 0$. Its gradient is $\mathrm{h}'(x) = \cos(10^{-N}x)$, which is 1 to within about 10^{-2N}.



The sine wave is still there, but it is stretched enormously: its first maximum is at $x = \frac{\pi}{2} \times 10^N$, with height 10^N, so the window $[-3,3]$ sees only the tiny straight-looking piece through the origin.

Checks

- Evaluating directly with $s = 1/(1 + \mathrm{e}^{-N\ln|x|})$: $s(0.999)$ and $s(1.001)$ are 0 and 1 to machine precision, and $s(\pm 1) = \frac{1}{2}$ exactly.
- With a modest N such as $N = 10$ or 200, numerical differentiation confirms $s'(1) = N/4$ and $\mathrm{g}'(1) = \frac{1}{2} + N/4$, and a fine grid shows g is increasing on $[-3,3]$. Increasing N only sharpens the steps.
- $10^{100}\sin(10^{-100}x)$ agrees with x to full double precision for random $x \in [-3,3]$, and the scaled-down analogue $M\sin(x/M)$ with $M = 1000$ obeys the bound $0 < x - M\sin(x/M) \le x^3/(6M^2)$.

Common slips

- Forgetting that N is even: x^N is huge and *positive* for $x < -1$, so f is 3 there, not something else.

- Describing the steps as genuine discontinuities. All three functions are continuous (indeed differentiable); the climbs are just too steep to draw.
- Treating h as an oscillating graph. Over $[-3, 3]$ it never gets anywhere near a turning point.

Beyond STEP

The functions $x^n/(1+x^n)$ converge *pointwise*, as $n \to \infty$ through even values, to a step function that is 0 inside $(-1, 1)$, $\frac{1}{2}$ at ± 1 and 1 outside: a sequence of smooth functions whose limit is discontinuous. This is the classic example showing why analysts need *uniform* convergence, a central idea in a first course on real analysis. The function h illustrates the other side: $M\sin(x/M) \to x$ uniformly on bounded intervals as $M \to \infty$. Abbott's *Understanding Analysis* treats both ideas very readably.

WORKED SOLUTION

STEP I 1993, Question 7

Pure • Roots of Cubics

STEP I 1993, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q7

The key idea

A cubic with positive leading coefficient rises from $-\infty$ to $+\infty$, so everything about its roots is decided by its two turning points. If the curve's local maximum is above the axis and its local minimum is below, the graph must cross three times; otherwise it crosses fewer times. For $x^3 + ax^2 + b$ the turning points are easy to find exactly, because the derivative factorises.

Solution

The turning points

For $\mathrm{f}(x) = x^3 + Ax^2 + B$,

$$\mathrm{f}'(x) = 3x^2 + 2Ax = x(3x + 2A),$$

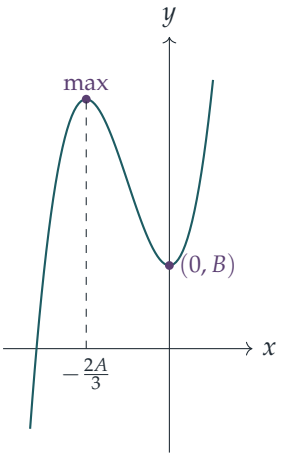
so the stationary points are at $x = 0$ and $x = -\frac{2A}{3}$. The values there are $\mathrm{f}(0) = B$ and

$$\mathrm{f}\left(-\tfrac{2A}{3}\right) = -\frac{8A^3}{27} + \frac{4A^3}{9} + B = \frac{4A^3}{27} + B.$$

Also $\mathrm{f}''(x) = 6x + 2A$, so $\mathrm{f}''(0) = 2A$ and $\mathrm{f}''(-\frac{2A}{3}) = -2A$. When $A > 0$ the point at $x = 0$ is a minimum and the one at $x = -\frac{2A}{3}$ a maximum; when $A < 0$ the roles swap.

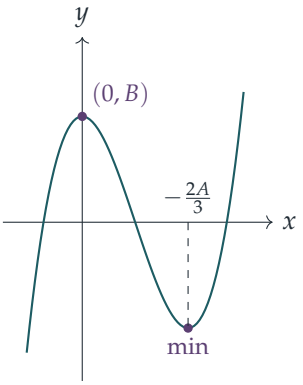
Sketch with $A > 0$, $B > 0$

The maximum is at $\left(-\frac{2A}{3},\, B + \frac{4A^3}{27}\right)$, to the left of the y-axis and higher than B; the minimum is at $(0, B)$, on the positive y-axis. Both turning points are above the x-axis, so the curve crosses it exactly once, to the left of the maximum. (The figure uses $A = 3$, $B = 2$: maximum $(-2, 6)$, minimum $(0, 2)$.)



Sketch with $A < 0, B > 0$

Now the maximum is at $(0, B)$ and the minimum is at $\left(-\frac{2A}{3}, B + \frac{4A^3}{27}\right)$, to the right of the y-axis. Since $A^3 < 0$, the minimum value is less than B, and it may be positive, zero or negative. There is always one root with $x < 0$, because $\mathrm{f}(0) = B > 0$ and $\mathrm{f} \to -\infty$ as $x \to -\infty$. If the minimum dips below the axis, there are two further roots with $x > 0$ (as in the figure, $A = -3$, $B = 2$: minimum $(2, -2)$, roots $1 - \sqrt{3}$, 1, $1 + \sqrt{3}$). If the minimum is on the axis, the curve touches it there. If the minimum is above the axis, the negative root is the only one.



Counting the roots

Now take $\mathrm{g}(x) = x^3 + ax^2 + b$. If $a = 0$, then g is strictly increasing (its derivative $3x^2$ vanishes only at 0), so it has exactly one real root. So suppose $a \neq 0$. From the work above, g has two distinct turning points, with values b and $b + \frac{4a^3}{27}$.

? **Ask yourself** How do the two turning-point values tell us the number of roots?

Write $x_1 < x_2$ for the turning points. The cubic is strictly increasing on $(-\infty, x_1]$, strictly decreasing on $[x_1, x_2]$ and strictly increasing on $[x_2, \infty)$, so it has at most one root on each piece. If $\mathrm{g}(x_1) > 0 > \mathrm{g}(x_2)$, the intermediate value theorem gives a root in each of $(-\infty, x_1)$, (x_1, x_2) and (x_2, ∞) (using $\mathrm{g} \to \mp\infty$ as $x \to \mp\infty$), so there are three distinct roots. If instead both values have the same sign, or either one is zero, at least one piece has no root in its interior and the roots number at most two. The maximum is always the larger of the two values, so "one positive, one negative" is the same as "their product is negative". Hence g has three distinct real roots exactly when

$$b\left(b + \frac{4a^3}{27}\right) < 0 \quad \Longleftrightarrow \quad 27b^2 + 4a^3 b < 0.$$

(Multiplying by $27 > 0$ keeps the inequality's direction.) This also covers $a = 0$, where $27b^2 + 4a^3 b = 27b^2 \geq 0$.

The first condition

Suppose $27b^2 + 3a^3 b < 0$. Then $3a^3 b < -27b^2 \leq 0$, so $a^3 b < 0$, and therefore

$$27b^2 + 4a^3 b = \left(27b^2 + 3a^3 b\right) + a^3 b < 0 + 0 = 0.$$

By the criterion above, the equation has three distinct real roots, as required.

The second condition

As printed, the second condition repeats the inequality $27b^2 + 4a^3 b < 0$, and with "$<$" the claim is false: $a = -3$, $b = 1$ gives $27 - 108 = -81 < 0$, yet $x^3 - 3x^2 + 1$ takes the values 1 at $x = 0$ and -3 at $x = 2$, so it has three distinct real roots. The intended inequality must be the reverse one. The criterion shows that if

$$27b^2 + 4a^3 b > 0$$

(indeed if $27b^2 + 4a^3 b \geq 0$), then the equation has fewer than three distinct real roots. This completes the question.

So the exact dividing line is $27b^2 + 4a^3 b = 0$. The condition with $3a^3$ is sufficient but not necessary for three roots: $a = -2$, $b = 1$ gives $27b^2 + 3a^3 b = 3 > 0$ but $27b^2 + 4a^3 b = -5 < 0$, and $x^3 - 2x^2 + 1 = (x - 1)(x^2 - x - 1)$ has the three roots 1 and $\frac{1 \pm \sqrt{5}}{2}$.

Checks

With $A = -3$, $B = 2$ we get $x^3 - 3x^2 + 2 = (x - 1)(x^2 - 2x - 2)$, whose roots $1, 1 \pm \sqrt{3}$ agree with the sketch, and $27 \cdot 4 + 4(-27)(2) = -108 < 0$, as the criterion predicts. Numerically, for about 80,000 random rational pairs (a, b) satisfying the first condition, the stationary values always had opposite signs. Over 200,000 pairs, "three distinct roots" matched $27b^2 + 4a^3 b < 0$ exactly.

Common slips

- Forgetting the case $a = 0$, where the two turning points merge into a point of inflection.
- Treating a zero turning-point value as giving three roots: it gives a repeated root, so only two distinct ones.
- Trying to prove the second statement exactly as printed instead of noticing that it contradicts the criterion.

Beyond STEP

The quantity $-(4a^3b + 27b^2)$ is the *discriminant* of $x^3 + ax^2 + b$, the cubic analogue of $b^2 - 4ac$. It equals the product of the squared differences of the roots, so it is positive for three distinct real roots, zero for a repeated root and negative when two roots are complex. Discriminants and the symmetric functions of roots are developed in Galois theory; Allenby's *Rings, Fields and Groups* is a good introduction.

WORKED SOLUTION

STEP I 1993, Question 8

Pure • Spheres and Vectors

STEP I 1993, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q8

The key idea

For part (i), put the origin and axes where the symmetry is. If the centre of the sphere sits on the z-axis and the plane is $z = 0$, the equations do almost all the work. Two spheres are then no harder, because subtracting their equations cancels every square and leaves the equation of a *plane*. In part (ii), coordinates again reduce everything to a scalar product. Put B at the origin, so that both pole tops are measured from E.

Solution

Part (i)

Sphere and plane. Let the sphere have centre O' and radius $r > 0$, and let N be the foot of the perpendicular from O' to the plane. Take N as origin, the plane as $z = 0$, and the z-axis along NO'. Then $O' = (0, 0, d)$, where $d \geq 0$ is the distance from the centre to the plane. The sphere is

$$x^2 + y^2 + (z - d)^2 = r^2,$$

so a point lies on both the sphere and the plane exactly when $z = 0$ and

$$x^2 + y^2 = r^2 - d^2.$$

- If $d < r$, this is the circle in the plane with centre N and radius $\sqrt{r^2 - d^2}$.
- If $d = r$, then $x^2 + y^2 = 0$ forces $x = y = 0$, so the intersection is the single point N (the plane is a tangent plane).
- If $d > r$, the right-hand side is negative, so there are no points and the intersection is empty.

These cases cover every possibility, which proves the first statement.

Two spheres. Let the spheres have radii $r, s > 0$ and distinct centres a distance $c > 0$

apart. Put the first centre at the origin and the second at $(0,0,c)$:

$$x^2+y^2+z^2=r^2, \qquad x^2+y^2+(z-c)^2=s^2.$$

Ask yourself How can two quadratic equations have a solution set that is just a circle?

Subtract the second equation from the first. All the squared terms cancel, leaving

$$2cz-c^2=r^2-s^2 \iff z=h, \qquad \text{where } h=\frac{r^2-s^2+c^2}{2c}.$$

Dividing by c is allowed because the centres are distinct. A point lies on both spheres if and only if it lies on the first sphere and satisfies the difference equation, so

$$(\text{sphere } 1)\cap(\text{sphere } 2)=(\text{sphere } 1)\cap(\text{plane } z=h).$$

By the first result this is a circle, a point or the empty set. (Explicitly, it is the circle $x^2+y^2=r^2-h^2$ in the plane $z=h$ when $|h|<r$, a single point when $|h|=r$, and empty when $|h|>r$.) The circle lies in a plane perpendicular to the line of centres, and its centre is on that line.

Part (ii)

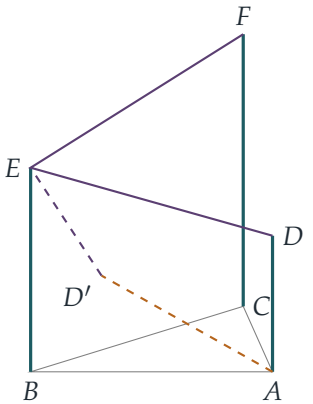
Assumption about the bases. The question does not say where A, B and C stand. We take them to be at the vertices of an equilateral triangle of side 2 m on level ground. This is the configuration that gives the stated angle (see the Checks).

Take B as origin with the z-axis vertical:

$$B=(0,0,0), \quad A=(2,0,0), \quad C=(1,\sqrt{3},0),$$

so that the tops of the poles are

$$D=(2,0,2), \quad E=(0,0,3), \quad F=(1,\sqrt{3},4).$$



In the diagram, D' (dashed) is the position of D after the vandals have moved the pole.

The original sculpture. Using vectors from E,

$$\overrightarrow{ED} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}, \qquad \overrightarrow{EF} = \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix},$$

so $\overrightarrow{ED} \cdot \overrightarrow{EF} = 2 + 0 - 1 = 1$ and $|\overrightarrow{ED}| = |\overrightarrow{EF}| = \sqrt{5}$. Therefore

$$\cos \angle DEF = \frac{1}{\sqrt{5}\sqrt{5}} = \frac{1}{5}, \qquad \text{so} \qquad \angle DEF = \cos^{-1} \tfrac{1}{5},$$

as required.

After the vandals. The pole AD still has length 2 and still stands at A. It now lies in the vertical plane through A and B (the plane $y = 0$), leaning towards B at $\frac{1}{4}\pi$ to the ground. Moving from A towards B is the direction of $-x$, so

$$D' = \left(2 - 2\cos \tfrac{\pi}{4},\ 0,\ 2\sin \tfrac{\pi}{4}\right) = \left(2 - \sqrt{2},\ 0,\ \sqrt{2}\right).$$

Since $2 - \sqrt{2} \approx 0.59$ lies between 0 and 2, the point D' really is above the segment AB. Now

$$\overrightarrow{ED'} = \begin{pmatrix} 2 - \sqrt{2} \\ 0 \\ \sqrt{2} - 3 \end{pmatrix}, \qquad \overrightarrow{ED'} \cdot \overrightarrow{EF} = (2 - \sqrt{2}) + (\sqrt{2} - 3) = -1,$$

and

$$|\overrightarrow{ED'}|^2 = (2 - \sqrt{2})^2 + (3 - \sqrt{2})^2 = (6 - 4\sqrt{2}) + (11 - 6\sqrt{2}) = 17 - 10\sqrt{2}.$$

Ask yourself The dot product is negative. What does that tell us about the angle?

A negative scalar product means the angle is obtuse, so we need the value of $\cos^{-1}$ that lies between $\frac{1}{2}\pi$ and π:

$$\cos \angle D'EF = \frac{-1}{\sqrt{5}\sqrt{17 - 10\sqrt{2}}} = \frac{-1}{\sqrt{85 - 50\sqrt{2}}} \approx -0.264542,$$

$$\angle D'EF \approx 1.839 \text{ radians (to four figures).}$$

Checks

- With an equilateral base of side s, the same method gives $\cos \angle DEF = \dfrac{s^2/2 - 1}{s^2 + 1}$. This equals $\frac{1}{5}$ only when $s = 2$, which supports the assumed base. A numerical check confirms $\cos \angle DEF = 0.2$ for $s = 2$.
- $|AD'| = 2$ and $\angle BAD' = \frac{1}{4}\pi$ were checked numerically. The new angle came out as 1.838525… radians (about 105.3°), agreeing with the exact form above.
- Part (i): random spheres were tested with points on the predicted circles; they lay on both surfaces to within 10^{-12}.

Common slips

- In the two-sphere case, claiming that the difference of the equations *is* the intersection. The plane only contains it; we still have to intersect the plane with one of the spheres.
- Placing D' on the far side of A, away from B. That point is not above the segment AB, and it gives a different angle.
- Giving the answer in degrees, or as the acute angle $\cos^{-1}(0.2645)$, after losing the minus sign.

Beyond STEP

Subtracting the equations of two spheres to get a plane is the idea behind the *radical plane* (in two dimensions, the radical axis) of two circles or spheres. The same trick turns GPS-style trilateration into a linear problem. More generally, the intersection of any quadric surface with a plane is a conic, and classifying such intersections is a good first exercise in linear algebra with quadratic forms. Strang's *Introduction to Linear Algebra* develops the tools, and Siklos's *Advanced Problems in Mathematics* has more STEP-style coordinate geometry.

WORKED SOLUTION

STEP I 1993, Question 9

Pure • First-Order Differential Equations

STEP I 1993, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q9

The key idea

This is a mixing problem: the rate of change of the amount of A in a vat is (rate in) minus (rate out), and because mixing is rapid the liquid leaving has the same concentration as the vat. The first thing to notice is that each vat's total volume never changes, so the concentrations are just "amount of A divided by a constant". Each vat then gives a first-order linear equation, and the output of the first vat becomes the forcing term for the second.

Solution

The first vat

Liquid enters the first vat at a total rate $a + (b - a) = b$ litres per second and leaves at b litres per second, so its volume stays at its initial value K litres for all $t \geq 0$.

Mixing is rapid, so at time t the contents are uniform, and the fraction of the liquid that is chemical A is $V(t)/K$. Over a short time δt, $a\,\delta t$ litres of A flow in, and $b\,\delta t$ litres of mixture flow out, of which $\frac{V}{K}\,b\,\delta t$ litres are A. Hence

$$\delta V \approx a\,\delta t - \frac{bV}{K}\,\delta t,$$

and dividing by δt and letting $\delta t \to 0$,

$$\dot{V}(t) = -\frac{bV(t)}{K} + a,$$

which is the required equation.

The equation is linear with constant coefficients. Writing it as $\dot{V} + \frac{b}{K}V = a$ and multiplying by the integrating factor $\mathrm{e}^{bt/K}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(V\mathrm{e}^{bt/K}\right) = a\,\mathrm{e}^{bt/K} \quad \Longrightarrow \quad V\mathrm{e}^{bt/K} = \frac{aK}{b}\mathrm{e}^{bt/K} + C.$$

Initially the vat holds only chemical B, so $V(0) = 0$, giving $C = -\frac{aK}{b}$. Therefore

$$V(t) = \frac{aK}{b}\left(1 - \mathrm{e}^{-bt/K}\right), \qquad t \geq 0,$$

as required.

The second vat

The second vat receives b litres per second from the first and empties at b litres per second, so its volume stays at L litres. Let $W(t)$ be the volume of A in it at time t, so $W(0) = 0$.

Ask yourself At what rate does chemical A arrive in the second vat?

It arrives at exactly the rate it leaves the first vat, namely $\frac{bV}{K} = a\left(1 - \mathrm{e}^{-bt/K}\right)$. It leaves at rate $\frac{bW}{L}$, by the same rapid-mixing argument. So

$$\dot{W} + \frac{b}{L}W = a - a\,\mathrm{e}^{-bt/K}.$$

Multiplying by the integrating factor $\mathrm{e}^{bt/L}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(W\mathrm{e}^{bt/L}\right) = a\,\mathrm{e}^{bt/L} - a\,\mathrm{e}^{bt(1/L - 1/K)}.$$

Because $L \neq K$, the exponent $b\left(\frac{1}{L} - \frac{1}{K}\right) = \frac{b(K-L)}{KL}$ is non-zero, so we can integrate the second term as an exponential:

$$W\mathrm{e}^{bt/L} = \frac{aL}{b}\mathrm{e}^{bt/L} - \frac{aKL}{b(K-L)}\mathrm{e}^{b(K-L)t/(KL)} + D.$$

Multiplying through by $\mathrm{e}^{-bt/L}$, and noting $\frac{b(K-L)}{KL} - \frac{b}{L} = -\frac{b}{K}$,

$$W = \frac{aL}{b} - \frac{aKL}{b(K-L)}\mathrm{e}^{-bt/K} + D\mathrm{e}^{-bt/L}.$$

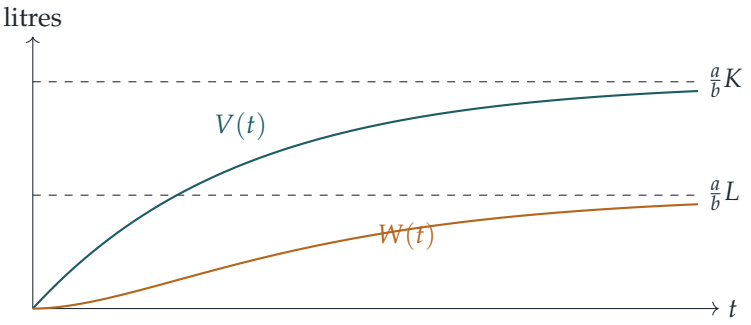
Setting $W(0) = 0$:

$$D = \frac{aKL}{b(K-L)} - \frac{aL}{b} = \frac{aL}{b} \cdot \frac{K - (K-L)}{K-L} = \frac{aL^2}{b(K-L)}.$$

So the second vat contains

$$\boxed{W(t) = \frac{aL}{b}\left(1 - \frac{K\mathrm{e}^{-bt/K} - L\mathrm{e}^{-bt/L}}{K-L}\right)}$$

litres of chemical A at time $t \geq 0$.



The sketch uses $a = 1$, $b = 2$, $K = 10$, $L = 5$. Notice that W starts with zero gradient: at first there is almost no A in the liquid arriving from the first vat.

Checks

- At $t = 0$ the bracket is $1 - \frac{K-L}{K-L} = 0$, as it should be. Also $\dot{W}(0) = 0$, matching the fact that A arrives at rate $bV(0)/K = 0$.
- As $t \to \infty$, $V \to \frac{a}{b}K$ and $W \to \frac{a}{b}L$: both vats end up with concentration a/b of A, which is the concentration of the incoming liquid.
- The formula is symmetric in K and L apart from the factor L in front, which is a nice consistency check on the algebra.
- With $a = 1$, $b = 2$, $K = 10$, $L = 5$, $t = 5$: $V = 5(1 - \mathrm{e}^{-1}) \approx 3.1606$ and $W = 2.5(1 - \mathrm{e}^{-1})^2 \approx 0.99894$. A numerical simulation of both vats (tracking volumes too) agrees with both formulas to about 10^{-13} for many random choices of a, b, K, L.

Common slips

- Using $b - a$ or a for the outflow of A: the outflow is the total outflow rate times the concentration.
- Taking the inflow of A to the second vat to be a: only the A that leaves the first vat reaches the second.
- Forgetting why $L \neq K$ is needed: if $L = K$ the forcing term $\mathrm{e}^{-bt/K}$ solves the homogeneous equation, and the answer instead contains a $t\,\mathrm{e}^{-bt/K}$ term.

Beyond STEP

A chain of well-mixed tanks is a standard compartment model, used for drug doses moving through the body and for radioactive decay chains. Written as a system $\dot{\mathbf{x}} = M\mathbf{x} + \mathbf{f}$, the decay rates b/K and b/L are the eigenvalues of M, and the case $L = K$ is a repeated eigenvalue, the same "resonance" that gives $t\,\mathrm{e}^{\lambda t}$ solutions. Boyce

and DiPrima's *Elementary Differential Equations* treats both mixing problems and linear systems in this way.

WORKED SOLUTION

STEP I 1993, Question 10

Mechanics • Equilibrium with Springs

STEP I 1993, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

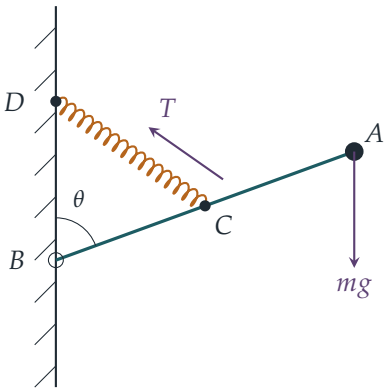
step.maths.org/questions/93-s1-q10

The key idea

Take moments about the hinge B: that removes the unknown reaction at B completely, leaving one equation linking the spring tension to the angle of the rod. The geometry is kinder than it looks, because $BD = BC = a$, so triangle BCD is isosceles and every length in it can be written in terms of half the angle at B.

Solution

We model the lamp as a particle, the rod as light and rigid, the hinge at B as smooth, and the spring as light and obeying Hooke's law. Let $\theta = \angle DBA$, the angle between the rod and the upward vertical BD. "Away from the wall" means $0 < \theta < \pi$: at $\theta = 0$ or $\theta = \pi$ the rod lies along the wall.



The length of the spring

Since $BC = BD = a$ and the angle between them is θ, triangle BCD is isosceles. Dropping the perpendicular from B to CD bisects both the angle θ and the side CD, so

$$CD = 2a\sin\tfrac{\theta}{2}, \qquad \text{distance from } B \text{ to the line } CD = a\cos\tfrac{\theta}{2}.$$

By Hooke's law the tension in the spring is

$$T = \frac{\lambda}{a}\left(2a\sin\tfrac{\theta}{2} - a\right) = \lambda\left(2\sin\tfrac{\theta}{2} - 1\right),$$

where a negative T would mean the spring is compressed and pushing.

Moments about B

The forces on the rod-and-lamp system are the weight mg at A, the spring force T at C along CD, and the reaction at B, which has no moment about B. The weight acts vertically at horizontal distance $2a\sin\theta$ from the wall, and turns the rod downwards (increasing θ). A tension pulls C towards D and turns the rod back towards the wall. For equilibrium,

$$T \cdot a\cos\tfrac{\theta}{2} = mg \cdot 2a\sin\theta = 4mga\sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2}.$$

For $0 < \theta < \pi$ we have $\cos\frac{\theta}{2} > 0$, so we may divide by it:

$$T = 4mg\sin\tfrac{\theta}{2}.$$

In particular $T > 0$: the spring must be stretched, which makes sense, since something has to hold the lamp up.

> **?** **Ask yourself** We now have two expressions for T. What does equating them give, and what must be true of the answer for it to describe a genuine position of the rod?

Write $s = \sin\frac{\theta}{2}$. Equating the two expressions for T,

$$\lambda(2s-1) = 4mgs \quad\Longrightarrow\quad s(2\lambda - 4mg) = \lambda \quad\Longrightarrow\quad s = \frac{\lambda}{2(\lambda - 2mg)},$$

provided $\lambda \neq 2mg$ (if $\lambda = 2mg$ the equation reads $0 = \lambda$, which is impossible). Since $0 < \theta < \pi$, a genuine position needs $0 < s < 1$.

- If $\lambda < 2mg$, then $s < 0$: no solution.
- If $\lambda > 2mg$, then $s > 0$, and

$$s < 1 \iff \lambda < 2\lambda - 4mg \iff \lambda > 4mg.$$

So when $\lambda > 4mg$ the value $s = \lambda/\left(2(\lambda - 2mg)\right)$ lies strictly between 0 and 1, giving exactly one angle $\theta \in (0, \pi)$ at which the moments balance. (Indeed $s > \frac{1}{2}$, so the spring

is indeed stretched.) The resultant force is then taken by the reaction at B, so the lamp can hang in equilibrium away from the wall, as required. Conversely, if $\lambda \leq 4mg$ there is no such position.

The angle is

$$\angle DBA = \theta = 2\arcsin\left(\frac{\lambda}{2(\lambda - 2mg)}\right),$$

or equivalently, since $\cos\theta = 1 - 2s^2$,

$$\cos\angle DBA = 1 - \frac{\lambda^2}{2(\lambda - 2mg)^2}.$$

Checks

- As $\lambda \to \infty$ the spring cannot stretch, so $CD \to a$ and triangle BCD becomes equilateral. The formula gives $s \to \frac{1}{2}$, so $\theta \to 60^\circ$. ✓
- As $\lambda \to 4mg$ from above, $s \to 1$ and $\theta \to 180^\circ$: the spring is too weak and the lamp sinks down to hang against the wall.
- With $\lambda = 6mg$: $s = \frac{3}{4}$, $\cos\theta = -\frac{1}{8}$, so $\theta \approx 97.2^\circ$. Substituting this angle into the full vector moment equation (with the spring force computed from coordinates) gives zero net moment. A numerical scan of the net moment over $0 < \theta < \pi$ finds a single root when $\lambda > 4mg$ (for example 112.9° at $\lambda = 5mg$) and none for $\lambda \leq 4mg$.

Common slips

- Using the whole rod length $2a$, rather than a, as the distance from B to the line of the spring force. The perpendicular distance is $a\cos\frac{\theta}{2}$.
- Resolving forces instead of taking moments: this brings in the two unknown components of the reaction at B for no gain.
- Forgetting to check that $0 < \sin\frac{\theta}{2} < 1$, which is exactly where the condition $\lambda > 4mg$ comes from.

Beyond STEP

This question asks only whether an equilibrium exists. The natural next question is whether it is *stable*: does a small nudge die away or grow? That is answered by writing down the potential energy $V(\theta)$ (gravity plus elastic energy) and looking at the sign of $V''(\theta)$ at the equilibrium, an idea that runs through first-year mechanics and later through the whole of dynamical systems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats energy methods and stability carefully.

WORKED SOLUTION

STEP I 1993, Question 11

Mechanics • Centre of Mass

STEP I 1993, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q11

The key idea

A rigid body hanging freely from one point is acted on by just two forces: the pull of the support and its weight. Two forces can balance only if they act along the same line, so the centre of mass must hang directly below the point of support. Everything therefore comes down to finding the centre of mass G of the bent wire, and then noticing that the body turns, between the two positions, through the angle between the lines GA and GB.

Solution

The centre of mass

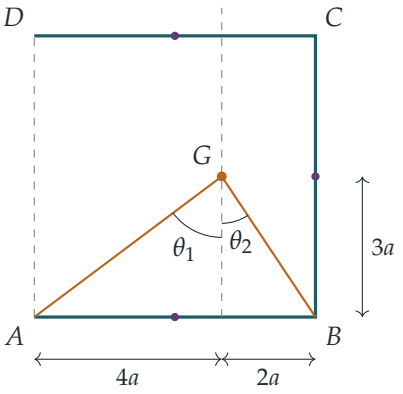
Model the wire as a uniform rigid rod bent into the three sides AB, BC, CD, each of the same weight W, hung from a smooth peg or a light string. To keep the numbers whole, take the side of the square to be $6a$ and set up axes in the plane of the wire with

$$A = (0,0), \quad B = (6a,0), \quad C = (6a,6a), \quad D = (0,6a).$$

Each side is a uniform rod, so its weight acts at its midpoint: $(3a,0)$ for AB, $(6a,3a)$ for BC and $(3a,6a)$ for CD. The three weights are equal, so G is the average of the three midpoints:

$$G = \left(\frac{3a+6a+3a}{3}, \frac{0+3a+6a}{3}\right) = (4a,\, 3a).$$

(By symmetry about the line through the midpoints of BC and AD, G had to lie at height $3a$; only its horizontal position needed working out.)



The dashed vertical line through G in the figure is parallel to BC (and to the missing side AD). Let θ_1 be the angle GA makes with it and θ_2 the angle GB makes with it, on the other side. Reading off the figure,

$$\tan\theta_1 = \frac{4a}{3a} = \frac{4}{3}, \qquad \tan\theta_2 = \frac{2a}{3a} = \frac{2}{3}.$$

(Indeed $AG = 5a$: a 3–4–5 triangle.)

Hanging from A, then from B

When the wire hangs from A in equilibrium, the only forces on it are the reaction at A and the total weight $3W$ acting at G. For these to balance, their moments about A must cancel, so the weight must pass through A: the line AG is vertical, with G below A. Since BC is parallel to the dashed line in the figure, BC is now tilted at θ_1 to the vertical.

In the same way, when the wire hangs from B, the line BG is vertical, and BC is tilted at θ_2 to the vertical, the other way.

Ask yourself Why is the angle between the two directions of BC equal to $\theta_1 + \theta_2$, rather than $\theta_1 - \theta_2$?

Going from the first position to the second, the wire is turned rigidly, so *every* line in it turns through the same angle. Equivalently, keep the wire fixed on the page and let "downwards" turn instead. In the first position downwards is along GA; in the second it is along GB. So the angle turned is the angle AGB between them. In the figure A and B lie on opposite sides of the dashed line through G, so

$$\angle AGB = \theta_1 + \theta_2.$$

Using the addition formula,

$$\tan(\theta_1 + \theta_2) = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1\tan\theta_2} = \frac{\frac{4}{3} + \frac{2}{3}}{1 - \frac{8}{9}} = \frac{2}{\frac{1}{9}} = 18.$$

Both θ_1 and θ_2 are acute, so $0 < \theta_1 + \theta_2 < 180°$, and since its tangent is positive the sum is in fact acute. Hence the angle between the two directions of BC is $\tan^{-1} 18$, as required.

Checks

- Numerically, $\theta_1 = \tan^{-1}\frac{4}{3} \approx 53.13^\circ$ and $\theta_2 = \tan^{-1}\frac{2}{3} \approx 33.69^\circ$, which add to 86.82°, and $\tan^{-1} 18 \approx 86.82^\circ$ too.
- Replacing the wire by 90 000 small equal pieces, averaging their positions to get $G = (4a, 3a)$, and rotating the whole body so that G lies below the support each time, gives the same angle between the two images of BC to nine decimal places.
- Sanity check on G: it lies nearer BC than the missing side AD, as it should, since all the wire on the right is there and none on the left.

Common slips

- Taking G at the centre of the square, $(3a, 3a)$. That would be right for all four sides, but the missing side AD pulls nothing to the left.
- Subtracting the two tilts instead of adding them: BC leans one way when hung from A and the other way when hung from B.

Beyond STEP

The principle "a freely suspended body hangs with its centre of mass below the support" is how you find the centre of mass of an awkward flat shape in practice: hang it from two points and see where the two plumb lines cross. In a first-year course the same idea appears through potential energy: the equilibrium positions are the stationary points of the height of G, and the stable one is the lowest. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats centres of mass and the statics of rigid bodies with many more examples like this.

WORKED SOLUTION

STEP I 1993, Question 12

Mechanics • Projectile Interception

STEP I 1993, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q12

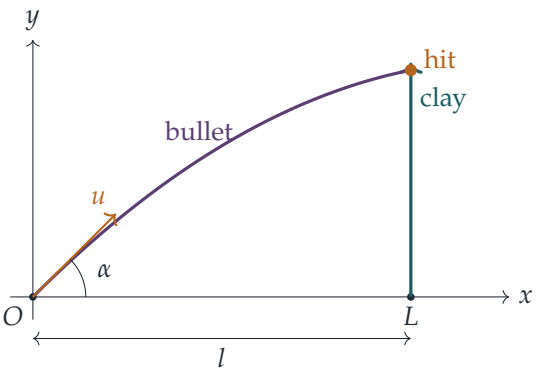
The key idea

A hit means the bullet and the clay are at the same place at the same moment. The bullet reaches the vertical line through the launcher at a time fixed by its horizontal motion alone, so we find that time first and then match the two heights. Gravity pulls both objects down by exactly the same amount, $\frac{1}{2}gt^2$, over the bullet's flight, so it cancels and leaves a tidy linear condition. When $T = 0$ this cancellation gives a neat way to read the whole problem.

Solution

The model

We treat the clay and the bullet as particles moving freely under gravity, with no air resistance, and assume the competitor fires in the vertical plane containing the launcher, towards it. Put the competitor at the origin O, with the x-axis horizontal towards the launcher L at $x = l$ and the y-axis vertically up. Measure time t from the moment the rifle is fired, so the clay has already been in the air for time T, and at time t it has been flying for $T + t$.



The condition for a hit

At time t after firing the bullet is at

$$x_B = (u\cos\alpha)\,t, \qquad y_B = (u\sin\alpha)\,t - \tfrac{1}{2}gt^2,$$

and the clay, which started up at speed v a time T earlier, is at $x = l$ with height

$$y_C = v(T+t) - \tfrac{1}{2}g(T+t)^2 = \left(vT - \tfrac{1}{2}gT^2\right) + (v - gT)\,t - \tfrac{1}{2}gt^2.$$

The bullet can only meet the clay on the line $x = l$, and it gets there when $(u\cos\alpha)\,t = l$, that is, at

$$t = \frac{l}{u\cos\alpha}.$$

(Here $\cos\alpha > 0$: a rifle pointing straight up never reaches the launcher, since $l > 0$.) A hit needs $y_B = y_C$ at this moment. The $-\frac{1}{2}gt^2$ terms cancel, so

$$(u\sin\alpha)\,t = \left(vT - \tfrac{1}{2}gT^2\right) + (v - gT)\,t.$$

Substituting $t = l/(u\cos\alpha)$ gives

$$\frac{l\sin\alpha}{\cos\alpha} = \left(vT - \tfrac{1}{2}gT^2\right) + \frac{(v - gT)\,l}{u\cos\alpha},$$

and multiplying through by $\cos\alpha$,

$$l\sin\alpha - \left(vT - \tfrac{1}{2}gT^2\right)\cos\alpha = \frac{v - gT}{u}\,l,$$

which is the required result.

The case $T = 0$

With $T = 0$ the condition becomes $l\sin\alpha = vl/u$, and since $l > 0$,

$$\sin\alpha = \frac{v}{u}.$$

? **Ask yourself** Why does this force $v < u$, and not just $v \le u$?

We need $\sin\alpha \le 1$, so $v \le u$. But $v = u$ would need $\alpha = \frac{\pi}{2}$, and then $\cos\alpha = 0$: the bullet goes straight up from O and never reaches the line $x = l$. So a hit is possible only if $v < u$. Then

$$\cos\alpha = \sqrt{1 - \frac{v^2}{u^2}} = \frac{\sqrt{u^2 - v^2}}{u},$$

taking the positive root because $0 < \alpha < \frac{\pi}{2}$, and the hit happens at

$$t = \frac{l}{u\cos\alpha} = \frac{l}{\sqrt{u^2 - v^2}}.$$

The clay is in the air from $t = 0$ until $vt - \frac{1}{2}gt^2 = 0$ again, that is, until $t = 2v/g$. Hitting it before it lands therefore requires

$$\frac{l}{\sqrt{u^2 - v^2}} < \frac{2v}{g} \quad \Longleftrightarrow \quad \frac{2v\sqrt{u^2 - v^2}}{g} > l,$$

where we multiplied by the positive quantity $\sqrt{u^2 - v^2}$. Together with $v < u$, this is what we had to show.

There's a nice way to see the $T = 0$ case. With $\sin\alpha = v/u$ the bullet's vertical velocity component is $u\sin\alpha = v$, the same as the clay's, and both feel the same gravity. So the bullet and the clay are at the *same height at every instant*, and the only question is whether the bullet's horizontal speed $\sqrt{u^2 - v^2}$ covers the distance l before the clay comes down.

Checks

- *Gravity-free limit.* If $g = 0$ and $T = 0$, both objects move in straight lines, and the condition $\sin\alpha = v/u$ is just the usual "aim off" condition for an interception at constant velocity.
- *Maximum range.* For fixed u, the bound $2v\sqrt{u^2 - v^2}/g$ is largest when $v = u/\sqrt{2}$, where it equals u^2/g, the maximum range of a projectile fired at speed u. That makes sense: with $v = u/\sqrt{2}$ we get $\alpha = 45^\circ$, and the hit happens just as the bullet would come back to the ground.
- *Numbers.* With $g = 9.8$, $v = 20$, $T = 0.7$, $u = 120$ and $l = 60$ (SI units), solving for the hitting angle numerically gives $\alpha \approx 0.2987$ rad, and both sides of the first identity agree. Stepping the two motions forward in time, the bullet and clay are both at height 17.13 m when the bullet reaches the launcher. For $T = 0$, $v = 20$ and $u = 100$, the bound is about 400 m with $\alpha \approx 11.5^\circ$.

Common slips

- Using the same time variable for both objects without allowing for the delay T: the clay's height must use $T + t$.
- Forgetting to rule out $v = u$, which gives $\alpha = 90^\circ$ and no hit at all.
- Checking only that the bullet reaches the launcher, and not that the clay is still in the air at that moment.

Beyond STEP

The cancellation of $\frac{1}{2}gt^2$ is a special case of a deep idea: in a frame falling freely with acceleration g, gravity disappears and every projectile moves in a straight line at constant velocity. This is Galilean relativity applied to an accelerating frame, and in

Einstein's hands it became the equivalence principle at the heart of general relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and non-inertial frames carefully.

WORKED SOLUTION

STEP I 1993, Question 13

Mechanics • Work and Power

STEP I 1993, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

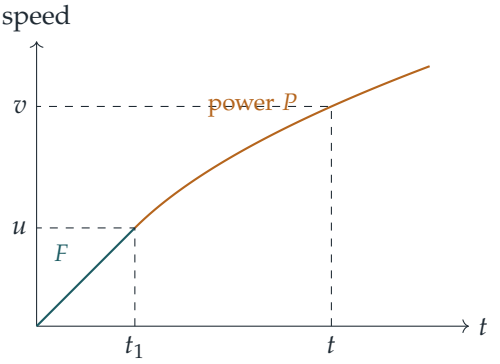
step.maths.org/questions/93-s1-q13

The key idea

The motion has two phases, and each needs its own form of Newton's second law. While the force is constant the acceleration is constant, so the *suvat* equations do everything. Once the engine works at constant power, the driving force is P/v, which changes with the speed; the trick is to write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}t}$ (for the time) or $v\frac{\mathrm{d}v}{\mathrm{d}x}$ (for the distance) so that the variables separate.

Solution

Model the engine and train together as a particle of mass M moving along a straight horizontal track with no resistance. The only horizontal force is the tractive force (weight and normal reaction are vertical and cancel). The figure sketches the speed against time: a straight line up to speed u, then a curve whose slope falls as the speed rises.



Phase 1: constant force

With constant force F the acceleration is F/M. Starting from rest, the speed reaches u at time

$$t_1 = \frac{Mu}{F} = \frac{Mu^2}{P},$$

using $F = P/u$. By $v^2 = u^2 + 2as$ (here from 0 to u), the distance covered is

$$s_1 = \frac{u^2}{2F/M} = \frac{Mu^2}{2F} = \frac{Mu^3}{2P}.$$

Phase 2: constant power

Power is force times velocity, so at speed w the tractive force is P/w. Note that at $w = u$ this is $P/u = F$, so the force does not jump when the engine changes mode. Newton's second law gives

$$M\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{P}{w}.$$

Time. Separating the variables, $Mw\,\mathrm{d}w = P\,\mathrm{d}t$. Integrating from speed u at time t_1 to speed v at time t,

$$\frac{M}{2}\left(v^2 - u^2\right) = P\,(t - t_1).$$

(This is just the work–energy principle: the engine does work $P(t - t_1)$, which all becomes kinetic energy.) So

$$t = \frac{Mu^2}{P} + \frac{M}{2P}\left(v^2 - u^2\right) = \frac{M}{2P}\left(2u^2 + v^2 - u^2\right) = \frac{M}{2P}\left(u^2 + v^2\right),$$

as required. Since $v \geq u$, the speed v is reached in phase 2 (or exactly at its start), which is why we may use phase 2's equation.

> **Ask yourself** For the distance we need x, not t, as the independent variable. How do we rewrite the acceleration?

By the chain rule, $\dfrac{\mathrm{d}w}{\mathrm{d}t} = \dfrac{\mathrm{d}w}{\mathrm{d}x}\dfrac{\mathrm{d}x}{\mathrm{d}t} = w\dfrac{\mathrm{d}w}{\mathrm{d}x}$. So the equation of motion becomes

$$Mw\frac{\mathrm{d}w}{\mathrm{d}x} = \frac{P}{w}, \qquad \text{that is,} \qquad Mw^2\,\mathrm{d}w = P\,\mathrm{d}x.$$

Distance. Integrating from $w = u$ to $w = v$, the distance s_2 travelled in phase 2 satisfies

$$\frac{M}{3}\left(v^3 - u^3\right) = P\,s_2, \qquad s_2 = \frac{M}{3P}\left(v^3 - u^3\right).$$

The total distance is therefore

$$\begin{aligned} s_1 + s_2 &= \frac{Mu^3}{2P} + \frac{M}{3P}\left(v^3 - u^3\right) \\ &= \frac{M}{6P}\left(3u^3 + 2v^3 - 2u^3\right) \\ &= \frac{M}{6P}\left(2v^3 + u^3\right). \end{aligned}$$

A misprint in the paper. The printed answer has u^2 in place of u^3. That can't be right: Mv^3/P is a length (power has units $\mathrm{kg\,m^2\,s^{-3}}$, and Mv^3 has units $\mathrm{kg\,m^3\,s^{-3}}$), whereas Mu^2/P is a time, so the u term must also be a cube for the two terms to be added. The correct distance is $\dfrac{M}{6P}\left(2v^3 + u^3\right)$, which is what we have shown.

Checks

- Take $v = u$: the time formula gives $Mu^2/P = Mu/F$ and the distance gives $Mu^3/(2P) = Mu^2/(2F)$, exactly the phase 1 results, as they must be.
- Units: Mv^2/P is energy over power, a time; Mv^3/P is a length. This is how the misprint shows itself.
- With $M = F = u = 1$ (so $P = 1$) and $v = 2$, the formulas give $t = 2.5$ and distance $17/6 \approx 2.833$. Stepping the motion numerically (force F below speed u, P/w above) reproduces both values, and agrees with the formulas to better than 10^{-7} for random M, F, u and $v \geq u$; the printed u^2 version disagrees.

Common slips

- Forgetting phase 1 and integrating the constant-power equation from rest. The phase 2 equation only holds once $w \geq u$; starting it from $w = 0$ would mean an infinite force.
- Using $\mathrm{d}w/\mathrm{d}t$ when integrating with respect to x: for distance the acceleration must be written as $w\,\mathrm{d}w/\mathrm{d}x$.
- Writing the phase 1 time as u/F: the acceleration is F/M, not F.

Beyond STEP

The constant-power phase is a first-order differential equation solved by separating variables, and the switch from t to x as independent variable is the same trick used for energy methods in general: $w\,\mathrm{d}w/\mathrm{d}x$ integrates to kinetic energy, which leads to the work–energy theorem and, later, to potentials and conserved quantities. The cap on force at low speed is also how real traction works, since a motor of fixed power cannot

supply unbounded force at rest. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops work, energy and power along these lines.

WORKED SOLUTION

STEP I 1993, Question 14

Probability and Statistics • Exponential Waiting Times

STEP I 1993, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q14

The key idea

What matters is not Mr Toad's speed but the *time* T he takes to cover the last kilometre, and $T = 1/V$. The density of V looks awkward, but its cumulative distribution function is very simple, and converting it into a statement about T shows that T is exponential with parameter α. After that, each green period is a difference of two exponentials, and adding up all the green periods gives a geometric series.

Solution

The distribution of the arrival time

Let V be the speed in km per minute, and let $T = 1/V$ be the time in minutes, measured from the moment the lights turn green, at which he reaches the lights. Spotting that

$$\frac{\mathrm{d}}{\mathrm{d}v}\,\mathrm{e}^{-\alpha/v} = \alpha v^{-2}\mathrm{e}^{-\alpha/v},$$

and that $\mathrm{e}^{-\alpha/v} \to 0$ as $v \to 0^+$, the distribution function of V is

$$\mathrm{P}(V \le v) = \int_0^v \alpha u^{-2}\mathrm{e}^{-\alpha/u}\,\mathrm{d}u = \mathrm{e}^{-\alpha/v} \qquad (v > 0).$$

(It tends to 1 as $v \to \infty$, so this is a genuine density.) Since $V > 0$, for $t > 0$ we have $T > t \iff 1/V > t \iff V < 1/t$, so

$$\mathrm{P}(T > t) = \mathrm{P}\left(V < \tfrac{1}{t}\right) = \mathrm{e}^{-\alpha t}.$$

So T has the exponential distribution with parameter α.

Ask yourself Why work with T rather than V?

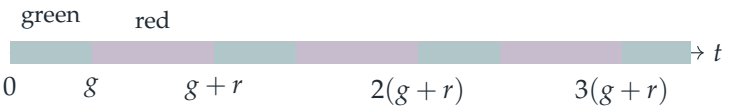
The lights are described in terms of time, so every event we care about is an interval of values of T. Translating the density of V into the density of T by hand means a change of variable; going through $\mathrm{P}(T > t)$ avoids that and gives the answer in one line.

One green period

At time 0 the lights turn green. They are green on $[0, g)$, red on $[g, g+r)$, green on $[g+r, (g+r)+g)$, and so on. The $(n+1)$th green period is $n(g+r) \le t < n(g+r)+g$. Because T is continuous, whether the endpoints are included makes no difference, and

$$\begin{aligned} \mathrm{P}\big(n(g+r) < T < n(g+r)+g\big) &= \mathrm{P}\big(T > n(g+r)\big) - \mathrm{P}\big(T > n(g+r)+g\big) \\ &= \mathrm{e}^{-\alpha n(g+r)} - \mathrm{e}^{-\alpha(n(g+r)+g)}, \end{aligned}$$

as required. (The printed paper puts the closing bracket in the wrong place in the second exponent; the exponent should be $-\alpha\big(n(g+r)+g\big)$, as here.) The working holds equally for $n = 0$, which is the first green period, before the lights have ever turned red.



The probability of a green light

He passes on green exactly when T lies in one of these green periods. They are disjoint, so we add their probabilities, including $n = 0$:

$$\begin{aligned} \mathrm{P}(\alpha) &= \sum_{n=0}^{\infty} \left(\mathrm{e}^{-\alpha n(g+r)} - \mathrm{e}^{-\alpha n(g+r)}\mathrm{e}^{-\alpha g}\right) = \left(1 - \mathrm{e}^{-\alpha g}\right) \sum_{n=0}^{\infty} \left(\mathrm{e}^{-\alpha(g+r)}\right)^n \\ &= \frac{1 - \mathrm{e}^{-\alpha g}}{1 - \mathrm{e}^{-\alpha(g+r)}}, \end{aligned}$$

summing the geometric series, which converges because $0 < \mathrm{e}^{-\alpha(g+r)} < 1$ (since $\alpha > 0$ and $g + r > 0$).

The limits

As $\alpha \to \infty$, both $\mathrm{e}^{-\alpha g} \to 0$ and $\mathrm{e}^{-\alpha(g+r)} \to 0$ (as $g > 0$, $g + r > 0$), so

$$\mathrm{P}(\alpha) \to \frac{1-0}{1-0} = 1.$$

This makes sense: a large α means a very fast Mr Toad, who almost always gets through before the first green period ends.

As $\alpha \to 0$ both numerator and denominator tend to 0, so we need more care. Write $1 - \mathrm{e}^{-x} = \mathrm{e}^{-x}(\mathrm{e}^{x} - 1)$ in each:

$$\mathrm{P}(\alpha) = \frac{\mathrm{e}^{-\alpha g}\left(\mathrm{e}^{\alpha g} - 1\right)}{\mathrm{e}^{-\alpha(g+r)}\left(\mathrm{e}^{\alpha(g+r)} - 1\right)} = \mathrm{e}^{\alpha r} \cdot \frac{\dfrac{\mathrm{e}^{\alpha g} - 1}{\alpha g}}{\dfrac{\mathrm{e}^{\alpha(g+r)} - 1}{\alpha(g+r)}} \cdot \frac{g}{g+r},$$

where we divided top and bottom by α and inserted the factors g and $g+r$. As $\alpha \to 0$, $e^{\alpha r} \to 1$, and both $\alpha g \to 0$ and $\alpha(g+r) \to 0$, so by $(e^x - 1)/x \to 1$ the two fractions each tend to 1. Therefore

$$P(\alpha) \to \frac{g}{r+g} \qquad \text{as } \alpha \to 0.$$

Here Mr Toad is very slow and his arrival time is spread over a great many cycles, so the chance of green is just the proportion of each cycle that is green.

Checks

- With $\alpha = 0.7$, $g = 1$, $r = 2$ the formula gives $P(\alpha) \approx 0.5737$. A simulation of 4×10^5 drives, sampling V by inverting $e^{-\alpha/v}$, gave 0.5744, and the individual green periods $n = 0, 1, 2$ matched too.
- With $g = 1$, $r = 2$: $P(10^{-4}) \approx 0.33337$, close to $\frac{1}{3}$, and $P(10) \approx 0.99995$.
- If $r = 0$ the light is always green and the formula gives $P(\alpha) = 1$ for every α, as it should.

Common slips

- Leaving out $n = 0$. The first green period, before any red, carries the most probability.
- Copying the printed exponent $-\alpha(n(g+r)) + g$, which is a misprint for $-\alpha\big(n(g+r)+g\big)$.
- Treating V itself as the arrival time, or trying to find $E(T)$. In fact $E(V)$ is infinite here, but nothing in the question needs it.

Beyond STEP

Turning a distribution for V into one for $T = g(V)$ through the distribution function is the standard method for transforming random variables. Here V has an inverse-exponential distribution. The limit $\alpha \to 0$ is a small example of a general idea: a random time spread over many periods lands in each phase in proportion to that phase's length, which is the start of renewal theory. Grimmett and Welsh, *Probability: An Introduction*, covers both transformations of random variables and the exponential distribution.

WORKED SOLUTION

STEP I 1993, Question 15

Probability and Statistics • Bayes' Theorem

STEP I 1993, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q15

The key idea

This is Bayes' theorem dressed up as a story. The trick is to condition on the *true* direction and ask how likely the native's string of answers is in each case. A Biscii always lies, so repeated answers from a Biscii always agree and always point the wrong way. Repetition therefore makes it more likely that you are talking to a Biscii, and that is why asking one more time can reverse your decision.

Solution

The Ascii on their own

An Ascii answers each question truthfully with probability $\frac{3}{4}$, independently of earlier answers. The question has only two possible answers (left or right), so two answers are the same exactly when both are true or both are false:

$$\mathrm{P}(\text{same answer twice}) = \left(\tfrac{3}{4}\right)^2 + \left(\tfrac{1}{4}\right)^2 = \tfrac{9}{16} + \tfrac{1}{16} = \tfrac{10}{16} = \tfrac{5}{8},$$

as required. The "both true" case contributes $\frac{9}{16}$ of this, so by the definition of conditional probability

$$\mathrm{P}(\text{telling the truth} \mid \text{same answer twice}) = \frac{9/16}{10/16} = \frac{9}{10},$$

which is the second result.

Two answers of "Left"

Let L be the event that left is the correct direction and R the event that right is. Let A and B be the events that the native is an Ascii or a Biscii, with $\mathrm{P}(A) = \frac{11}{16}$ and $\mathrm{P}(B) = \frac{5}{16}$, chosen independently of the geography. The captain knows nothing about the direction to begin with, so we take $\mathrm{P}(L) = \mathrm{P}(R) = \frac{1}{2}$. Let E_2 be the event that the native answers "Left" twice.

Ask yourself Why is it easier to work out $\mathrm{P}(E_2 \mid L)$ than $\mathrm{P}(L \mid E_2)$?

Once we know the true direction, each tribe's behaviour is completely described, so the probability of the answers is easy to write down. Bayes' theorem then turns these round.

If left is correct, an Ascii says "Left" twice by telling the truth twice, with probability $\frac{9}{16}$. A Biscii would lie and say "Right", so cannot produce E_2. Hence

$$\mathrm{P}(E_2 \mid L) = \tfrac{11}{16} \cdot \tfrac{9}{16} + \tfrac{5}{16} \cdot 0 = \tfrac{99}{256}.$$

If right is correct, an Ascii says "Left" twice by lying twice, with probability $\frac{1}{16}$, and a Biscii certainly says "Left" twice. Hence

$$\mathrm{P}(E_2 \mid R) = \tfrac{11}{16} \cdot \tfrac{1}{16} + \tfrac{5}{16} \cdot 1 = \tfrac{11+80}{256} = \tfrac{91}{256}.$$

The priors of L and R are equal, so they cancel in Bayes' theorem:

$$\mathrm{P}(L \mid E_2) = \frac{\frac{1}{2} \cdot \frac{99}{256}}{\frac{1}{2} \cdot \frac{99}{256} + \frac{1}{2} \cdot \frac{91}{256}} = \frac{99}{99+91} = \frac{99}{190}.$$

This is greater than $\frac{1}{2}$, so left is the more likely direction and the captain should go left, with probability $\frac{99}{190}$ of being right.

Three answers of "Left"

Let E_3 be the event of three answers of "Left". The same reasoning applies, now with three independent answers from an Ascii:

$$\mathrm{P}(E_3 \mid L) = \tfrac{11}{16}\left(\tfrac{3}{4}\right)^3 + \tfrac{5}{16} \cdot 0 = \tfrac{11}{16} \cdot \tfrac{27}{64} = \tfrac{297}{1024},$$
$$\mathrm{P}(E_3 \mid R) = \tfrac{11}{16}\left(\tfrac{1}{4}\right)^3 + \tfrac{5}{16} \cdot 1 = \tfrac{11}{1024} + \tfrac{320}{1024} = \tfrac{331}{1024}.$$

Since $\mathrm{P}(E_3 \mid R) > \mathrm{P}(E_3 \mid L)$ and the priors are equal, right is now the more likely direction. Precisely,

$$\mathrm{P}(R \mid E_3) = \frac{331}{297+331} = \frac{331}{628} > \frac{1}{2}.$$

So the captain should go *right*, and he is then correct with probability $\frac{331}{628}$.

Ask yourself How can a third "Left" make left *less* likely?

Each extra "Left" multiplies the Ascii's contribution to $\mathrm{P}(E \mid L)$ by $\frac{3}{4}$. In $\mathrm{P}(E \mid R)$ the Ascii term shrinks by a factor of $\frac{1}{4}$, but the Biscii term stays at $\frac{5}{16}$. A long run of identical answers is just what a Biscii produces, so the evidence increasingly says "this is a liar". A truthful Ascii is consistent too, but only with probability $(\frac{3}{4})^n$, which dies away.

Checks

- After a single "Left", the likelihoods are $\frac{11}{16} \cdot \frac{3}{4} = \frac{33}{64}$ (given L) and $\frac{11}{16} \cdot \frac{1}{4} + \frac{5}{16} = \frac{31}{64}$ (given R), so $\mathrm{P}(L \mid \text{Left}) = \frac{33}{64}$. The sequence $\frac{33}{64} \approx 0.516$, $\frac{99}{190} \approx 0.521$, $\frac{297}{628} \approx 0.473$ rises and then falls, which matches the explanation above.
- With many answers, $\mathrm{P}(L \mid E_n) \to 0$, because the Biscii term dominates.
- We checked all four probabilities by exact enumeration with fractions, and by simulating 10^6 conversations: the simulated values were 0.5214 (exact $\frac{99}{190} \approx 0.5211$) and 0.5266 (exact $\frac{331}{628} \approx 0.5271$).

Common slips

- Forgetting to state the assumption $\mathrm{P}(L) = \mathrm{P}(R) = \frac{1}{2}$. Without some prior on the direction the question cannot be answered.
- Using the $\frac{9}{10}$ from the first part on the whole population. It is a statement about an Ascii only; a Biscii also gives the same answer twice, every time.
- Treating the native's three answers as independent *whoever* the native is. They are independent only once we know the tribe (and the true direction), which is why we condition first.

Beyond STEP

The calculation is an example of Bayesian updating: the ratio $\mathrm{P}(E \mid R)/\mathrm{P}(E \mid L)$, called the likelihood ratio or Bayes factor, measures how strongly the evidence favours one hypothesis. Because the tribe is a hidden variable shared by all the answers, the answers are only *conditionally* independent, and evidence that seems to point one way can end up pointing the other. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of conditional probability and of exactly this kind of paradox.

WORKED SOLUTION

STEP I 1993, Question 16

Probability and Statistics • Geometrical Probability

STEP I 1993, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.20, *Buffon's Needle*: https://beyondstep.keithmcnulty.org/ch12/buffon.html

The key idea

This is Buffon's needle. Whether the pin crosses the nearest line depends only on how far its tip reaches *towards* that line, and that reach is the projection of a half-pin onto the perpendicular direction. Fix the distance of the centre first, find the conditional probability from the angle alone, then average over the distance. The integral in the first part is exactly the one that this averaging produces, which is why the question starts there.

Solution

The integral

Put $y = \cos^{-1} t$, so $t = \cos y$ and $\mathrm{d}t = -\sin y \,\mathrm{d}y$. When $t = 0$, $y = \frac{\pi}{2}$, and when $t = 1$, $y = 0$. Hence

$$\int_0^1 \cos^{-1} t \,\mathrm{d}t = \int_{\pi/2}^0 y\,(-\sin y)\,\mathrm{d}y = \int_0^{\pi/2} y \sin y \,\mathrm{d}y.$$

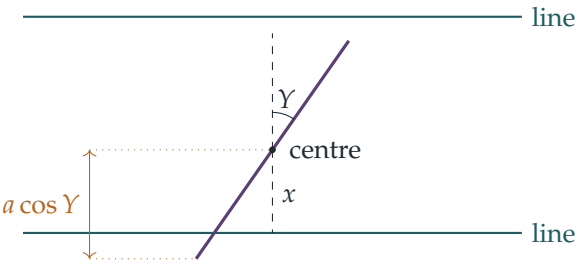
Integrating by parts with $u = y$ and $\mathrm{d}v = \sin y \,\mathrm{d}y$,

$$\begin{aligned}\int_0^{\pi/2} y \sin y \,\mathrm{d}y &= \Big[-y\cos y\Big]_0^{\pi/2} + \int_0^{\pi/2} \cos y \,\mathrm{d}y \\ &= 0 + \Big[\sin y\Big]_0^{\pi/2} = 1,\end{aligned}$$

as required.

(Or otherwise: the integral is the area between $y = \cos^{-1} t$ and the axes, which is the same region as the area under $t = \cos y$ for $0 \le y \le \frac{\pi}{2}$, namely $\int_0^{\pi/2} \cos y \,\mathrm{d}y = 1$.)

The probability given $X = x$



Each half of the pin has length a and makes angle Y with the perpendicular to the lines, so the ends of the pin lie a perpendicular distance $a\cos Y$ on either side of the centre. The nearest line is at distance x from the centre, so the pin crosses it exactly when

$$a\cos Y \;\geq\; x.$$

(Equality, where the tip just touches the line, has probability zero, so it doesn't matter whether we count it.)

? **Ask yourself** How do we turn $a\cos Y \geq x$ into a condition on Y alone?

Since cos is decreasing on $[0, \frac{\pi}{2}]$, it depends on whether x/a is attainable.

- If $x \geq a$, then $a\cos Y \leq a \leq x$ with equality only when $Y = 0$ and $x = a$, so the probability is 0.
- If $0 \leq x < a$, then $\cos Y \geq x/a$ is equivalent to $Y \leq \cos^{-1}(x/a)$, where $\cos^{-1}(x/a) \in (0, \frac{\pi}{2}]$. As Y is uniform on $[0, \frac{\pi}{2}]$,

$$\mathrm{P}(\text{crosses} \mid X = x) = \frac{\cos^{-1}(x/a)}{\pi/2} = \frac{2}{\pi}\cos^{-1}\frac{x}{a}.$$

Here we used the independence of X and Y: knowing $X = x$ leaves Y uniform on $[0, \frac{\pi}{2}]$.

The probability for a general throw

Now let $a < b$. The density of X is $1/b$ on $[0, b]$. Because $a < b$, the interval $[0, a)$ where crossing is possible lies inside $[0, b]$, and the conditional probability is zero on $[a, b]$. By the law of total probability,

$$\mathrm{P}(\text{crosses}) = \int_0^b \mathrm{P}(\text{crosses} \mid X = x)\,\frac{1}{b}\,\mathrm{d}x = \frac{1}{b}\int_0^a \frac{2}{\pi}\cos^{-1}\frac{x}{a}\,\mathrm{d}x.$$

Substituting $x = at$, $\mathrm{d}x = a\,\mathrm{d}t$, and using the first part,

$$\mathrm{P}(\text{crosses}) = \frac{2a}{\pi b}\int_0^1 \cos^{-1} t\,\mathrm{d}t = \frac{2a}{\pi b},$$

as required. Since $a < b$, the pin is shorter than the gap $2b$ and can never reach a second line, so "crosses a line" and "crosses the nearest line" are the same event.

Checks

- As $a \to b$ the answer tends to $2/\pi \approx 0.637$, which is less than 1, as it must be: a pin as long as the gap still often lies nearly parallel to the lines.
- The answer is proportional to a: doubling a short pin doubles the chance of a crossing, which matches the fact that the expected number of crossings is linear in the length.
- A simulation of the actual geometry (centre uniform across a strip, angle uniform) with $a = 1$, $b = 2$ gave 0.3183 over 4×10^5 throws, against $1/\pi \approx 0.31831$. Midpoint quadrature gave $\int_0^1 \cos^{-1} t \, \mathrm{d}t = 1.0000000$.

Common slips

- Using $a \sin Y$ instead of $a \cos Y$: here Y is measured from the *perpendicular* to the lines, not from the lines themselves.
- Integrating the conditional probability over all of $[0, b]$ with the formula $\frac{2}{\pi} \cos^{-1}(x/a)$, which is undefined for $x > a$; the conditional probability is 0 there.
- Forgetting the density $1/b$ when averaging over X.

Beyond STEP

Buffon posed this in the eighteenth century, and it is the classic example of *geometric probability*: probabilities computed as areas in a space of parameters, here the rectangle $[0, b] \times [0, \frac{\pi}{2}]$. Since the answer involves π, throwing pins gives a (very slow) Monte Carlo estimate of π. A beautiful extension, "Buffon's noodle", shows by linearity of expectation that the expected number of crossings for a curve of any shape depends only on its length. Grimmett and Welsh, *Probability: An Introduction*, covers continuous joint distributions and conditioning of this kind.

WORKED SOLUTION

STEP II 1993, Question 1

Pure • Combinatorial Game Strategy

STEP II 1993, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q1

The key idea

Everything rests on one counting fact. Both players have exactly 9 to share out, so Betty cannot match or beat Adam on every field and win outright on one: that would cost her at least $9+1$. So whenever Betty wins, she must win exactly two fields and lose the third. Once you have this, the first part becomes a recipe, and the two-triple part becomes a short case analysis.

Solution

Throughout, "Betty wins field i" means $b_i > a_i$, "Adam wins field i" means $a_i > b_i$, and a field with $a_i = b_i$ is tied.

A lemma: how Betty can win

Suppose Betty's triple beats Adam's triple (a_1, a_2, a_3), where $a_1 + a_2 + a_3 = 9$.

- She cannot win all three fields, since that would need $b_i \geq a_i + 1$ for each i, giving $b_1 + b_2 + b_3 \geq 12$.
- Suppose she wins exactly one field. Since she has more points than Adam, Adam scores 0, so the other two fields are ties or Betty wins. Then $b_i \geq a_i$ for every i, with strict inequality at least once, and so $b_1 + b_2 + b_3 \geq 9 + 1 = 10$. That's impossible.

She has more points than Adam, so she scores at least one. Hence she wins *exactly two* fields. So Betty beats (a_1, a_2, a_3) if and only if, for some pair of fields i, j,

$$b_i \geq a_i + 1, \qquad b_j \geq a_j + 1,$$

and she loses the remaining field k. Note that losing field k is automatic, since $b_k = 9 - b_i - b_j \leq 9 - a_i - a_j - 2 = a_k - 2$.

Betty can always win

Label the fields so that $a_1 \geq a_2 \geq a_3$. Then $3a_1 \geq a_1 + a_2 + a_3 = 9$, so $a_1 \geq 3$. Betty gives up the field where Adam is strongest and beats him by one on the other two:

$$(b_1, b_2, b_3) = (a_1 - 2,\ a_2 + 1,\ a_3 + 1).$$

These are non-negative integers (as $a_1 - 2 \geq 1$) with sum 9. Betty wins fields 2 and 3 and loses field 1, because $a_1 - 2 < a_1$. She scores 2 to Adam's 1, so she wins.

Adam can make sure he never wins

Take $(a_1, a_2, a_3) = (9, 0, 0)$. Adam cannot score on fields 2 or 3, because $b_2, b_3 \geq 0$. So he scores at most one point, and only by winning field 1, which means $b_1 \leq 8$. In that case $b_2 + b_3 = 9 - b_1 \geq 1$, so Betty has a positive entry on field 2 or field 3 and scores at least one point. Either Adam scores 0, or he scores 1 and Betty scores at least 1. In neither case does he score more than Betty, so he never wins, whatever Betty does.

(A brute-force search shows that $(9, 0, 0)$ and its rearrangements are the *only* triples with this property. Against any other triple, some foolish choice by Betty hands Adam the game.)

Two triples: $(5, 3, 1)$ and $(3, 1, 5)$

We need to show that no triple (b_1, b_2, b_3) beats both. By the lemma, Betty needs to win two fields against each triple.

Against $(5, 3, 1)$. Winning fields 1 and 2 costs $6 + 4 = 10 > 9$, which is impossible. So she must win field 3 together with field 1 or field 2:

$$b_3 \geq 2 \quad \text{and} \quad (b_1 \geq 6 \ \text{ or } \ b_2 \geq 4). \tag{A}$$

Against $(3, 1, 5)$. Winning fields 1 and 3 costs $4 + 6 = 10 > 9$, which is impossible. So she must win field 2 together with field 1 or field 3:

$$b_2 \geq 2 \quad \text{and} \quad (b_1 \geq 4 \ \text{ or } \ b_3 \geq 6). \tag{B}$$

> **Ask yourself** Both conditions involve a choice of fields. How do we combine them without missing a case?

Split on the second choice in (B).

- If $b_3 \geq 6$, then $b_1 + b_2 \leq 3$, so neither $b_1 \geq 6$ nor $b_2 \geq 4$ can hold, which contradicts (A).
- If $b_1 \geq 4$, then together with $b_2 \geq 2$ from (B) and $b_3 \geq 2$ from (A), adding the alternative in (A) gives either

  $$b_1 + b_2 + b_3 \geq 6 + 2 + 2 = 10 \qquad \text{or} \qquad b_1 + b_2 + b_3 \geq 4 + 4 + 2 = 10.$$

  Both are impossible, since the sum is 9.

Every case leads to a contradiction, so no single triple beats both $(5,3,1)$ and $(3,1,5)$. Adam wins with this pair, as required.

Checks

There are only $\binom{11}{2} = 55$ triples, so everything can be checked by brute force. A computer confirms that every one of Adam's 55 triples is beaten by Betty's recipe above, and that every winning reply wins exactly two fields. It also lists the 13 triples that beat $(5,3,1)$ and the 13 that beat $(3,1,5)$, and finds no triple common to the two lists. For a concrete case, $(4,2,3)$ beats $(3,1,5)$ by 2–1, but against $(5,3,1)$ it wins only field 3 and loses fields 1 and 2, so it loses 1–2.

Common slips

- Trying $(3,3,3)$ for Adam's "never win" triple. Betty can play $(2,2,5)$ and lose 1–2, so Adam *can* win with it.
- Forgetting that a win for Betty with only one field won needs Adam to score nothing, and then not using the sum 9 to rule it out.
- In the two-triple part, checking only a few natural replies such as $(6,0,3)$ or $(0,2,7)$ instead of proving that *no* reply works.

Beyond STEP

Colonel Blotto is a classic of game theory. Because Betty moves second with full information she always wins, so the interesting version has both players move at once. Then there is usually no best pure strategy, and the solution is a *mixed strategy*: a probability distribution over triples, as guaranteed by von Neumann's minimax theorem. The two-triple variant in this question is a first step in that direction, since Adam's best defence is to spread his choice over several triples. For problems in the same spirit, see Siklos, *Advanced Problems in Mathematics*. For probability tools that support mixed strategies, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP II 1993, Question 2

Pure • Trigonometric and Algebraic Integrals

STEP II 1993, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q2

The key idea

In part (i) a product of cosines is awkward to integrate, but a sum is easy, so we turn the product into a sum with the factor formula. Everything then depends on one fact: $\cos kx$ integrates to zero over a full period unless $k = 0$. The special cases are exactly the ones where $m - n$ or $m + n$ is zero. In part (ii) we combine the square root into a single fraction and rationalise, which splits the integrand into two standard pieces.

Solution

Part (i)

The factor formula $2\cos A\cos B = \cos(A-B) + \cos(A+B)$ gives

$$\cos(mx)\cos(nx) = \tfrac{1}{2}\left[\cos\big((m-n)x\big) + \cos\big((m+n)x\big)\right].$$

So we need $\int_0^{2\pi} \cos(kx)\,\mathrm{d}x$ for an integer k:

$$\int_0^{2\pi} \cos(kx)\,\mathrm{d}x = \begin{cases} \left[\dfrac{\sin kx}{k}\right]_0^{2\pi} = \dfrac{\sin 2k\pi}{k} = 0, & k \neq 0, \\ \displaystyle\int_0^{2\pi} 1\,\mathrm{d}x = 2\pi, & k = 0. \end{cases}$$

Here we used that k is an integer, so $\sin 2k\pi = 0$. We can't divide by k when $k = 0$, which is why that case is separate.

? **Ask yourself** How many different cases are there, and do they overlap?

Each of $k = m - n$ and $k = m + n$ is either zero or not, and both are zero only when $m = n = 0$. That gives three cases:

- $m \neq n$ and $m \neq -n$, that is $|m| \neq |n|$: both integrals vanish and the answer is 0.

- Exactly one of $m-n$, $m+n$ is zero, that is $|m| = |n| \neq 0$: one term contributes $\frac{1}{2} \cdot 2\pi$ and the other 0, so the answer is π.
- $m = n = 0$: both terms contribute $\frac{1}{2} \cdot 2\pi$. (Directly, the integrand is 1.) The answer is 2π.

In summary,

$$\int_0^{2\pi} \cos(mx)\cos(nx)\,\mathrm{d}x = \begin{cases} 0, & |m| \neq |n|, \\ \pi, & |m| = |n| \neq 0, \\ 2\pi, & m = n = 0. \end{cases}$$

Note that $m = -n$ belongs with $m = n$, since cos is even and $\cos(-nx) = \cos(nx)$.

Part (ii)

The integrand $\sqrt{1+1/x} = \sqrt{(x+1)/x}$ is real only when $(x+1)/x \geq 0$, that is for $x > 0$ or $x \leq -1$. We deal with $x > 0$ first.

The case $x > 0$. Multiply top and bottom inside the root by $x+1$:

$$\sqrt{\frac{x+1}{x}} = \sqrt{\frac{(x+1)^2}{x(x+1)}} = \frac{x+1}{\sqrt{x^2+x}},$$

using $x+1 > 0$ to take the square root of $(x+1)^2$. Now split the numerator so that one part is half the derivative of x^2+x:

$$\frac{x+1}{\sqrt{x^2+x}} = \frac{1}{2} \cdot \frac{2x+1}{\sqrt{x^2+x}} + \frac{1}{2} \cdot \frac{1}{\sqrt{x^2+x}}.$$

? **Ask yourself** How do we integrate $1/\sqrt{x^2+x}$?

We complete the square: $x^2 + x = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4}$. Then the standard result $\int (u^2 - a^2)^{-1/2}\,\mathrm{d}u = \ln\left(u + \sqrt{u^2 - a^2}\right)$, valid for $u > a > 0$, applies with $u = x + \frac{1}{2}$ and $a = \frac{1}{2}$:

$$\int \frac{\mathrm{d}x}{\sqrt{x^2+x}} = \ln\left(x + \tfrac{1}{2} + \sqrt{x^2+x}\right) + \text{const} = \ln\left(2x + 1 + 2\sqrt{x^2+x}\right) + \text{const}.$$

The first piece is $\int \frac{1}{2}(2x+1)(x^2+x)^{-1/2}\,\mathrm{d}x = \sqrt{x^2+x}$. So, for $x > 0$,

$$\int \sqrt{1 + \frac{1}{x}}\,\mathrm{d}x = \sqrt{x^2+x} + \frac{1}{2}\ln\left(2x + 1 + 2\sqrt{x^2+x}\right) + C.$$

Since $2x + 1 + 2\sqrt{x^2+x} = \left(\sqrt{x} + \sqrt{x+1}\right)^2$, this can also be written as

$$\sqrt{x(x+1)} + \ln\left(\sqrt{x} + \sqrt{x+1}\right) + C.$$

The substitution $x = \sinh^2 t$ gives this second form directly. The integrand becomes $2\cosh^2 t = 1 + \cosh 2t$, which integrates to $t + \sinh t \cosh t$.

The case $x \le -1$. Now $x+1 \le 0$, so $\sqrt{(x+1)^2} = -(x+1)$ and

$$\sqrt{\frac{x+1}{x}} = -\frac{x+1}{\sqrt{x^2+x}}.$$

The same split works, but now $x + \frac{1}{2} < 0$, so we need $\int (u^2 - a^2)^{-1/2}\,\mathrm{d}u = \ln\left|u + \sqrt{u^2 - a^2}\right|$, valid for $|u| > a$. For $x < -1$,

$$\int \sqrt{1 + \frac{1}{x}}\,\mathrm{d}x = -\sqrt{x^2+x} - \frac{1}{2}\ln\left|2x + 1 + 2\sqrt{x^2+x}\right| + C.$$

Because $\left(2x+1+2\sqrt{x^2+x}\right)\left(2x+1-2\sqrt{x^2+x}\right) = 1$, the logarithm can also be written as $+\frac{1}{2}\ln\left|2x+1-2\sqrt{x^2+x}\right|$.

Checks

- Part (i): with $m = n = 1$ the integrand is $\cos^2 x$, whose average is $\frac{1}{2}$, so the integral is $\frac{1}{2} \cdot 2\pi = \pi$, as predicted. A numerical (Simpson's rule) evaluation for every pair $-6 \le m, n \le 6$ matched the three-case formula.
- Part (ii): differentiate the answer. The derivative of $\sqrt{x^2+x}$ is $\dfrac{2x+1}{2\sqrt{x^2+x}}$. The derivative of $\frac{1}{2}\ln(\cdots)$ is $\dfrac{1}{2\sqrt{x^2+x}}$. They add to $\dfrac{x+1}{\sqrt{x^2+x}}$, as required.
- Numerically, $\int_1^3 \sqrt{1+1/x}\,\mathrm{d}x \approx 2.48547$, both by Simpson's rule and from the formula: $\sqrt{12} - \sqrt{2} + \frac{1}{2}\ln\dfrac{7+4\sqrt{3}}{3+2\sqrt{2}}$.

Common slips

- Forgetting the case $m = -n \neq 0$, which gives π just as $m = n \neq 0$ does.
- Writing $\dfrac{\sin kx}{k}$ when $k = 0$; that case needs separate treatment.
- Writing $\sqrt{(x+1)^2} = x+1$ without checking the sign, which gives the wrong answer for $x < -1$.

Beyond STEP

Part (i) says that the functions $1, \cos x, \cos 2x, \ldots$ are *orthogonal* on $[0, 2\pi]$. This is the fact behind Fourier series: if $f(x) = \sum a_n \cos nx$, multiplying by $\cos mx$ and integrating picks out the single coefficient a_m. The same idea, with an integral playing the part of a dot product, underlies much of linear algebra on spaces of functions. Strang's *Introduction to Linear Algebra* covers orthogonality, and Boyce and DiPrima's *Elementary Differential Equations* shows Fourier series in action solving the heat equation.

WORKED SOLUTION

STEP II 1993, Question 3

Pure • Riccati Differential Equations

STEP II 1993, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q3

The key idea

Both equations are nonlinear because of the y^2 term, so none of the standard first-order methods applies directly. The substitutions are designed to kill that y^2 term. In (i), writing y as a multiple of u'/u makes the squared terms cancel and leaves a linear second-order equation with constant coefficients. In (ii), $y = 1/x$ happens to be an exact solution, and writing $y = 1/x + 1/v$ turns the equation into a linear first-order equation for v, which an integrating factor solves.

Solution

Part (i)

Put $y = -\dfrac{u'}{3u}$, where $u' = \dfrac{\mathrm{d}u}{\mathrm{d}x}$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{u''}{3u} + \frac{(u')^2}{3u^2}, \qquad \text{and} \qquad 3y^2 = 3 \cdot \frac{(u')^2}{9u^2} = \frac{(u')^2}{3u^2}.$$

So the awkward terms cancel:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 3y^2 = -\frac{u''}{3u}.$$

The equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} - y - 3y^2 = -2$ therefore becomes

$$-\frac{u''}{3u} + \frac{u'}{3u} = -2.$$

Multiplying by $-3u$ (where $u \neq 0$, which is needed anyway for y to be defined) gives

$$u'' - u' - 6u = 0.$$

The auxiliary equation is $m^2 - m - 6 = (m-3)(m+2) = 0$, so $m = 3$ or $m = -2$, and

$$u = A\mathrm{e}^{3x} + B\mathrm{e}^{-2x}$$

for constants A and B, not both zero. Then $u' = 3A\mathrm{e}^{3x} - 2B\mathrm{e}^{-2x}$ and

$$y = -\frac{u'}{3u} = \frac{2B\mathrm{e}^{-2x} - 3A\mathrm{e}^{3x}}{3\left(A\mathrm{e}^{3x} + B\mathrm{e}^{-2x}\right)}.$$

Ask yourself A first-order equation should have one arbitrary constant. Why do we have two?

Only the ratio of A and B matters: multiplying u by a non-zero constant doesn't change u'/u. If $A \neq 0$, divide top and bottom by $A\mathrm{e}^{3x}$ and write $K = B/A$:

$$y = \frac{2K\mathrm{e}^{-5x} - 3}{3\left(1 + K\mathrm{e}^{-5x}\right)} = \frac{2K - 3\mathrm{e}^{5x}}{3\left(\mathrm{e}^{5x} + K\right)},$$

where K is an arbitrary constant. The case $A = 0$ gives the constant solution $y = \frac{2}{3}$, which is the limit of this family as $K \to \infty$. Taking $K = 0$ gives the other constant solution, $y = -1$. As a check, the constant solutions satisfy $-y - 3y^2 = -2$, that is, $(3y - 2)(y + 1) = 0$.

Part (ii)

Put $y = \dfrac{1}{x} + \dfrac{1}{v}$. Then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{x^2} - \frac{v'}{v^2},$$

and so

$$\begin{aligned} x^2\frac{\mathrm{d}y}{\mathrm{d}x} &= -1 - \frac{x^2v'}{v^2}, \\ xy &= 1 + \frac{x}{v}, \\ x^2y^2 &= x^2\left(\frac{1}{x^2} + \frac{2}{xv} + \frac{1}{v^2}\right) = 1 + \frac{2x}{v} + \frac{x^2}{v^2}. \end{aligned}$$

Adding these three lines, the equation $x^2y' + xy + x^2y^2 = 1$ becomes

$$1 - \frac{x^2v'}{v^2} + \frac{3x}{v} + \frac{x^2}{v^2} = 1, \qquad \text{that is,} \qquad -\frac{x^2v'}{v^2} + \frac{3x}{v} + \frac{x^2}{v^2} = 0.$$

Multiplying by v^2/x^2 (with $x \neq 0$, as the equation requires) gives the linear equation

$$\frac{\mathrm{d}v}{\mathrm{d}x} - \frac{3}{x}v = 1.$$

The integrating factor is $\exp\left(-\int \frac{3}{x}\,\mathrm{d}x\right) = x^{-3}$ (on either of the intervals $x > 0$ or $x < 0$; we can drop the modulus sign because a constant multiple of an integrating factor works just as well). Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{v}{x^3}\right) = \frac{1}{x^3} \quad \Longrightarrow \quad \frac{v}{x^3} = -\frac{1}{2x^2} + c \quad \Longrightarrow \quad v = cx^3 - \tfrac{1}{2}x = \frac{x\left(Kx^2 - 1\right)}{2},$$

where c is an arbitrary constant and $K = 2c$. Therefore

$$y = \frac{1}{x} + \frac{2}{x\left(Kx^2 - 1\right)} = \frac{Kx^2 + 1}{x\left(Kx^2 - 1\right)},$$

with K an arbitrary constant. Taking $K = 0$ gives $y = -1/x$. The substitution assumed $1/v$ exists, so it misses $y = 1/x$ itself ("$v = \infty$"). This is also a solution, since $x^2\left(-\frac{1}{x^2}\right) + 1 + 1 = 1$, and it is the limit of the family as $K \to \infty$.

Checks

- In (i), at $x = 0$ the solution is $y(0) = \dfrac{2K - 3}{3(1 + K)}$. As $x \to +\infty$, $y \to -1$, and as $x \to -\infty$, $y \to \frac{2}{3}$ (when $K \neq 0$). So solutions run from one constant solution to the other, which is what we'd expect.
- In (ii), take $K = 1$ and $x = 2$: $y = \frac{5}{6}$ and $y'(2) = -\frac{31}{36}$. Then $x^2y' + xy + x^2y^2 = -\frac{31}{9} + \frac{5}{3} + \frac{25}{9} = 1$, as required.
- We checked both general solutions numerically, by finite differences, at hundreds of random points and values of K. The residual in the original equation was zero to within rounding error.

Common slips

- Leaving two constants A, B in the answer to (i) without noticing that only their ratio matters.
- Forgetting the factor 2 in the cross term of x^2y^2 in (ii), which turns $3v/x$ into $2v/x$ and gives the wrong integrating factor.
- Losing the solution $y = 1/x$ (or $y = \frac{2}{3}$ in (i)) that the substitution can't reach.

Beyond STEP

Equations of the form $y' = p(x) + q(x)y + r(x)y^2$ are called Riccati equations. Both tricks here are general. The substitution $y = -u'/(ru)$ always turns a Riccati equation into a linear second-order equation. And if you know one particular solution y_1, then $y = y_1 + 1/v$ always gives a linear first-order equation for v. This link between Riccati equations and linear second-order equations comes up again in quantum mechanics and control theory. Boyce and DiPrima, *Elementary Differential Equations*, has exercises on Riccati equations.

WORKED SOLUTION

STEP II 1993, Question 4

Pure • Skew Lines and Closest Approach

STEP II 1993, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q4

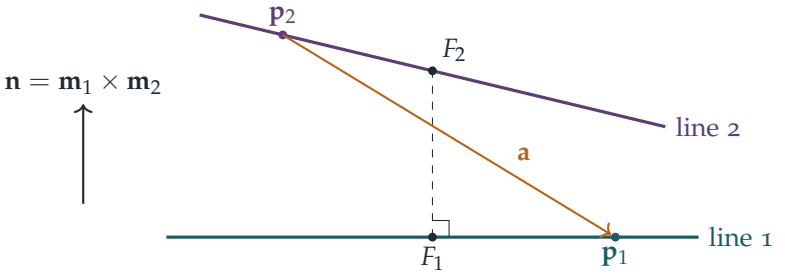
The key idea

Any vector joining a point of one line to a point of the other has the same component along the common normal $\mathbf{m}_1 \times \mathbf{m}_2$, because moving along either line adds only multiples of $\mathbf{m}_1$ and $\mathbf{m}_2$, which are perpendicular to that normal. That fixed component is the shortest distance. For the aircraft, the trick is to work in the frame of A_2: the vector from A_2 to A_1 moves along a *single* straight line, so the closest approach is the distance from a point to a line.

Solution

The shortest distance between the lines

Write $\mathbf{a} = \mathbf{p}_1 - \mathbf{p}_2$ and $\mathbf{n} = \mathbf{m}_1 \times \mathbf{m}_2$, which is non-zero because the lines are not parallel, and is perpendicular to both $\mathbf{m}_1$ and $\mathbf{m}_2$.



The sketch shows the two lines (skew, so they do not meet), the vector $\mathbf{a}$ from $\mathbf{p}_2$ to $\mathbf{p}_1$, and the common perpendicular F_1F_2, which is parallel to $\mathbf{n}$.

A general vector from a point of line 2 to a point of line 1 is

$$\mathbf{d} = \mathbf{a} + t_1\mathbf{m}_1 - t_2\mathbf{m}_2.$$

Since $\mathbf{m}_1 \cdot \mathbf{n} = \mathbf{m}_2 \cdot \mathbf{n} = 0$, its component along the unit normal $\hat{\mathbf{n}} = \mathbf{n}/|\mathbf{n}|$ is

$$\mathbf{d} \cdot \hat{\mathbf{n}} = \mathbf{a} \cdot \hat{\mathbf{n}},$$

the same for every choice of t_1, t_2. A vector is at least as long as any of its components, so $|\mathbf{d}| \geq |\mathbf{a} \cdot \hat{\mathbf{n}}|$, with equality exactly when $\mathbf{d}$ is parallel to $\mathbf{n}$. Equality does happen: $\mathbf{m}_1, \mathbf{m}_2, \mathbf{n}$ are not coplanar, so they form a basis and we can write $\mathbf{a} = \alpha\mathbf{m}_1 + \beta\mathbf{m}_2 + \gamma\mathbf{n}$; choosing $t_1 = -\alpha$, $t_2 = \beta$ leaves $\mathbf{d} = \gamma\mathbf{n}$. That choice gives the feet F_1, F_2 in the sketch. In the picture, the shortest distance is the "height" of $\mathbf{a}$ in the direction of $\mathbf{n}$. Hence the shortest distance is

$$|\mathbf{a} \cdot \hat{\mathbf{n}}| = \frac{|(\mathbf{p}_1 - \mathbf{p}_2) \cdot (\mathbf{m}_1 \times \mathbf{m}_2)|}{|\mathbf{m}_1 \times \mathbf{m}_2|}.$$

Part (i)

Here $\mathbf{a} = (1, 1, 1)$. The formula is unchanged if we replace $\mathbf{m}_1$ and $\mathbf{m}_2$ by any non-zero multiples, since the factors cancel top and bottom, so we may use $(4, 3, 0)$ and $(0, -3, 1)$:

$$(4, 3, 0) \times (0, -3, 1) = (3, -4, -12), \qquad |(3, -4, -12)| = \sqrt{9 + 16 + 144} = 13.$$

Then $(1, 1, 1) \cdot (3, -4, -12) = -13$, so the shortest distance is $13/13 = 1$.

Part (ii)

At time t the aircraft are at $\mathbf{p}_1 + v_1 t\,\mathbf{m}_1$ and $\mathbf{p}_2 + v_2 t\,\mathbf{m}_2$, so the vector from A_2 to A_1 is

$$\mathbf{r}(t) = \mathbf{a} + t\mathbf{w}, \qquad \mathbf{w} = v_1\mathbf{m}_1 - v_2\mathbf{m}_2.$$

Since $\mathbf{m}_1, \mathbf{m}_2$ are not parallel, $\mathbf{w} \neq \mathbf{0}$ (unless both speeds are zero). We take the flight to include all times, before and after $t = 0$. Then

$$|\mathbf{r}(t)|^2 = |\mathbf{a}|^2 + 2t\,\mathbf{a} \cdot \mathbf{w} + t^2|\mathbf{w}|^2,$$

a quadratic in t with positive leading coefficient. Completing the square,

$$|\mathbf{r}(t)|^2 = |\mathbf{w}|^2\Big(t + \frac{\mathbf{a} \cdot \mathbf{w}}{|\mathbf{w}|^2}\Big)^2 + |\mathbf{a}|^2 - \frac{(\mathbf{a} \cdot \mathbf{w})^2}{|\mathbf{w}|^2},$$

so the minimum, at $t = -\mathbf{a} \cdot \mathbf{w}/|\mathbf{w}|^2$, is

$$d^2 = \frac{|\mathbf{p}_1 - \mathbf{p}_2|^2\,|v_1\mathbf{m}_1 - v_2\mathbf{m}_2|^2 - [(\mathbf{p}_1 - \mathbf{p}_2) \cdot (v_1\mathbf{m}_1 - v_2\mathbf{m}_2)]^2}{|v_1\mathbf{m}_1 - v_2\mathbf{m}_2|^2},$$

as required. Geometrically, d is the distance from the origin to the line $\mathbf{r} = \mathbf{a} + t\mathbf{w}$, which is $|\mathbf{a}| \sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{w}$.

Part (iii)

First a bound that needs no algebra: at every moment A_1 is on line 1 and A_2 on line 2, so $d \geq 1$ by part (i). Equality would need both aircraft to be at the feet F_1, F_2 *at the same time*.

Ask yourself Can the pilot of A_2 actually arrange that?

Find the feet. Requiring $\mathbf{a} + t_1\mathbf{m}_1 - t_2\mathbf{m}_2$ to be perpendicular to $\mathbf{m}_1$ and $\mathbf{m}_2$, with $\mathbf{a}\cdot\mathbf{m}_1 = \frac{7}{5}$, $\mathbf{a}\cdot\mathbf{m}_2 = -\frac{2}{\sqrt{10}}$ and $\mathbf{m}_1\cdot\mathbf{m}_2 = -\frac{9}{5\sqrt{10}}$, and solving the two linear equations gives

$$t_1 = -\tfrac{20}{13}, \qquad t_2 = \tfrac{\sqrt{10}}{13}.$$

A_1 is at F_1 at time $t_1/v_1 < 0$, but A_2 is at F_2 at time $t_2/v_2 > 0$ when $v_2 > 0$. These can never agree: F_1 lies behind $\mathbf{p}_1$ while F_2 lies ahead of $\mathbf{p}_2$. So with A_2 flying in the direction $\mathbf{m}_2$, the value $d = 1$ can't be reached. To see what can be reached, we use part (ii).

Put $v_2 = \sqrt{10}\,\lambda v_1$, so λ is a scaled speed ratio. Then

$$\mathbf{w} = v_1\big(\tfrac{1}{5}(4,3,0) - \lambda(0,-3,1)\big) = \tfrac{v_1}{5}(4,\ 3+15\lambda,\ -5\lambda),$$

and the factor $\frac{v_1}{5}$ cancels from the formula for d^2. With $|\mathbf{a}|^2 = 3$,

$$\mathbf{a}\cdot(4,3+15\lambda,-5\lambda) = 7+10\lambda, \qquad |(4,3+15\lambda,-5\lambda)|^2 = 25+90\lambda+250\lambda^2,$$

so

$$d^2 = 3 - \frac{(7+10\lambda)^2}{25+90\lambda+250\lambda^2} = \frac{26+130\lambda+650\lambda^2}{25+90\lambda+250\lambda^2} = \frac{26}{5}\cdot\frac{1+5\lambda+25\lambda^2}{5+18\lambda+50\lambda^2}.$$

Differentiating the last fraction N/D, the numerator of the derivative simplifies:

$$\begin{aligned} N'D - ND' &= (5+50\lambda)(5+18\lambda+50\lambda^2) - (1+5\lambda+25\lambda^2)(18+100\lambda) \\ &= 7+150\lambda+200\lambda^2 = (1+20\lambda)(7+10\lambda). \end{aligned}$$

So, as a function of every real λ, d^2 has a maximum at $\lambda = -\frac{7}{10}$ (where $\mathbf{a}\perp\mathbf{w}$ and $d = \sqrt{3}$) and a minimum at $\lambda = -\frac{1}{20}$, where

$$d^2 = \frac{26}{5}\cdot\frac{1-\frac{1}{4}+\frac{1}{16}}{5-\frac{9}{10}+\frac{1}{8}} = \frac{26}{5}\cdot\frac{13/16}{169/40} = 1.$$

This is the bound $d = 1$, attained when $v_2 = -\frac{v_1}{2\sqrt{10}}$, which agrees with t_2/t_1 from the feet.

So the aircraft can come within $d = 1$, the shortest distance between the lines, and no closer. To achieve it, the pilot of A_2 flies along line 2 in the direction $-\mathbf{m}_2$ at speed $\frac{v_1}{2\sqrt{10}}$: in the question's notation, $v_2 = -\frac{v_1}{2\sqrt{10}}$, a velocity along $\mathbf{m}_2$ rather than a speed. Then both aircraft reach the feet of the common perpendicular at the same moment. This reproduces the answer to part (i), which is presumably what the examiners intended. Notice that the answer depends only on the ratio v_2/v_1.

A remark on positive speeds. If v_2 must be a genuine speed, so $v_2 > 0$ and $\lambda > 0$, then $(1+20\lambda)(7+10\lambda) > 0$ and d^2 is strictly increasing in v_2:

$$d^2 \to \frac{26}{25} \quad \text{as } v_2 \to 0^+, \qquad d^2 \to \frac{13}{5} \quad \text{as } v_2 \to \infty.$$

In that reading there is no best speed: the pilot does better the slower they fly, and the closest approach can be made as near as we like to $\frac{\sqrt{26}}{5} \approx 1.020$, the distance from $\mathbf{p}_2$ to line 1, but never reaches it.

Checks

- The common perpendicular in (i): with $t_1 = -\frac{20}{13}$, $t_2 = \frac{\sqrt{10}}{13}$ the joining vector is $\frac{1}{13}(-3, 4, 12)$, which is parallel to $(3, -4, -12)$ and has length exactly 1.
- A computer minimised $|\mathbf{r}(t)|$ by direct search for random speeds and matched the formula in (ii) to 10^{-6}. It also scanned v_2/v_1 over $[-2, 2]$: the minimum is $d^2 = 1.0000$ at $v_2/v_1 \approx -0.1581 = -1/(2\sqrt{10})$, and d^2 rises from 1.04 for $v_2 > 0$.
- Limiting cases: $v_2 = 0$ makes A_2 a fixed point at $\mathbf{p}_2$, and $3 - (\frac{7}{5})^2 = \frac{26}{25}$ is indeed the squared distance from $\mathbf{p}_2$ to line 1.

Common slips

- Answering (iii) with "1, from part (i)" without checking that both aircraft can reach the feet of the common perpendicular at the same moment.
- Forgetting that the formula in (ii) is unchanged by scaling $\mathbf{w}$, and so carrying v_1 through pages of algebra.
- Dropping the modulus in the line-distance formula and getting -1.

Beyond STEP

Part (ii) is a small example of *orthogonal projection*: d is the length of the component of $\mathbf{a}$ perpendicular to $\mathbf{w}$, and $|\mathbf{a}|^2|\mathbf{w}|^2 - (\mathbf{a} \cdot \mathbf{w})^2 = |\mathbf{a} \times \mathbf{w}|^2$ is the Gram determinant, the squared area of the parallelogram on $\mathbf{a}$ and $\mathbf{w}$. Least-squares fitting in a first linear algebra course rests on the same projection idea. Strang's *Introduction to Linear Algebra* develops it beautifully.

WORKED SOLUTION

STEP II 1993, Question 5

Pure • Polar Curves and Arc Length

STEP II 1993, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q5

The key idea

The right angle at R lets us read off every length in the small triangle PQR from the right-angled triangle ORQ, using only OQ and the small angle $\delta\theta$. Taking the limit as $Q \to P$ turns the chord PQ into the tangent. For the spiral, $\mathrm{d}r/\mathrm{d}\theta$ is a constant multiple of r, so $r\,\mathrm{d}\theta/\mathrm{d}r$ is the same everywhere, and the arc-length integrand is just an exponential.

Solution

The angle between the curve and the radius

The point R lies on the line OP (produced), and $\angle ORQ$ is a right angle. In the right-angled triangle ORQ the hypotenuse is $OQ = r + \delta r$ and the angle at O is $\delta\theta$, so

$$QR = (r+\delta r)\sin\delta\theta, \qquad OR = (r+\delta r)\cos\delta\theta.$$

Since $OP = r$, we have $PR = OR - OP = (r+\delta r)\cos\delta\theta - r$. The triangle PRQ is right-angled at R, so

$$\tan\angle QPR = \frac{QR}{PR} = \frac{(r+\delta r)\sin\delta\theta}{(r+\delta r)\cos\delta\theta - r}.$$

For small $\delta\theta$ we have $\sin\delta\theta = \delta\theta + O(\delta\theta^3)$ and $\cos\delta\theta = 1 + O(\delta\theta^2)$, so

$$\tan\angle QPR = \frac{r\,\delta\theta + \delta r\,\delta\theta + \cdots}{\delta r + O(\delta\theta^2)} = \frac{r\,\dfrac{\delta\theta}{\delta r} + \delta\theta + \cdots}{1 + O\left(\dfrac{\delta\theta^2}{\delta r}\right)},$$

where the dots are terms that are small compared with those shown. As $Q \to P$, both δr and $\delta\theta$ tend to 0 with $\delta\theta/\delta r \to \mathrm{d}\theta/\mathrm{d}r$. So $\delta\theta^2/\delta r = \delta\theta\cdot(\delta\theta/\delta r) \to 0$, and also $\delta\theta \to 0$. Hence

$$\tan\angle QPR \to r\frac{\mathrm{d}\theta}{\mathrm{d}r}.$$

Ask yourself Why is the limit of $\angle QPR$ the angle at which the curve cuts OP?

The angle QPR is the angle between the chord PQ and the radius line through P. As $Q \to P$ along the curve, the chord PQ turns into the tangent at P. So in the limit $\angle QPR$ becomes the angle between the tangent and OP, which is what "the angle at which the curve cuts OP" means. Therefore that angle is

$$\tan^{-1}\left(r\frac{\mathrm{d}\theta}{\mathrm{d}r}\right),$$

as required.

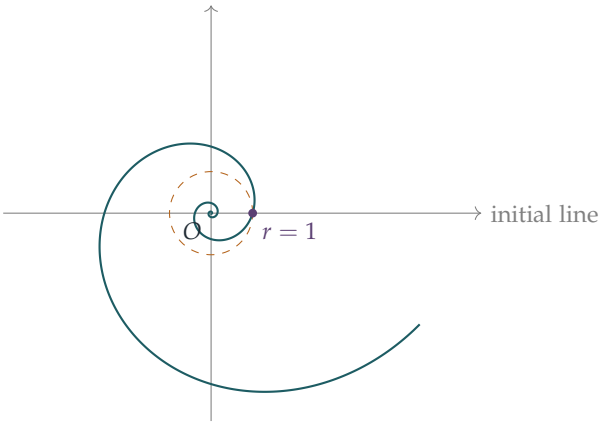
The equiangular spiral

For $r = \mathrm{e}^{\theta\cot\alpha}$,

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \cot\alpha\,\mathrm{e}^{\theta\cot\alpha} = r\cot\alpha, \qquad \text{so} \qquad r\frac{\mathrm{d}\theta}{\mathrm{d}r} = \frac{r}{r\cot\alpha} = \tan\alpha.$$

Because $0 < \alpha < \frac{1}{2}\pi$, the angle α lies in the range of $\tan^{-1}$, so $\tan^{-1}(\tan\alpha) = \alpha$. This doesn't depend on θ, so the spiral cuts every radius line at the angle α, at every point.

Sketch. Since $\cot\alpha > 0$, r increases steadily with θ. Each full turn multiplies r by the fixed factor $\mathrm{e}^{2\pi\cot\alpha}$. As $\theta \to -\infty$, $r \to 0$, so the curve winds round the origin infinitely often without ever reaching it. As $\theta \to +\infty$, it spirals outwards anticlockwise. It crosses the initial line at $r = 1$ when $\theta = 0$. The sketch uses $\cot\alpha = 0.3$.



The dashed circle is $r = 1$. The part of the spiral inside it is the infinite inward coil.

Arc lengths

Using the given formula and $\mathrm{d}r/\mathrm{d}\theta = r\cot\alpha$,

$$\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)^2 = r^2 + r^2\cot^2\alpha = r^2\operatorname{cosec}^2\alpha, \qquad \text{so} \qquad \frac{\mathrm{d}s}{\mathrm{d}\theta} = \frac{\mathrm{e}^{\theta\cot\alpha}}{\sin\alpha},$$

taking the positive root, since s increases with θ and $\sin\alpha > 0$.

The complete turn starting at $r = 1$ and going outwards runs from $\theta = 0$ to $\theta = 2\pi$. Its length is

$$\int_0^{2\pi} \frac{\mathrm{e}^{\theta\cot\alpha}}{\sin\alpha}\,\mathrm{d}\theta = \frac{1}{\sin\alpha}\cdot\frac{\mathrm{e}^{2\pi\cot\alpha}-1}{\cot\alpha} = \sec\alpha\left(\mathrm{e}^{2\pi\cot\alpha}-1\right).$$

The part with $r \le 1$ is $\theta \le 0$, that is, $-\infty < \theta \le 0$. Its length is the improper integral

$$\int_{-\infty}^{0} \frac{\mathrm{e}^{\theta\cot\alpha}}{\sin\alpha}\,\mathrm{d}\theta = \lim_{T\to\infty}\frac{1-\mathrm{e}^{-T\cot\alpha}}{\sin\alpha\cot\alpha} = \sec\alpha,$$

since $\cot\alpha > 0$ makes $\mathrm{e}^{-T\cot\alpha} \to 0$. So even though this part winds round the origin infinitely many times, its total length is finite, namely $\sec\alpha$.

Checks

- As $\alpha \to \frac{1}{2}\pi$ we have $\cot\alpha \to 0$, and the spiral becomes the unit circle, which every radius cuts at a right angle. The length of one turn is $\sec\alpha(\mathrm{e}^{2\pi\cot\alpha}-1) \approx \sec\alpha\cdot 2\pi\cot\alpha = 2\pi\operatorname{cosec}\alpha \to 2\pi$, the circumference.
- A nice geometric check of $\sec\alpha$: the length from any point to the origin along the spiral is $r\sec\alpha$, which is the length of the tangent segment from that point to the line through O perpendicular to OP.
- Numerically, with $\alpha = 0.9$, summing 2×10^5 chords gives one turn of length 233.8349 and the inner part of length 1.608726. These agree with $\sec\alpha(\mathrm{e}^{2\pi\cot\alpha}-1)$ and $\sec\alpha$ to at least eight significant figures.

Common slips

- Writing $PR = \delta r$ and $QR = r\,\delta\theta$ without explaining why. Both are only true to first order, and the limit argument is what the question is testing.
- Integrating over $[0, 2\pi]$ for the inner part instead of over $(-\infty, 0]$. The part with $r \le 1$ consists of infinitely many turns, not one.

Beyond STEP

The equiangular (logarithmic) spiral is self-similar: scaling it by $\mathrm{e}^{2\pi\cot\alpha}$, or by any factor, gives the same curve rotated. In complex terms it is $z = \mathrm{e}^{(\cot\alpha+\mathrm{i})\theta}$, the image of a straight line under the exponential map, and the constant angle follows from the fact that exp is conformal (it preserves angles). Needham's *Visual Complex Analysis* develops this picture beautifully. The finite length of an infinitely coiled curve is a first taste of the improper integrals and convergence questions met in a first analysis course.

WORKED SOLUTION

STEP II 1993, Question 6

Pure • Matrix Algebra Counterexamples

STEP II 1993, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q6

This question has a full guided treatment in *Beyond STEP*, Section 7.1, *Rules That Fail for Matrices*: https://beyondstep.keithmcnulty.org/ch07/matrix-rules.html

The key idea

Each assertion is something that would be true for real numbers, so the question is really asking which habits from ordinary algebra survive for matrices. Two things break: matrix multiplication need not commute, and a product of two non-zero matrices can be zero. A single well-chosen pair of very simple matrices settles (i) and (ii), the adjugate matrix settles (iii), and a repeated "root" with a nilpotent twist settles (iv).

Solution

Throughout, $\mathbf{I}$ is the 2×2 identity and $\mathbf{0}$ the zero matrix.

Part (i)

False. Take

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

both non-zero. Then

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}.$$

? **Ask yourself** How would you find a counterexample like this, rather than just guess one?

We want $\mathbf{AB} = \mathbf{0}$, so every column of $\mathbf{B}$ must be killed by $\mathbf{A}$. The matrix $\mathbf{A}$ above kills any vector of the form $(0, y)^{\mathsf{T}}$, so we may take the columns of $\mathbf{B}$ to be of that form.

Then we just check whether $\mathbf{BA}$ happens to vanish; with the most obvious choice it doesn't.

Part (ii)

False. Expanding carefully, keeping the order of every product,

$$(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B}) = \mathbf{A}^2 + \mathbf{AB} - \mathbf{BA} - \mathbf{B}^2,$$

so the assertion holds exactly when $\mathbf{AB} = \mathbf{BA}$. The matrices of part (i) don't commute, so they give a counterexample: there $\mathbf{AB} - \mathbf{BA} = -\mathbf{BA} \neq \mathbf{0}$. Explicitly,

$$(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B}) = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \qquad \mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

using $\mathbf{A}^2 = \mathbf{A}$ and $\mathbf{B}^2 = \mathbf{0}$.

Part (iii)

True. We prove both directions.

If $\mathbf{AX} = \mathbf{0}$ has a non-zero solution, then $\det \mathbf{A} = 0$. We prove the contrapositive. Suppose $\det \mathbf{A} \neq 0$. Then $\mathbf{A}^{-1}$ exists, and $\mathbf{AX} = \mathbf{0}$ gives $\mathbf{X} = \mathbf{A}^{-1}\mathbf{AX} = \mathbf{A}^{-1}\mathbf{0} = \mathbf{0}$. So the only solution is the zero matrix.

If $\det \mathbf{A} = 0$, *then there is a non-zero solution.* Write

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

(so $\mathbf{X}$ is the adjugate of $\mathbf{A}$). Multiplying out,

$$\mathbf{AX} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (\det \mathbf{A})\,\mathbf{I} = \mathbf{0}.$$

The entries of $\mathbf{X}$ are those of $\mathbf{A}$ rearranged, with some signs changed, and $\mathbf{A} \neq \mathbf{0}$, so $\mathbf{X} \neq \mathbf{0}$. This completes the proof. Notice that the hypothesis $\mathbf{A} \neq \mathbf{0}$ is used here: it is what guarantees that the adjugate is non-zero.

Part (iv)

False. For numbers, a quadratic has at most two roots, but the proof of that fact uses "if $pq = 0$ then $p = 0$ or $q = 0$", which part (i) has just shown fails for matrices. So look for a quadratic that factorises as a perfect square. Take $\mathbf{A} = -2\mathbf{I}$ and $\mathbf{B} = \mathbf{I}$, both non-zero. Since $\mathbf{I}$ commutes with everything,

$$\mathbf{X}^2 - 2\mathbf{X} + \mathbf{I} = (\mathbf{X} - \mathbf{I})^2,$$

so we need $(\mathbf{X} - \mathbf{I})^2 = \mathbf{0}$. Any matrix $\mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$ will do for $\mathbf{X} - \mathbf{I}$, and there are infinitely many. For each real t let

$$\mathbf{X}_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Then

$$\mathbf{X}_t^2 - 2\mathbf{X}_t + \mathbf{I} = \begin{pmatrix} 1 & 2t \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 2 & 2t \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{0}.$$

Each $\mathbf{X}_t$ is non-zero, and different values of t give different matrices. So this equation has infinitely many solutions, not at most two.

? **Ask yourself** Is there anything special about $\mathbf{A} = -2\mathbf{I}$, $\mathbf{B} = \mathbf{I}$?

Not really. With $\mathbf{A} = -2\lambda\mathbf{I}$ and $\mathbf{B} = \lambda^2\mathbf{I}$ (for any $\lambda \neq 0$), every $\mathbf{X} = \lambda\mathbf{I} + \mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$ works. The nilpotent matrices themselves form a large family: $\begin{pmatrix} pq & -p^2 \\ q^2 & -pq \end{pmatrix}$ squares to $\mathbf{0}$ for all real p, q. The essential point is only that the quadratic factorises as a product of two matrices, and such a product can vanish in many more ways than for numbers.

Checks

Multiply out each counterexample by hand: in (i), $\mathbf{AB}$ has every entry 0 while $\mathbf{BA}$ has a 1 in the bottom-left corner. The identity $(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B}) = \mathbf{A}^2 - \mathbf{B}^2 + \mathbf{AB} - \mathbf{BA}$ was tested on 1000 random integer pairs, and $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = \mathbf{0}$ with $\mathrm{adj}\,\mathbf{A} \neq \mathbf{0}$ on 2000 random singular matrices. In (iv), try $t = 3$: $\mathbf{X}_3^2 = \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$, and $\mathbf{X}_3^2 - 2\mathbf{X}_3 + \mathbf{I} = \mathbf{0}$.

Common slips

- Expanding $(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B})$ as if $\mathbf{AB} = \mathbf{BA}$. Writing "false" in (ii) without an explicit counterexample earns little: the question asks for one.
- In (iii), proving only one direction of the "if and only if".
- In (iv), "solving" the quadratic with the formula or by factorising and concluding $\mathbf{X} = \mathbf{I}$ is the only root. Factorising is fine, but $\mathbf{P}^2 = \mathbf{0}$ does not imply $\mathbf{P} = \mathbf{0}$.

Beyond STEP

The 2×2 real matrices form a *non-commutative ring with zero divisors*, and each part of this question tests a law that holds in a field but fails in such a ring. Part (iii) is the 2×2 case of a central theorem of linear algebra: a square matrix has a non-trivial null space exactly when its determinant vanishes. Matrices with $\mathbf{N}^2 = \mathbf{0}$, like those in (iv), are called nilpotent and are the key ingredient in the Jordan normal form. Strang's *Introduction to Linear Algebra* covers the linear algebra, and Allenby's *Rings, Fields and Groups* the ring theory.

WORKED SOLUTION

STEP II 1993, Question 7

Pure • Infinite Descent Modulo 5

STEP II 1993, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q7

The key idea

The right-hand side is a multiple of 5, so working modulo 5 turns the equation into a small finite check: a square can only be 0, 1 or 4 $\pmod 5$. Once a and b are forced to be multiples of 5, the equation forces c to be one too, and dividing everything by 5 gives a *smaller* solution. A non-zero integer can't be divided by 5 for ever, so this is Fermat's method of infinite descent.

Solution

Divisibility of a and b by 5

Squaring each residue modulo 5:

$x \pmod 5$	0	1	2	3	4
$x^2 \pmod 5$	0	1	4	4	1
$2x^2 \pmod 5$	0	2	3	3	2

So $b^2 \equiv 0, 1$ or 4, and $2a^2 \equiv 0, 2$ or $3 \pmod 5$. Since $5c^2 \equiv 0 \pmod 5$, we need

$$2a^2 + b^2 \equiv 0 \pmod 5.$$

Now we list every combination:

$2a^2 + b^2$	$b^2 \equiv 0$	$b^2 \equiv 1$	$b^2 \equiv 4$
$2a^2 \equiv 0$	**0**	1	4
$2a^2 \equiv 2$	2	3	1
$2a^2 \equiv 3$	3	4	2

The only entry that is 0 is the one with $2a^2 \equiv 0$ and $b^2 \equiv 0$. From the first table, $x^2 \equiv 0 \pmod 5$ only when $x \equiv 0$ (equivalently, 5 is prime, so $5 \mid x^2$ implies $5 \mid x$). Hence $a \equiv 0$ and $b \equiv 0 \pmod 5$: both a and b are divisible by 5, as required.

The only solution is $a = b = c = 0$

Write $a = 5p$ and $b = 5q$ with p, q integers. Substituting,

$$25(2p^2 + q^2) = 5c^2 \quad \Longrightarrow \quad c^2 = 5(2p^2 + q^2).$$

So $5 \mid c^2$, and since 5 is prime, $5 \mid c$. Write $c = 5r$. Then $25r^2 = 5(2p^2 + q^2)$, that is,

$$2p^2 + q^2 = 5r^2.$$

This is the original equation again. So if (a, b, c) is a solution, then all three are divisible by 5 and $(a/5,\ b/5,\ c/5)$ is also a solution in integers.

Ask yourself Why does "we can keep dividing by 5" force everything to be zero?

Apply the step repeatedly. Starting from a solution (a, b, c), after n steps we have the integer solution $(a/5^n,\ b/5^n,\ c/5^n)$, so 5^n divides each of a, b and c, for *every* $n \geq 1$.

But a non-zero integer m can only be divided by 5 a limited number of times: if $5^n \mid m$ and $m \neq 0$ then $5^n \leq |m|$, which fails once n is large. So none of a, b, c can be non-zero.

Equivalently, as a descent argument: suppose a solution existed with $c \neq 0$, and pick one with $|c|$ as small as possible and positive. Then $(a/5, b/5, c/5)$ is a solution with $0 < |c/5| < |c|$, a contradiction. So $c = 0$, and then $2a^2 + b^2 = 0$ forces $a = b = 0$, since both terms are non-negative.

Hence the only integer solution is $a = b = c = 0$.

Checks

- Run through all 25 pairs of residues (a, b) mod 5: only $(0, 0)$ makes $2a^2 + b^2 \equiv 0$.
- A computer search over all $|a|, |b| \leq 300$ finds no solution of $2a^2 + b^2 = 5c^2$ other than $a = b = c = 0$.
- Contrast with $a^2 + b^2 = 5c^2$, which *does* have non-trivial solutions such as $(1, 2, 1)$: there $1 + 4 \equiv 0 \pmod 5$, so the residue argument in the first part breaks down, exactly as it should.

Common slips

- Concluding $5 \mid a$ from $5 \mid a^2$ without a reason. It's true because 5 is prime (or by the table), but it needs saying.
- Forgetting to show that c is divisible by 5 as well. Without it, $(a/5, b/5, c/5)$ need not be a solution in integers, and the descent doesn't start.
- Stopping after "we can divide by 5 again". The proof isn't complete until you say why repeated division forces zero.

Beyond STEP

This is Fermat's *infinite descent*, the same method that proves $\sqrt{2}$ is irrational and that $x^4 + y^4 = z^2$ has no non-trivial solutions. Equations like $2a^2 + b^2 = 5c^2$ are studied systematically in number theory: Legendre's theorem gives exact conditions, in terms of quadratic residues, for when $Ax^2 + By^2 = Cz^2$ has a non-trivial integer solution. Here the obstruction is that -2 is not a square modulo 5. Davenport's *The Higher Arithmetic* covers congruences, quadratic residues and descent.

WORKED SOLUTION

STEP II 1993, Question 8

Pure • AM-GM Inequality by Induction

STEP II 1993, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q8

This question has a full guided treatment in *Beyond STEP*, Section 3.2, *AM–GM, Backwards*: https://beyondstep.keithmcnulty.org/ch03/am-gm-backwards.html

The key idea

The two-term inequality is just a square being non-negative. To go from 2^m terms to 2^{m+1}, split the list into two halves of length 2^m, use the induction hypothesis on each half, and then use the two-term inequality on the two half-means. That proves AM–GM only when the number of terms is a power of 2; the final part fills the gap with a neat trick (due to Cauchy): pad the list with copies of its own mean, which changes neither the mean nor anything essential about the product.

Solution

Two terms

Expanding,

$$\left(\frac{a_1+a_2}{2}\right)^2 - a_1a_2 = \frac{a_1^2+2a_1a_2+a_2^2-4a_1a_2}{4} = \left(\frac{a_1-a_2}{2}\right)^2 \geq 0,$$

so $a_1a_2 \leq \left(\frac{a_1+a_2}{2}\right)^2$, with equality exactly when $a_1 = a_2$. Notice that this holds for *any* two positive numbers; we shall use it below for numbers other than a_1 and a_2.

The induction

Let $P(m)$ be the statement that $(*)$ holds for every list of 2^m positive numbers. The case $m = 1$ is the two-term inequality just proved.

Now suppose $P(m)$ holds, and take 2^{m+1} positive numbers $a_1, \ldots, a_{2^{m+1}}$. Write $N = 2^m$

and split them into two halves of N terms each, with means

$$X = \frac{a_1 + \cdots + a_N}{N}, \qquad Y = \frac{a_{N+1} + \cdots + a_{2N}}{N}.$$

Applying $P(m)$ to each half,

$$a_1 \cdots a_N \le X^N \qquad \text{and} \qquad a_{N+1} \cdots a_{2N} \le Y^N.$$

All the quantities here are positive, so we may multiply these two inequalities:

$$a_1 \cdots a_{2N} \le X^N Y^N = (XY)^N.$$

Now X and Y are positive, so the two-term inequality gives $XY \le \left(\frac{X+Y}{2}\right)^2$. Since $t \mapsto t^N$ is increasing for $t > 0$, raising both sides to the power N keeps the direction:

$$(XY)^N \le \left(\frac{X+Y}{2}\right)^{2N}.$$

Finally, $\dfrac{X+Y}{2} = \dfrac{(a_1 + \cdots + a_N) + (a_{N+1} + \cdots + a_{2N})}{2N}$ is the mean of all $2N = 2^{m+1}$ numbers. Putting the inequalities together,

$$a_1 \cdots a_{2^{m+1}} \le \left(\frac{a_1 + \cdots + a_{2^{m+1}}}{2^{m+1}}\right)^{2^{m+1}},$$

which is $P(m+1)$. By induction, $(*)$ holds for all positive integers m.

Ask yourself Is it legitimate to multiply two inequalities together like this?

If $0 < p \le q$ and $0 < r \le s$ then $pr \le qr \le qs$, using positivity of r and then of q. Without positivity this can fail (for example $-3 \le 1$ and $-3 \le 1$, but $9 > 1$), which is why the condition $a_i > 0$ matters.

Padding to a power of 2

The printed definition has $b_1 = a_2$, which must be a misprint for $b_1 = a_1$: we take $b_i = a_i$ for $1 \le i \le n$ and $b_i = A$ for $n < i \le 2^m$. Each b_i is positive (since A is a mean of positive numbers). Their sum is

$$b_1 + \cdots + b_{2^m} = (a_1 + \cdots + a_n) + (2^m - n)A = nA + (2^m - n)A = 2^m A,$$

so the mean of the b_i is exactly A. Their product is $a_1 \cdots a_n \, A^{2^m - n}$. Applying $(*)$ to the b_i therefore gives

$$a_1 \cdots a_n \, A^{2^m - n} \le A^{2^m},$$

as required.

The AM–GM inequality

Given any positive integer n, choose m with $2^m > n$ (possible since $2^m \to \infty$), so the previous step applies. Dividing by A^{2^m-n}, which is positive, gives

$$a_1 \cdots a_n \le A^n.$$

Both sides are positive and $t \mapsto t^{1/n}$ is increasing on $t > 0$, so taking nth roots,

$$(a_1 \cdots a_n)^{1/n} \le A = \frac{a_1 + \cdots + a_n}{n},$$

which is the AM–GM inequality. (If n happens to be a power of 2 we could instead quote $(*)$ directly, but choosing $2^m > n$ means no separate case is needed.)

Checks

Try $n = 3$ with $a = (1, 2, 6)$: $A = 3$, and with $m = 2$ the padded list is $(1, 2, 6, 3)$, whose mean is 3 and whose product is $36 \le 3^4 = 81$; dividing by 3 gives $12 \le 27 = A^3$, so $\sqrt[3]{12} \approx 2.29 \le 3$. Also, when all the a_i are equal every inequality used becomes an equality, as it should. A computer testing $(*)$ for $m \le 6$ and the final inequality for $n \le 40$ on thousands of random lists, in exact rational arithmetic, found no failures.

Common slips

- Applying the induction hypothesis to all 2^{m+1} terms at once: it only covers lists of length 2^m, so you must split into halves.
- Forgetting to check that the padded list still has mean A; this is the whole point of padding with A rather than anything else.
- Taking nth roots or dividing without noting that the quantities are positive.

Beyond STEP

This "forward–backward" induction is Cauchy's proof of AM–GM: prove it for powers of 2, then come back down to every n. AM–GM is also a special case of Jensen's inequality for the concave function ln, and a whole family of power-mean, Hölder and Minkowski inequalities grows out of the same ideas; these are basic tools in first-year analysis. Bartle and Sherbert, *Introduction to Real Analysis*, uses inequalities like these from the start, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

WORKED SOLUTION

STEP II 1993, Question 9

Pure • Arguments of Complex Numbers

STEP II 1993, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q9

The key idea

With the convention $0 \leq \arg z < 2\pi$, "positive imaginary part" means "argument strictly between 0 and π". Adding complex numbers is adding vectors, and a sum of vectors that all point into the same open half-plane stays inside it; rotating so that z_1 or z_3 lies along the real axis turns the inequalities into exactly this fact. For the last part, the argument of a product is the sum of the arguments (up to a multiple of 2π), and the inequalities just proved trap the difference strictly between 0 and 2π.

Solution

Positive imaginary part

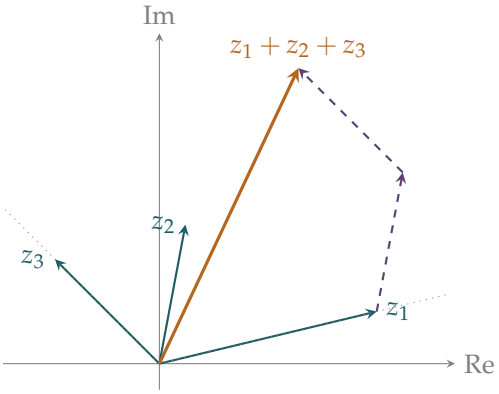
Write $z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r > 0$ and $0 \leq \theta < 2\pi$. Then $\operatorname{Im} z = r\sin\theta > 0$ exactly when $\sin\theta > 0$, and on $[0, 2\pi)$ this happens exactly when

$$0 < \arg z < \pi.$$

So z lies in the upper half-plane, and its argument is never 0 or π (those would make z real).

The diagram

Write $\theta_k = \arg z_k$, so $0 < \theta_1 < \theta_2 < \theta_3 < \pi$. Draw z_1, z_2, z_3 as arrows from the origin; then place them head to tail to reach $z_1 + z_2 + z_3$.



Each step of the path turns anticlockwise relative to the previous one (because $\theta_1 < \theta_2 < \theta_3$), and every step points into the upper half-plane. The path starts along the ray at angle θ_1 and then bends to the left of it, so its endpoint lies strictly anticlockwise of that ray; looking back from the other end, the path finishes with a step at angle θ_3, and everything before it lies clockwise of the ray at angle θ_3. So the sum lies in the wedge between the rays at angles θ_1 and θ_3.

To make this precise, multiply by $\mathrm{e}^{-\mathrm{i}\theta_1}$, which rotates everything clockwise by θ_1:

$$\mathrm{e}^{-\mathrm{i}\theta_1}(z_1 + z_2 + z_3) = r_1 + r_2\mathrm{e}^{\mathrm{i}(\theta_2-\theta_1)} + r_3\mathrm{e}^{\mathrm{i}(\theta_3-\theta_1)},$$

where $r_k = |z_k| > 0$. The first term is real and positive, and the other two have angles in $(0, \pi)$, so positive imaginary part. Hence the right-hand side has positive imaginary part, and its argument α satisfies $0 < \alpha < \pi$. Therefore $z_1 + z_2 + z_3$ has an argument $\theta_1 + \alpha$, and since $\theta_1 < \theta_1 + \alpha < \theta_1 + \pi < 2\pi$ this is the argument in the range $[0, 2\pi)$. So $\arg(z_1 + z_2 + z_3) > \theta_1$.

Similarly, multiplying by $\mathrm{e}^{-\mathrm{i}\theta_3}$ gives $r_1\mathrm{e}^{\mathrm{i}(\theta_1-\theta_3)} + r_2\mathrm{e}^{\mathrm{i}(\theta_2-\theta_3)} + r_3$, where the first two angles lie in $(-\pi, 0)$, so this has negative imaginary part and can be written $\rho\,\mathrm{e}^{-\mathrm{i}\beta}$ with $0 < \beta < \pi$. Then $z_1 + z_2 + z_3 = \rho\,\mathrm{e}^{\mathrm{i}(\theta_3-\beta)}$, where $-\pi < \theta_3 - \beta < \theta_3 < \pi$. The sum has positive imaginary part, so its argument lies in $(0, \pi)$ and differs from $\theta_3 - \beta$ by a multiple of 2π; as both numbers lie in $(-\pi, \pi)$, an interval of length 2π, they are equal. So $\arg(z_1 + z_2 + z_3) = \theta_3 - \beta < \theta_3$, and altogether

$$\arg z_1 < \arg(z_1 + z_2 + z_3) < \arg z_3.$$

The product and the sum

Let $\varphi = \arg(z_1 + z_2 + z_3)$, so $\theta_1 < \varphi < \theta_3$ by the previous part. Multiplying complex numbers adds their arguments, so

$$z_1z_2z_3 = r_1r_2r_3\,\mathrm{e}^{\mathrm{i}(\theta_1+\theta_2+\theta_3)},$$

and $\arg(z_1z_2z_3)$ is the number in $[0, 2\pi)$ that differs from $\theta_1 + \theta_2 + \theta_3$ by a multiple of 2π.

Ask yourself The sum $\theta_1+\theta_2+\theta_3$ can be anywhere in $(0,3\pi)$, so we don't know which multiple of 2π to subtract. How can we avoid needing to know?

We don't need to know: we only need that if $\arg(z_1z_2z_3)=\varphi$ then

$$D=\theta_1+\theta_2+\theta_3-\varphi$$

is a multiple of 2π. Now bound D using $\theta_1<\varphi<\theta_3$:

$$D>\theta_1+\theta_2+\theta_3-\theta_3=\theta_1+\theta_2>0,$$
$$D<\theta_1+\theta_2+\theta_3-\theta_1=\theta_2+\theta_3<2\pi,$$

using $\theta_k>0$ in the first line and $\theta_k<\pi$ in the second. So $0<D<2\pi$, and no multiple of 2π lies strictly between 0 and 2π. Hence $\arg(z_1z_2z_3)$ is never equal to $\arg(z_1+z_2+z_3)$.

Checks

For the diagram's example, $z_1=2.5+0.6\mathrm{i}$, $z_2=0.3+1.6\mathrm{i}$, $z_3=-1.2+1.2\mathrm{i}$, the arguments are about 0.236, 1.385 and 2.356, and $\arg(z_1+z_2+z_3)=\arg(1.6+3.4\mathrm{i})\approx 1.131$, which lies between the first and last. In a test of 200 000 random triples (moduli between e^{-4} and e^{4}), both inequalities and $0<D<2\pi$ held every time, and the product and sum never pointed in the same direction.

To see why the upper half-plane matters, try $z_1=z_2=z_3=\mathrm{e}^{2\pi\mathrm{i}/3}$ with the ordering relaxed to equality: the product is 1 and the sum points along $\mathrm{e}^{2\pi\mathrm{i}/3}$, consistent with $D=4\pi/3$. Letting all three angles approach π pushes D towards 2π, so the bound $\theta_k<\pi$ is doing real work.

Common slips

- Writing $\arg(z_1z_2z_3)=\theta_1+\theta_2+\theta_3$ outright: this is false whenever the sum is 2π or more. Only equality modulo 2π is available.
- Arguing that $\arg(z_1+z_2+z_3)$ is some kind of average of the θ_k. It isn't (the moduli matter); what is true is only that it lies strictly between θ_1 and θ_3.

Beyond STEP

The first fact is the statement that an open half-plane (indeed any wedge of angle less than π) is a *convex cone*: positive combinations of vectors in it stay in it. Convex cones are central in linear programming and optimisation, and the multivaluedness of arg that the last part works around leads to branch cuts and the complex logarithm in a first course on complex analysis. Needham's *Visual Complex Analysis* is a beautiful geometric treatment of both ideas.

WORKED SOLUTION

STEP II 1993, Question 10

Pure • Quadratic Forms and Conics

STEP II 1993, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q10

The key idea

A quadratic equation in x and y is messy for two reasons: the xy cross term and the linear terms. The matrix $\mathbf{P}$ deals with the first, because $\mathbf{PAP}$ is diagonal. The shift $\mathbf{a}$ deals with the second: we choose it to move the origin to the centre of the conic, so that every term linear in $\mathbf{x}_1$ cancels. What's left is a pure "sum of squares = constant".

Solution

Diagonalising

First,

$$\mathbf{AP} = \begin{pmatrix} -1 & 8 \\ 8 & 11 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 15 & -10 \\ 30 & 5 \end{pmatrix},$$

so

$$\mathbf{PAP} = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 15 & -10 \\ 30 & 5 \end{pmatrix} = \begin{pmatrix} 75 & 0 \\ 0 & -25 \end{pmatrix},$$

which is diagonal, as required. Note also that $\mathbf{P}$ is symmetric, so $\mathbf{P}^{\mathrm{T}} = \mathbf{P}$. We'll use this in a moment.

Substituting

Put $\mathbf{x} = \mathbf{P}\mathbf{x}_1 + \mathbf{a}$. Then $\mathbf{x}^{\mathrm{T}} = \mathbf{x}_1^{\mathrm{T}}\mathbf{P} + \mathbf{a}^{\mathrm{T}}$, and

$$\begin{aligned}\mathbf{x}^{\mathrm{T}}\mathbf{A}\mathbf{x} &= \mathbf{x}_1^{\mathrm{T}}\mathbf{PAP}\mathbf{x}_1 + \mathbf{x}_1^{\mathrm{T}}\mathbf{PA}\mathbf{a} + \mathbf{a}^{\mathrm{T}}\mathbf{AP}\mathbf{x}_1 + \mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a} \\ &= \mathbf{x}_1^{\mathrm{T}}\mathbf{PAP}\mathbf{x}_1 + 2\mathbf{a}^{\mathrm{T}}\mathbf{AP}\mathbf{x}_1 + \mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a}.\end{aligned}$$

The two middle terms are equal because each is a 1×1 matrix, so equals its own transpose, and $(\mathbf{x}_1^{\mathrm{T}}\mathbf{PA}\mathbf{a})^{\mathrm{T}} = \mathbf{a}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{P}^{\mathrm{T}}\mathbf{x}_1 = \mathbf{a}^{\mathrm{T}}\mathbf{AP}\mathbf{x}_1$, using that $\mathbf{A}$ and $\mathbf{P}$ are symmetric. Also $\mathbf{b}^{\mathrm{T}}\mathbf{x} = \mathbf{b}^{\mathrm{T}}\mathbf{P}\mathbf{x}_1 + \mathbf{b}^{\mathrm{T}}\mathbf{a}$. So the equation becomes

$$\mathbf{x}_1^{\mathrm{T}}\mathbf{PAP}\mathbf{x}_1 + \left(2\mathbf{a}^{\mathrm{T}}\mathbf{A} + \mathbf{b}^{\mathrm{T}}\right)\mathbf{P}\mathbf{x}_1 + \left(\mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a} + \mathbf{b}^{\mathrm{T}}\mathbf{a} - 11\right) = 0.$$

Ask yourself What does "a suitable vector $\mathbf{a}$" have to do?

The target $3x_1^2 - y_1^2 = c$ has no terms in x_1 or y_1 alone, so the middle bracket must vanish. Since $\mathbf{A}$ is symmetric, $2\mathbf{a}^{\mathrm{T}}\mathbf{A} + \mathbf{b}^{\mathrm{T}} = (2\mathbf{A}\mathbf{a} + \mathbf{b})^{\mathrm{T}}$, so we need

$$\mathbf{A}\mathbf{a} = -\tfrac{1}{2}\mathbf{b} = \begin{pmatrix} -9 \\ -3 \end{pmatrix}.$$

Here $\det \mathbf{A} = -11 - 64 = -75 \neq 0$, so there is exactly one solution:

$$\mathbf{a} = -\frac{1}{75}\begin{pmatrix} 11 & -8 \\ -8 & -1 \end{pmatrix}\begin{pmatrix} -9 \\ -3 \end{pmatrix} = -\frac{1}{75}\begin{pmatrix} -75 \\ 75 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

(Check: $\mathbf{A}\mathbf{a} = \begin{pmatrix} -1-8 \\ 8-11 \end{pmatrix} = \begin{pmatrix} -9 \\ -3 \end{pmatrix}$.)

The constant

With this $\mathbf{a}$, we have $\mathbf{a}^{\mathrm{T}}\mathbf{A}\mathbf{a} = \mathbf{a}^{\mathrm{T}}(-\tfrac{1}{2}\mathbf{b}) = -\tfrac{1}{2}\mathbf{b}^{\mathrm{T}}\mathbf{a}$, so the constant term is

$$\tfrac{1}{2}\mathbf{b}^{\mathrm{T}}\mathbf{a} - 11 = \tfrac{1}{2}(18 - 6) - 11 = -5.$$

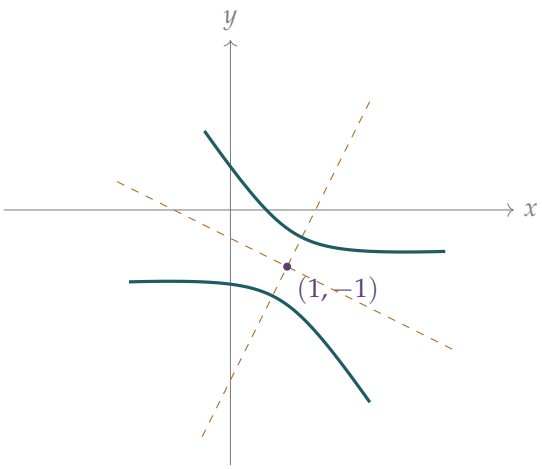
The quadratic term is

$$\mathbf{x}_1^{\mathrm{T}}\begin{pmatrix} 75 & 0 \\ 0 & -25 \end{pmatrix}\mathbf{x}_1 = 75x_1^2 - 25y_1^2.$$

So the equation becomes $75x_1^2 - 25y_1^2 - 5 = 0$. Dividing by 25,

$$3x_1^2 - y_1^2 = \tfrac{1}{5}, \qquad \text{so } c = \tfrac{1}{5}.$$

Written out, the substitution is $x = x_1 + 2y_1 + 1$, $y = 2x_1 - y_1 - 1$. The original equation is $-x^2 + 16xy + 11y^2 + 18x + 6y - 11 = 0$, so the curve is a hyperbola with centre $(1, -1)$. The columns of $\mathbf{P}$, $(1, 2)$ and $(2, -1)$, are perpendicular, so they give its axes of symmetry.



The dashed lines are the axes of symmetry, in the directions $(1,2)$ and $(2,-1)$.

Ask yourself Why isn't $\mathbf{P}$ a rotation?

Its columns are perpendicular, but each has length $\sqrt{5}$, so $\mathbf{P}^2 = 5\mathbf{I}$ rather than $\mathbf{I}$. That's why the diagonal entries of $\mathbf{PAP}$ are 75 and -25: they are 5 times the eigenvalues 15 and -5 of $\mathbf{A}$. The scale factor doesn't matter here, because it just multiplies the quadratic part by 5.

Checks

- The trace and determinant of $\mathbf{A}$ are 10 and -75, so its eigenvalues are 15 and -5. These match $75/5$ and $-25/5$.
- Take $x_1 = y_1 = 0$. Then $(x,y) = (1,-1)$ and the original left-hand side is $-1-16+11+18-6-11 = -5$, which agrees with $75x_1^2 - 25y_1^2 - 5$.
- Take $x_1 = 1$, $y_1 = 1$. Then $(x,y) = (4,0)$, and $-16+72-11 = 45 = 75-25-5$.

We also checked the identity LHS $= 25\left(3x_1^2 - y_1^2 - \frac{1}{5}\right)$ exactly at 1000 random rational points.

Common slips

- Forgetting that the cross terms $\mathbf{x}_1^{\mathrm{T}}\mathbf{PAa}$ and $\mathbf{a}^{\mathrm{T}}\mathbf{APx}_1$ are equal, which loses the factor 2 and gives $\mathbf{Aa} = -\mathbf{b}$.
- Stopping at $75x_1^2 - 25y_1^2 = 5$ and giving $c = 5$, without dividing through to match the required form.

Beyond STEP

This is the principal axes theorem in miniature. Every real symmetric matrix can be diagonalised by an orthogonal matrix (here $\mathbf{P}/\sqrt{5}$). So every conic, and every quadric surface in three dimensions, can be put into standard form by a rotation (or reflection) and a translation. The same idea classifies stationary points of functions of several variables through the Hessian matrix. Strang's *Introduction to Linear Algebra* has a good chapter on symmetric matrices and positive definiteness.

WORKED SOLUTION

STEP II 1993, Question 11

Mechanics • Motion with Quadratic Resistance

STEP II 1993, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

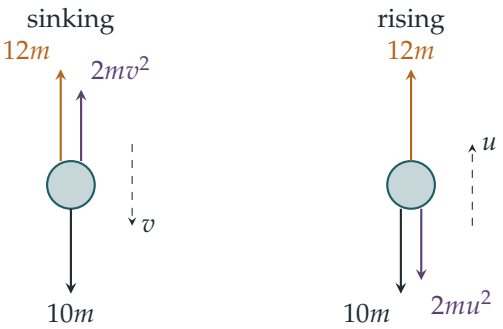
step.maths.org/questions/93-s2-q11

The key idea

Once the body is in the liquid, the buoyancy beats gravity, so there is a constant net upward force as well as the v^2 resistance. The resistance always opposes the motion, so its direction flips when the body turns round, and the sinking and rising phases need separate equations of motion. Each one separates neatly. Use $\frac{\mathrm{d}v}{\mathrm{d}t}$ when we want times and $v\frac{\mathrm{d}v}{\mathrm{d}x}$ when we want distances.

Solution

We model the body as a particle. In the liquid its weight is $10m$ N downwards and the buoyancy is $\frac{6}{5} \times 10m = 12m$ N upwards, so together they give a net force of $2m$ N *upwards*, whichever way the body is moving. The resistance $2mv^2$ N acts against the velocity.



The sinking phase

Take downwards as positive, with x the depth below the surface and v the downward speed. Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = 10m - 12m - 2mv^2 \qquad \Longrightarrow \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = -2(1+v^2).$$

Suppose the body enters with speed $V > 0$. Separating the variables and integrating from entry ($t = 0$, $v = V$) to the moment it stops sinking ($t = T_1$, $v = 0$):

$$\int_V^0 \frac{\mathrm{d}v}{1+v^2} = -2\int_0^{T_1} \mathrm{d}t \qquad \Longrightarrow \qquad T_1 = \tfrac{1}{2}\arctan V.$$

Since $\arctan V < \frac{\pi}{2}$ for every finite V, we get $T_1 < \frac{1}{4}\pi$: the body stops sinking less than $\frac{1}{4}\pi$ seconds after entering, however fast it was dropped in. (Also $\frac{\mathrm{d}v}{\mathrm{d}t} \le -2$ throughout, so it certainly does stop.)

Depth reached when $V = 1$

Now $T_1 = \frac{1}{2}\arctan 1 = \frac{\pi}{8}$. For the depth, write the acceleration as $v\,\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = -2(1+v^2) \qquad \Longrightarrow \qquad \int_1^0 \frac{v\,\mathrm{d}v}{1+v^2} = -2\int_0^D \mathrm{d}x,$$

so $-\frac{1}{2}\ln 2 = -2D$, and the greatest depth is

$$D = \tfrac{1}{4}\ln 2 \text{ metres},$$

as required.

The rising phase

At depth D the body is momentarily at rest, but the net upward force of $2m$ N is still there, so it starts to rise. Now the resistance acts downwards. Take upwards as positive, with y the height risen from the lowest point and u the upward speed:

$$m\frac{\mathrm{d}u}{\mathrm{d}t} = 12m - 10m - 2mu^2 \qquad \Longrightarrow \qquad \frac{\mathrm{d}u}{\mathrm{d}t} = 2(1-u^2).$$

Ask yourself Why can't we keep the sinking equation and just let v go negative?

The term $-2mv^2$ always points upwards, whatever the sign of v. Once the body is rising, the resistance must point downwards, so the equation changes. (A single equation would need $-2mv|v|$.) Notice too that $u < 1$ throughout the ascent: u starts at 0 and $\frac{\mathrm{d}u}{\mathrm{d}t} > 0$ only while $u < 1$, so u can only approach the terminal speed 1 ms^{-1}. This keeps $1 - u^2$ positive below.

Speed at the surface. Using $u\,\frac{\mathrm{d}u}{\mathrm{d}y} = 2(1-u^2)$ and integrating from $y = 0$, $u = 0$ to $y = D$, $u = U$:

$$\int_0^U \frac{u\,\mathrm{d}u}{1-u^2} = 2\int_0^D \mathrm{d}y \qquad \Longrightarrow \qquad -\tfrac{1}{2}\ln(1-U^2) = 2D = \tfrac{1}{2}\ln 2.$$

So $1 - U^2 = \frac{1}{2}$, and the body returns to the surface with speed $U = \dfrac{1}{\sqrt{2}}\ \mathrm{ms}^{-1}$.

Time to rise. Using $\frac{\mathrm{d}u}{\mathrm{d}t} = 2(1-u^2)$ with partial fractions $\dfrac{1}{1-u^2} = \dfrac{1}{2}\left(\dfrac{1}{1+u} + \dfrac{1}{1-u}\right)$:

$$T_2 = \frac{1}{2}\int_0^U \frac{\mathrm{d}u}{1-u^2} = \frac{1}{4}\left[\ln\frac{1+u}{1-u}\right]_0^U = \frac{1}{4}\ln\frac{1+\frac{1}{\sqrt{2}}}{1-\frac{1}{\sqrt{2}}} = \frac{1}{4}\ln\frac{\sqrt{2}+1}{\sqrt{2}-1},$$

multiplying top and bottom by $\sqrt{2}$ in the last step.

Total time. The body reaches the surface again at

$$T_1 + T_2 = \frac{\pi}{8} + \frac{1}{4}\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right) \text{ seconds}$$

after entering the pool, as required.

Checks

- Since $\dfrac{\sqrt{2}+1}{\sqrt{2}-1} = (\sqrt{2}+1)^2$, the rising time is $T_2 = \frac{1}{2}\ln(1+\sqrt{2}) \approx 0.4407\,\mathrm{s}$, and the total is $\approx 0.3927 + 0.4407 = 0.8334\,\mathrm{s}$.
- Stepping the motion numerically (Runge–Kutta, with the resistance written as $-2v|v|$) gives a stopping time $0.39270\,\mathrm{s}$, a depth $0.17329\,\mathrm{m} = \frac{1}{4}\ln 2$, a return speed $0.70711\ \mathrm{ms}^{-1}$ and a return time $0.83339\,\mathrm{s}$, all matching.
- The return speed is less than the entry speed, as it must be: the resistance does negative work on both legs.

Common slips

- Keeping the resistance upwards on the way back up, which would give a constant net upward force of $2m + 2mu^2$ and a much faster rise.
- Forgetting that the buoyancy outweighs gravity, so the net force is $-2m$ (downward positive), not $+2m$.
- Using $\frac{\mathrm{d}v}{\mathrm{d}t}$ for the depth or $v\frac{\mathrm{d}v}{\mathrm{d}x}$ for the times, and then getting stuck on the integral.

Beyond STEP

Quadratic drag makes the equation of motion nonlinear, yet it is still separable, which is why arctan appears going down and $\tanh^{-1}$ going up (indeed $T_2 = \frac{1}{2}\tanh^{-1} U$). The ascent approaches a terminal speed of $1\ \mathrm{ms}^{-1}$, where drag balances the net upward force. A first-year course treats such models systematically, including the phase-line picture of an equation like $\dot{u} = 2(1-u^2)$. Kleppner and Kolenkow, *An Introduction*

to Mechanics, covers velocity-dependent forces, and Boyce and DiPrima, *Elementary Differential Equations*, covers separable and autonomous equations.

WORKED SOLUTION

STEP II 1993, Question 12

Mechanics • Friction and Rigid Bodies

STEP II 1993, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q12

The key idea

The sphere is wedged between the wall and the plane, so it can't move off without losing contact with one of them. If it slips at both contacts at once, its centre stays put and it simply spins about its centre. Working out which way it would spin tells us which way each friction force acts. After that it's three equations (two resolutions and one moment equation) in the unknowns N_1, N_2 and m. For the second part, the right moment point makes the answer fall out in one line.

Solution

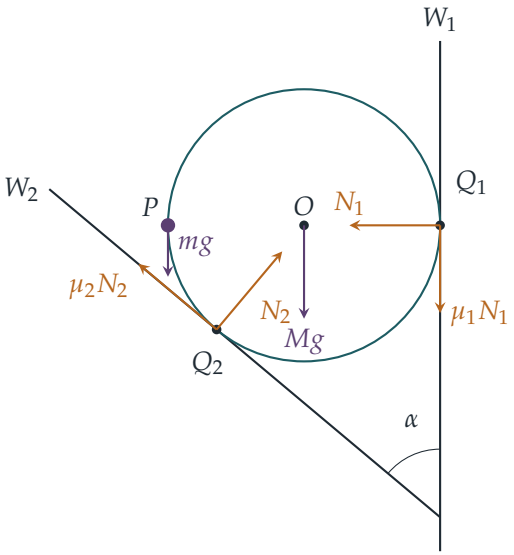
The model and the directions of friction

We model the sphere as a rigid uniform sphere and the particle as a point mass fixed at P. The contacts at Q_1 and Q_2 are rough. Let the normal reactions be N_1 (horizontal, away from the wall) and N_2 (perpendicular to the plane, through the centre O).

Since PQ_1 is a horizontal diameter, the particle hangs at horizontal distance r from O, on the side away from the wall. Its weight therefore has a moment mgr about O, turning the sphere so that P goes down, and the forces at the contacts must balance that.

? **Ask yourself** Which way does friction act at each contact?

If the sphere slips at both points it stays in contact with both, so O is fixed and the sphere rotates about O in the sense in which P falls. Then Q_1, the point opposite P, moves *up* the wall, and Q_2 moves *down* the plane towards the corner. Friction opposes the slipping, so it acts *down* the wall at Q_1 and *up* the plane at Q_2. Since slipping is about to happen at both points, friction is limiting at both: $F_1 = \mu_1 N_1$ and $F_2 = \mu_2 N_2$. Both friction forces have moments about O that oppose the particle's moment, as they should.



Slipping at both contacts

The plane makes angle α with the vertical, so its normal makes angle α with the horizontal. Then N_2 has components $N_2\cos\alpha$ towards the wall and $N_2\sin\alpha$ upwards, and the friction $\mu_2 N_2$ up the plane has components $\mu_2 N_2\sin\alpha$ away from the wall and $\mu_2 N_2\cos\alpha$ upwards.

Resolving horizontally:

$$N_1 = N_2\cos\alpha - \mu_2 N_2\sin\alpha = N_2(\cos\alpha - \mu_2\sin\alpha). \tag{1}$$

Resolving vertically:

$$N_2\sin\alpha + \mu_2 N_2\cos\alpha - \mu_1 N_1 = (M+m)g. \tag{2}$$

Taking moments about O: both normal reactions pass through O, and so does Mg. The two friction forces act along tangents, at distance r from O. So

$$mgr = \mu_1 N_1 r + \mu_2 N_2 r \quad\Longrightarrow\quad mg = \mu_1 N_1 + \mu_2 N_2. \tag{3}$$

Substituting (1) into (3) and into (2):

$$\begin{aligned} mg &= N_2\big(\mu_2 + \mu_1\cos\alpha - \mu_1\mu_2\sin\alpha\big),\\ (M+m)g &= N_2\big(\sin\alpha + \mu_2\cos\alpha - \mu_1\cos\alpha + \mu_1\mu_2\sin\alpha\big).\end{aligned}$$

Subtracting the first from the second gives

$$\begin{aligned} Mg &= N_2\big(\sin\alpha + \mu_2\cos\alpha - 2\mu_1\cos\alpha + 2\mu_1\mu_2\sin\alpha - \mu_2\big)\\ &= N_2\big((2\mu_1\mu_2+1)\sin\alpha + (\mu_2 - 2\mu_1)\cos\alpha - \mu_2\big).\end{aligned}$$

Dividing the equation for mg by this one cancels N_2 and g:

$$m = \frac{\mu_2 + \mu_1\cos\alpha - \mu_1\mu_2\sin\alpha}{(2\mu_1\mu_2+1)\sin\alpha + (\mu_2 - 2\mu_1)\cos\alpha - \mu_2}\,M,$$

as required.

For this to make physical sense we need $N_1 > 0$, which by (1) means $\mu_2 < \cot\alpha$. If μ_2 is larger than that, friction at Q_2 would pull the sphere off the wall, and we get the situation in the final part.

Rolling about Q_2

Now the sphere doesn't slip at Q_2. Instead it is about to turn about Q_2 (rolling up the plane), so its centre moves away from the wall and contact at Q_1 is about to be lost. At the point of rolling, $N_1 = 0$, and so the friction at Q_1 is zero too.

? **Ask yourself** What do we do with the friction at Q_2, which is now unknown?

We take moments about Q_2, so that both the reaction and the friction at Q_2 drop out. That leaves only the two weights. Measure horizontal distances from Q_2. Since OQ_2 is perpendicular to the plane, it makes angle α with the horizontal, so O is a horizontal distance $r\cos\alpha$ from Q_2, on the wall side. The particle is a further r beyond O, away from the wall, so it is a horizontal distance $r - r\cos\alpha$ from Q_2, on the other side. For the moments to balance,

$$Mg\,r\cos\alpha = mg\,r(1-\cos\alpha) \quad\Longrightarrow\quad m = \frac{M\cos\alpha}{1-\cos\alpha} = \frac{M}{\sec\alpha - 1},$$

dividing top and bottom by $\cos\alpha$. That is the result.

Checks

- The two answers should agree on the boundary between the two regimes, $N_1 = 0$, that is $\mu_2 = \cot\alpha$. Putting $\mu_2 = \cos\alpha/\sin\alpha$ into the first formula, the numerator becomes $\cot\alpha$ and the denominator becomes $(1-\cos\alpha)/\sin\alpha$, whatever μ_1 is. So $m = M\cos\alpha/(1-\cos\alpha)$, which matches the second answer.
- With $\alpha = 60^\circ$ the second answer gives $m = M$. With $\alpha = 60^\circ$, $\mu_1 = 0.2$, $\mu_2 = 0.3$ and $M = 1$, the first gives $m \approx 0.561$. We checked the force and moment balance for these values, and for 2000 random cases, with a computer.
- If both contacts were smooth ($\mu_1 = \mu_2 = 0$), the first formula gives $m = 0$: nothing can resist the particle's moment, as we'd expect.

Common slips

- Getting a friction direction wrong. Decide how the sphere would turn, then which way each contact point would slide.
- Forgetting that both reactions pass through the centre, so they don't appear in the moment equation about O.

- In the final part, keeping the wall's friction. Once the sphere is about to leave the wall, $N_1 = 0$, so the friction there is zero.

Beyond STEP

Problems like this, with several contacts that may slip or roll, are the start of the mechanics of rigid bodies with constraints. Each contact gives an inequality ($|F| \leq \mu N$, $N \geq 0$), and you have to work out which of them hold with equality. The same case analysis turns up in robotics, where it decides whether a robot hand can grip something. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers statics and the rotation of rigid bodies carefully.

WORKED SOLUTION

STEP II 1993, Question 13

Mechanics • Simple Harmonic Motion

STEP II 1993, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q13

The key idea

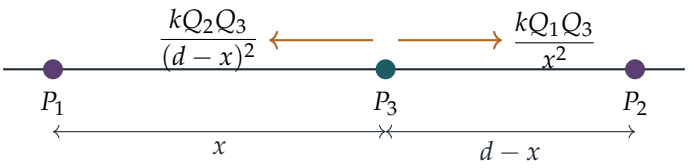
Put a coordinate on the moving bead and write down the net force on it: two inverse-square repulsions pulling in opposite directions. Equilibrium is where they balance. For the oscillation, displace the bead by a small y, use the given approximation on *both* forces, and the constant terms cancel by the equilibrium condition, leaving a restoring force proportional to y. That is exactly the equation of simple harmonic motion.

(The question says P_3 moves "between P_1 and P_3"; this is a slip in the paper for "between P_1 and P_2".)

Solution

Model. The beads are particles; the wire is smooth and horizontal, so the weight of P_3 and the normal reaction are both perpendicular to the wire and play no part in the motion along it. The only forces along the wire are the two electrical repulsions on P_3 (the fixed beads are held in place, so we need not worry about forces on them).

Measure x from P_1 towards P_2, so P_3 is at distance x from P_1 and $d - x$ from P_2, with $0 < x < d$.



P_1 pushes P_3 towards P_2 (positive direction) and P_2 pushes it back, so Newton's second law along the wire gives

$$m\ddot{x} = \frac{kQ_1Q_3}{x^2} - \frac{kQ_2Q_3}{(d-x)^2}. \qquad (*)$$

Equilibrium position

At equilibrium $x = a$ and the right-hand side of $(*)$ vanishes. Since k and Q_3 are positive we can divide by kQ_3:

$$\frac{Q_1}{a^2} = \frac{Q_2}{(d-a)^2} \quad \Longrightarrow \quad \left(\frac{d-a}{a}\right)^2 = \frac{Q_2}{Q_1}.$$

Both a and $d-a$ are positive (the bead is between the fixed beads), so we take the positive square root:

$$\frac{d-a}{a} = \frac{\sqrt{Q_2}}{\sqrt{Q_1}} \quad \Longrightarrow \quad \sqrt{Q_1}\,(d-a) = \sqrt{Q_2}\,a \quad \Longrightarrow \quad a = \frac{d\sqrt{Q_1}}{\sqrt{Q_1}+\sqrt{Q_2}},$$

as required. It will help to write $S = \sqrt{Q_1} + \sqrt{Q_2}$ and $b = d - a$, so that

$$a = \frac{d\sqrt{Q_1}}{S}, \qquad b = \frac{d\sqrt{Q_2}}{S}.$$

Notice the bead sits nearer the *weaker* charge, which makes sense.

Small oscillations

Let $x = a + y$ with y small, so $d - x = b - y$ and $\ddot{x} = \ddot{y}$. Using the given approximation (and the same approximation with y replaced by $-y$ and a by b),

$$\frac{1}{(a+y)^2} \approx \frac{1}{a^2} - \frac{2y}{a^3}, \qquad \frac{1}{(b-y)^2} \approx \frac{1}{b^2} + \frac{2y}{b^3}.$$

Substituting into $(*)$:

$$\begin{aligned} m\ddot{y} &\approx kQ_3\left[Q_1\left(\frac{1}{a^2} - \frac{2y}{a^3}\right) - Q_2\left(\frac{1}{b^2} + \frac{2y}{b^3}\right)\right] \\ &= kQ_3\left(\frac{Q_1}{a^2} - \frac{Q_2}{b^2}\right) - 2kQ_3\left(\frac{Q_1}{a^3} + \frac{Q_2}{b^3}\right)y. \end{aligned}$$

Ask yourself What happens to the terms that don't involve y?

They are exactly the equilibrium condition, so $\frac{Q_1}{a^2} - \frac{Q_2}{b^2} = 0$ and they cancel. This always happens when you expand about an equilibrium: the zeroth-order force is zero by definition.

Now simplify the coefficient of y using the expressions for a and b:

$$\frac{Q_1}{a^3} = \frac{Q_1 S^3}{d^3 Q_1^{3/2}} = \frac{S^3}{d^3\sqrt{Q_1}}, \qquad \frac{Q_2}{b^3} = \frac{S^3}{d^3\sqrt{Q_2}},$$

so

$$\frac{Q_1}{a^3} + \frac{Q_2}{b^3} = \frac{S^3}{d^3}\cdot\frac{\sqrt{Q_2}+\sqrt{Q_1}}{\sqrt{Q_1Q_2}} = \frac{S^4}{d^3\sqrt{Q_1Q_2}}.$$

Hence

$$\ddot{y} \approx -\omega^2 y, \qquad \omega^2 = \frac{2kQ_3S^4}{md^3\sqrt{Q_1Q_2}} > 0.$$

This is the simple harmonic motion equation, so for small displacements P_3 performs approximate SHM about $x = a$. (Released from rest at $y = y_0$, the motion is $y \approx y_0 \cos \omega t$.) The period is

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{md^3\sqrt{Q_1Q_2}}{2kQ_3S^4}} = \frac{2\pi d}{S^2}\sqrt{\frac{md\sqrt{Q_1Q_2}}{2kQ_3}}$$
$$= \frac{\pi d}{S^2}\sqrt{\frac{4md\sqrt{Q_1Q_2}}{2kQ_3}} = \frac{\pi d}{(\sqrt{Q_1}+\sqrt{Q_2})^2}\sqrt{\frac{2md\sqrt{Q_1Q_2}}{kQ_3}},$$

which is the required result.

Checks

- **Symmetry:** with $Q_1 = Q_2$ we get $a = d/2$, the midpoint, as it must be. The period formula is symmetric in Q_1 and Q_2, which it should be too (swapping the charges just reflects the picture).
- **Dimensions:** kQQ'/r^2 is a force, so $kQ_3\sqrt{Q_1Q_2}$ has dimensions of force $\times$ length^2. Then $md\sqrt{Q_1Q_2}/(kQ_3) \cdot d^2/(\sqrt{Q_1}+\sqrt{Q_2})^4$ has dimensions mass $\times$ length/force $=$ time^2. Good.
- **Numbers:** with $Q_1 = 4$, $Q_2 = 1$, $d = 3$ and $k = Q_3 = m = 1$, we get $a = 2$ (check: $4/2^2 = 1/1^2$) and $T = 2\pi/\sqrt{3} \approx 3.628$. Integrating $(*)$ numerically from a tiny displacement, for many random choices of the constants, reproduces the period formula to better than one part in 10^4.

Common slips

- Expanding only one of the two forces: both depend on y, and both contribute to the restoring force (with the same sign).
- Getting the sign of the y term in $1/(b-y)^2$ wrong: replacing y by $-y$ in the given approximation gives $+2y/b^3$.
- Taking the negative square root in the equilibrium condition, which would put the bead outside the segment P_1P_2.

Beyond STEP

What we did is *linearisation*: near a stable equilibrium of a potential $V(x)$, the motion is approximately SHM with $\omega^2 = V''(a)/m$. Here $V = kQ_3\big(Q_1/x + Q_2/(d-x)\big)$, and you can check that $V''(a)$ gives the same ω^2. This idea, extended to several coordinates,

leads to normal modes of vibration, a central topic in first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats small oscillations about equilibrium clearly.

WORKED SOLUTION

STEP II 1993, Question 14

Mechanics • Bouncing and Restitution

STEP II 1993, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q14

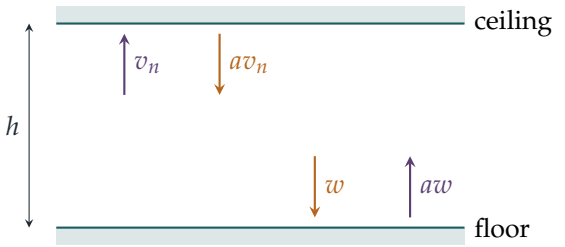
The key idea

Between impacts the ball moves freely under gravity, so $v^2 = u^2 \pm 2gh$ links its speeds at the floor and the ceiling, and each impact multiplies the speed by a. Following one full cycle (ceiling, floor, back to the ceiling) gives a linear recurrence for the kinetic energy at the ceiling, which we solve by shifting to its fixed point. The ball reaches the ceiling for the nth time exactly when that energy is positive, and taking logarithms turns this into the bound.

Solution

We model the ball as a particle moving vertically, with no air resistance, and take $0 < a < 1$ (the bound has $\ln a$ in the denominator, and if $a = 1$ the ball would hit the ceiling forever). Write v_n for the speed just before the nth ceiling impact, and

$$E_n = \frac{v_n^2}{2gh}, \qquad \text{so that} \qquad \tfrac{1}{2}mv_n^2 = mgh\,E_n.$$



The first ceiling impact

The ball leaves the floor with $u^2 = 2kgh$ and rises a height h against gravity, so

$$v_1^2 = 2kgh - 2gh = 2gh(k-1), \qquad E_1 = k - 1 > 0,$$

which is positive because $k > 1$; so the ball certainly reaches the ceiling once.

One full cycle

Just after the nth ceiling impact the ball moves down with speed av_n (Newton's experimental law at the fixed ceiling). It falls a height h, so it reaches the floor with speed w, where

$$w^2 = a^2 v_n^2 + 2gh.$$

It rebounds upwards with speed aw and rises a height h. If it gets back to the ceiling, its speed there satisfies

$$v_{n+1}^2 = a^2 w^2 - 2gh = a^4 v_n^2 + 2gh\,(a^2 - 1).$$

Dividing by $2gh$,

$$E_{n+1} = a^4 E_n + (a^2 - 1).$$

This is valid whenever the right-hand side is positive; if it is zero or negative, the ball never gets back to the ceiling.

? **Ask yourself** How do we solve a recurrence of the form $E_{n+1} = a^4 E_n + c$?

Look for the constant solution E with $E = a^4 E + (a^2 - 1)$:

$$E = \frac{a^2 - 1}{1 - a^4} = \frac{a^2 - 1}{(1 - a^2)(1 + a^2)} = -\frac{1}{1 + a^2}.$$

Subtracting this fixed point removes the constant term: putting $F_n = E_n + \dfrac{1}{1 + a^2}$ gives $F_{n+1} = a^4 F_n$. Hence $F_n = a^{4(n-1)} F_1$, and since $F_1 = (k - 1) + \dfrac{1}{1 + a^2}$,

$$E_n = a^{4n-4}\left(k - 1 + \frac{1}{a^2 + 1}\right) - \frac{1}{a^2 + 1} = a^{4n-4}(k - 1) + \frac{a^{4n-4} - 1}{a^2 + 1}.$$

Multiplying by mgh, the kinetic energy just before the nth ceiling impact is

$$mgh\left(a^{4n-4}(k - 1) + \frac{a^{4n-4} - 1}{a^2 + 1}\right),$$

as required.

The number of ceiling impacts

The ball hits the ceiling for the nth time only if it has a positive speed on arriving there, that is, only if $E_n > 0$. (If it hits the ceiling an nth time it has hit it on every earlier occasion too, so the formula above applies.) Multiplying $E_n > 0$ by $a^2 + 1 > 0$,

$$a^{4n-4}\big((k - 1)(a^2 + 1) + 1\big) > 1 \quad \Longleftrightarrow \quad a^{4n-4}\big(a^2(k - 1) + k\big) > 1.$$

Both sides are positive, so we may take logarithms:

$$(4n - 4)\ln a + \ln\big[a^2(k - 1) + k\big] > 0.$$

Ask yourself Which way does the inequality go when we divide by $4\ln a$?

Since $0 < a < 1$, we have $\ln a < 0$, so dividing by $4\ln a$ reverses the inequality:

$$n - 1 < -\frac{\ln\left[a^2(k-1)+k\right]}{4\ln a}, \qquad \text{that is,} \qquad n < 1 - \frac{\ln\left[a^2(k-1)+k\right]}{4\ln a}.$$

So if the ball hits the ceiling N times, then the inequality holds with $n = N$ and

$$N < 1 - \frac{\ln[a^2(k-1)+k]}{4\ln a}.$$

In particular the number of ceiling impacts is at most this value, which is what we had to show. (In fact E_n decreases with n, so N is exactly the number of positive integers strictly below the bound.)

Checks

- $n = 1$: the formula gives $mgh(k-1)$, matching $\frac{1}{2}m(2kgh) - mgh$ from energy conservation.
- Large n: $E_n \to -\frac{1}{1+a^2} < 0$, so the ball does eventually stop reaching the ceiling, as it must when energy is lost at every impact.
- Numerically, with $k = 6$, $a = 0.9$, $h = 2\,\text{m}$, $m = 1\,\text{kg}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$, a step-by-step simulation gives ceiling kinetic energies 98, 60.57, 36.02 J, agreeing with the formula. The ball hits the ceiling 6 times, and the bound is about 6.48.

Common slips

- Forgetting that the speed is multiplied by a at *both* the ceiling and the floor, which gives a^2 instead of a^4 per cycle.
- Dividing by $\ln a$ without reversing the inequality.
- Applying restitution to the kinetic energy rather than to the speed: the energy is multiplied by a^2 at each impact, not by a.

Beyond STEP

The recurrence $E_{n+1} = a^4 E_n + c$ is a first-order linear difference equation. The trick of subtracting the fixed point is the discrete version of finding a particular integral for a linear ODE, and it reappears in the study of iterated maps and their stability. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and energy loss more generally.

WORKED SOLUTION

STEP II 1993, Question 15

Probability and Statistics • Central Limit Theorem

STEP II 1993, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q15

This question has a full guided treatment in *Beyond STEP*, Section 13.6, *Errors That Add Up*: https://beyondstep.keithmcnulty.org/ch13/errors-add-up.html

The key idea

The total error is a sum of many small, independent errors, one for each number. So work out the mean and variance of the error made on a *single* number, add them up over all the numbers, and use the Central Limit Theorem to treat the total as approximately normal. The whole contrast between the two computers comes from the means: one machine's errors cancel on average, while the other's all push the same way.

Solution

The error on one number

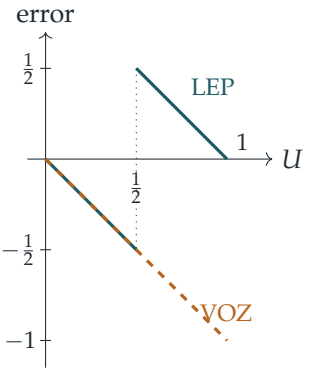
Let a number be $a = m + U$, where $m = \lfloor a \rfloor$ is an integer and U is its fractional part, with $U \sim \mathrm{U}[0,1)$. We define the error as (approximated value) $-$ (true value); the sign convention doesn't affect the magnitude.

LEP (rounding). If $U < \frac{1}{2}$ the number is replaced by m, an error of $-U$; if $U \geq \frac{1}{2}$ it is replaced by $m+1$, an error of $1-U$. So the error is

$$X = \begin{cases} -U, & 0 \leq U < \frac{1}{2}, \\ 1-U, & \frac{1}{2} \leq U < 1, \end{cases}$$

which takes each value in $(-\frac{1}{2}, \frac{1}{2}]$ with constant density: X is uniform on $[-\frac{1}{2}, \frac{1}{2}]$ (the sawtooth in the figure).

VOZ (truncation). The number is replaced by m, so the error is $Y = -U$, uniform on $(-1, 0]$.



A uniform distribution on an interval of length 1 has variance $\frac{1}{12}$, wherever the interval sits. So

$$\mathrm{E}(X) = 0, \quad \mathrm{Var}(X) = \tfrac{1}{12}; \qquad \mathrm{E}(Y) = -\tfrac{1}{2}, \quad \mathrm{Var}(Y) = \tfrac{1}{12}.$$

(For the record, $\mathrm{Var}(U) = \int_0^1 u^2\,\mathrm{d}u - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.)

Adding 1500 numbers

The error in the answer is the sum of the errors on the individual numbers, since the computer adds the approximated values exactly. The fractional parts are independent, so the individual errors are independent and their variances add. For $n = 1500$ numbers,

$$\mathrm{Var}(\text{total error}) = \frac{1500}{12} = 125, \qquad \text{standard deviation } \sqrt{125} = 5\sqrt{5} \approx 11.18.$$

By the Central Limit Theorem, a sum of 1500 independent, identically distributed errors is very nearly normal.

LEP. The total error E_L is approximately $\mathrm{N}(0, 125)$, so

$$\begin{aligned}\mathrm{P}(|E_L| > 15) &\approx \mathrm{P}\Big(|Z| > \frac{15}{\sqrt{125}}\Big) = \mathrm{P}(|Z| > 1.342)\\ &= 2\big(1 - \Phi(1.342)\big) \approx 2(1 - 0.9102) \approx 0.18.\end{aligned}$$

No continuity correction is needed: the errors are continuous random variables, not integers.

VOZ. The total error E_V is approximately $\mathrm{N}(-750, 125)$. The error is only at most 15 in size if $E_V \geq -15$, which is

$$\frac{-15 - (-750)}{\sqrt{125}} = \frac{735}{11.18} \approx 66$$

standard deviations above the mean. That probability is utterly negligible, so

$$\mathrm{P}(|E_V| > 15) \approx 1.$$

Truncation always errs downwards, by $\frac{1}{2}$ on average, so after 1500 numbers VOZ is out by about 750.

Ask yourself Both machines make errors of the same variance. Why are the answers so different?

The spread of the total error is the same, ± 11 or so, but its *centre* is not. LEP's errors are centred at 0, so the total wanders around 0. VOZ's errors all have mean $-\frac{1}{2}$, so the bias grows in proportion to n, while the random spread grows only like $\sqrt{n}$. The bias wins easily.

How many additions for LEP?

Suppose LEP adds n numbers. Its total error is approximately $\mathrm{N}(0, \frac{n}{12})$, and we need

$$\mathrm{P}(|E_L| < 10) = 2\Phi\Big(\frac{10}{\sqrt{n/12}}\Big) - 1 \;\geq\; 0.9 \quad\Longleftrightarrow\quad \Phi\Big(\frac{10}{\sqrt{n/12}}\Big) \geq 0.95.$$

Since Φ is increasing and $\Phi(1.6449) = 0.95$, this is

$$\frac{10}{\sqrt{n/12}} \geq 1.6449 \iff n \leq 12\Big(\frac{10}{1.6449}\Big)^2 \approx 443.5.$$

The probability decreases as n increases, so it stays at least 0.9 for $n \leq 443$ and first drops below 0.9 at $n = 444$. So LEP can add **443** numbers. (If you count an "addition" as one use of the $+$ sign, adding 443 numbers takes 442 of them; either reading is fine if you say which you mean.)

Checks

- *Is* 1.342 *believable?* A total error of 15 is a little over one standard deviation (11.18), and about 18% of a normal distribution lies more than 1.34 standard deviations from the mean. That's about right.
- *Numerically.* A simulation of 10^5 batches of 1500 rounded numbers gave $\mathrm{P}(|E_L| > 15) \approx 0.179$. The exact distribution of the sum of uniforms gives 0.1797, against 0.1797 from the normal approximation. For the second part the exact probability is 0.90019 at $n = 443$ and 0.89981 at $n = 444$, confirming the cut-off.

Common slips

- Taking LEP's error to be uniform on $[0, \frac{1}{2}]$: rounding can go up or down, so the error is uniform on $[-\frac{1}{2}, \frac{1}{2}]$.
- Forgetting VOZ's mean of $-\frac{1}{2}$ per number and so giving it the same answer as LEP.
- Using $\Phi(z) = 0.9$ instead of 0.95 in the last part: the event $|E| < 10$ is two-sided, so each tail gets 0.05.

Beyond STEP

This is the classic analysis of rounding error in floating-point arithmetic: unbiased rounding makes the accumulated error grow like $\sqrt{n}$, while a biased rule makes it grow like n. That is why the IEEE 754 standard uses "round half to even" rather than always rounding halves upwards. The sum of n uniforms has an exact distribution (the Irwin–Hall distribution), and seeing how quickly it becomes normal is a good way into the Central Limit Theorem; Grimmett and Welsh, *Probability: An Introduction*, proves the theorem and discusses it.

WORKED SOLUTION

STEP II 1993, Question 16

Probability and Statistics • Poisson Processes

STEP II 1993, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q16

The key idea

There are two independent sources of randomness here: the buses, which give a geometric waiting time, and the passengers, who form a Poisson process. Condition on when the first bus comes. Once you know that, the number waiting is just a Poisson count over a known length of time. Part (iii) then uses the law of total probability, and the real question is how to evaluate the resulting infinite sum to two significant figures with only tables and a calculator.

Solution

Measure time in minutes after 8:00. Let T be the time at which the first bus arrives, and let X_t be the number of passengers who arrive in $(0, t]$. The passenger process is Poisson with rate 4 per minute, so $X_t \sim \text{Po}(4t)$. Buses can come only at $t = 1, 2, 3, \ldots$ ("just after eight" means the bus that may come at 8:00 has already been and gone), and they are independent of the passengers.

Part (i)

The first bus arrives at n minutes past 8 exactly when no bus comes at minutes $1, 2, \ldots, n-1$ and a bus does come at minute n. These events are independent, so

$$\text{P}(T = n) = \left(\tfrac{3}{4}\right)^{n-1} \tfrac{1}{4}, \qquad n = 1, 2, 3, \ldots$$

So T is geometric with parameter $\frac{1}{4}$. (As a check, these probabilities sum to $\frac{1}{4} \big/ \left(1 - \frac{3}{4}\right) = 1$.)

Part (ii)

The people waiting at 8:05 are exactly those who arrived in $(0, 5]$. Nobody has left, because no bus has come before 8:05. So we need the distribution of X_5 given $T = 5$.

Ask yourself Does knowing that the first bus came at 8:05 change the distribution of the number of passengers?

No. The event $T = 5$ depends only on the buses, and the buses are independent of the passengers. So conditioning on $T = 5$ leaves the distribution of X_5 unchanged, and $X_5 \sim \text{Po}(4 \times 5) = \text{Po}(20)$. Hence

$$\text{P}(m \text{ people waiting} \mid T = 5) = \frac{\text{e}^{-20}\, 20^m}{m!}, \qquad m = 0, 1, 2, \ldots$$

Part (iii)

The bus fails to take everyone exactly when more than 25 people are waiting, that is, when $X_T \geq 26$. Let F be this event. Conditioning on T and using independence as in (ii),

$$\text{P}(F) = \sum_{n=1}^{\infty} \text{P}(T = n)\, \text{P}(X_n \geq 26) = \sum_{n=1}^{\infty} \left(\tfrac{3}{4}\right)^{n-1} \tfrac{1}{4}\, \text{P}\big(\text{Po}(4n) \geq 26\big).$$

We now need a plan that turns this infinite sum into a finite amount of work, with an error well below 0.005. We want two significant figures, and the answer will turn out to be of order 0.1.

Step 1: work with the complement, so the tail is easy. Write $w_n = \left(\tfrac{3}{4}\right)^{n-1} \tfrac{1}{4}$ and $q_n = \text{P}(\text{Po}(4n) \leq 25)$. Since $\sum w_n = 1$,

$$\text{P}(F) = 1 - \sum_{n=1}^{\infty} w_n\, q_n.$$

This form is better because q_n is tiny when n is large. For $n \geq 11$ the mean $4n \geq 44$ is far above 25. Since q_n decreases as n increases (more time means more passengers),

$$\sum_{n \geq 11} w_n q_n \leq q_{11} \sum_{n \geq 11} w_n = q_{11} \left(\tfrac{3}{4}\right)^{10}.$$

A normal approximation gives $q_{11} \approx \Phi\big((25.5 - 44)/\sqrt{44}\big) = \Phi(-2.79) \approx 0.003$, so this tail is below 0.0002 and can be dropped. That leaves ten terms.

Step 2: the first few terms. For $n = 1$ and $n = 2$ (means 4 and 8), getting 26 or more passengers is so unlikely that $q_1 = q_2 = 1$ to well beyond the accuracy needed, so these contribute $w_1 + w_2 = \tfrac{1}{4} + \tfrac{3}{16}$.

Step 3: the middle terms from tables. For $n = 3, 4, 5$ (means 12, 16, 20), read $q_n = \text{P}(\text{Po}(4n) \leq 25)$ directly from cumulative Poisson tables, which normally cover means up to 20. If the tables also list means 24 and 28, use them for $n = 6, 7$ as well.

Step 4: larger means by the normal approximation. For the remaining n up to 10 (means up to 40), use $\mathrm{Po}(\mu) \approx \mathrm{N}(\mu, \mu)$ with a continuity correction:

$$q_n \approx \Phi\left(\frac{25.5 - 4n}{\sqrt{4n}}\right),$$

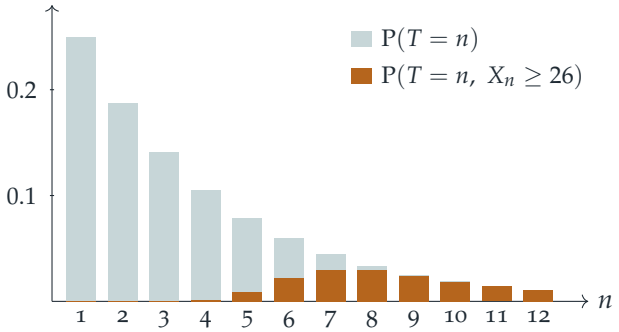
reading Φ from normal tables. These means are large enough for the approximation to be good. More to the point, the weights w_n are already small ($w_8 \approx 0.033$, $w_{10} \approx 0.019$), so an error of a few thousandths in q_n changes $\mathrm{P}(F)$ by less than 0.001.

Step 5: assemble. Compute $w_1, \ldots, w_{10}$ on the calculator and form

$$\mathrm{P}(F) \approx 1 - \sum_{n=1}^{10} w_n q_n,$$

then round to two significant figures. Each neglected or approximated piece is at most about 0.001, so the rounding is safe unless the total lands very close to a rounding boundary. If it does, the method shows exactly which terms to refine.

The question doesn't ask us to carry this out, but doing so gives $\mathrm{P}(F) \approx 0.19$. The chart below shows where the probability comes from: most of it comes from first buses at 8:06 to 8:10, when the expected queue of 24 to 40 is close to or above capacity.



Checks

- In (i), the probabilities sum to 1, and $\mathrm{P}(T = 1) = \frac{1}{4}$ as it should.
- In (ii), the mean number waiting is 20, which is four per minute for five minutes.
- In (iii), the exact sum (computed by machine) is 0.1904. A Monte Carlo simulation of 2×10^5 mornings gives 0.189. The table-and-normal method above gives 0.1912. All three round to 0.19.

Common slips

- Getting the boundary wrong by one. The bus fails only if 26 or more people are waiting, so the probability we need is $1 - \mathrm{P}(\mathrm{Po}(4n) \leq 25)$, not $1 - \mathrm{P}(\mathrm{Po}(4n) \leq 24)$.

- Treating the queue length as $\mathrm{Po}(4\,\mathrm{E}[T]) = \mathrm{Po}(16)$. The waiting time is random, so you must average the Poisson probabilities over T, not plug in the average time.
- Leaving the sum infinite without saying why it can be cut off. You must show that the neglected tail is too small to affect the second significant figure.

Beyond STEP

The number waiting, X_T, is a Poisson count with a random mean: a *mixture* of Poisson distributions. Because T is geometric, X_T turns out to have a neat closed-form generating function, and conditioning arguments like this one are the basis of queueing theory and of the compound and mixed distributions met in a first probability course. The question of how to cut off a sum and bound the error is the start of numerical analysis. Grimmett and Welsh, *Probability: An Introduction*, covers Poisson processes and generating functions at the right level.

WORKED SOLUTION

STEP III 1993, Question 1

Pure • Arc Length and Volume

STEP III 1993, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q1

The key idea

The double-angle formula $\cos 2\theta = 1 - 2\sin^2\theta$ turns the parametric curve straight into a Cartesian one, and the range of θ tells us exactly which piece of the parabola we get. For the length, $\sqrt{1+16x^2}$ is the classic signal for a hyperbolic substitution, because $1 + \sinh^2 u = \cosh^2 u$. The volume is an ordinary volume of revolution once we notice that the ends of P sit on the axis of rotation.

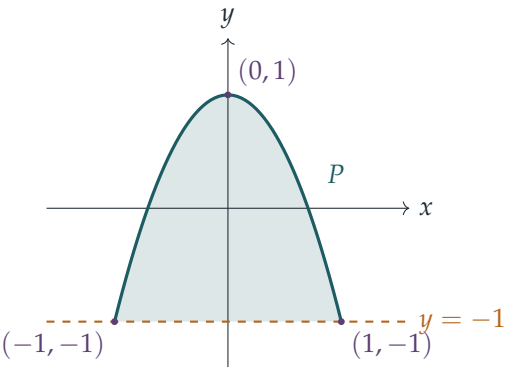
Solution

The curve and its sketch

Since $\cos 2\theta = 1 - 2\sin^2\theta$ and $x = \sin\theta$,

$$y = 1 - 2x^2.$$

So every point of P lies on the parabola $y = 1 - 2x^2$. As θ runs from $-\pi/2$ to $\pi/2$, $x = \sin\theta$ increases steadily from -1 to 1, taking each value exactly once. So P is precisely the arc of the parabola with $-1 \le x \le 1$. It runs from $(-1,-1)$ up to its vertex $(0,1)$ and back down to $(1,-1)$, and it is symmetric about the y-axis.



The length of P

Working in Cartesian form, $\frac{\mathrm{d}y}{\mathrm{d}x} = -4x$, so

$$\text{length} = \int_{-1}^{1} \sqrt{1+16x^2}\,\mathrm{d}x = 2\int_{0}^{1} \sqrt{1+16x^2}\,\mathrm{d}x,$$

using the fact that the integrand is even. (The parametric formula $\int \sqrt{\dot{x}^2+\dot{y}^2}\,\mathrm{d}\theta$ gives the same integral after substituting $x = \sin\theta$.)

> **Ask yourself** Which substitution makes $\sqrt{1+16x^2}$ collapse?

We want $16x^2$ to be a $\sinh^2$, so put $4x = \sinh u$. Then $4\,\mathrm{d}x = \cosh u\,\mathrm{d}u$ and $\sqrt{1+16x^2} = \cosh u$ (positive, since $\cosh u > 0$). The limits $x = 0, 1$ become $u = 0$ and $u = \alpha$, where $\alpha = \sinh^{-1} 4$. Note that $\sinh\alpha = 4$ and $\cosh\alpha = \sqrt{1+16} = \sqrt{17}$. Using $\cosh^2 u = \frac{1}{2}(1+\cosh 2u)$,

$$\begin{aligned} 2\int_0^1 \sqrt{1+16x^2}\,\mathrm{d}x &= 2\int_0^\alpha \cosh u \cdot \tfrac{1}{4}\cosh u\,\mathrm{d}u = \tfrac{1}{4}\int_0^\alpha (1+\cosh 2u)\,\mathrm{d}u \\ &= \tfrac{1}{4}\Big[u + \tfrac{1}{2}\sinh 2u\Big]_0^\alpha = \tfrac{1}{4}\big(\alpha + \sinh\alpha\cosh\alpha\big) \\ &= \tfrac{1}{4}\big(\sinh^{-1}4 + 4\sqrt{17}\big). \end{aligned}$$

So the length of P is $\sqrt{17} + \frac{1}{4}\sinh^{-1}4$, as required.

The volume

Both ends of P lie on the line $y = -1$, so when P is rotated about that line it sweeps out a closed surface. The solid it encloses is exactly what we get by rotating the shaded region between P and $y = -1$. A slice at position x is a disc of radius

$$y - (-1) = 2 - 2x^2 = 2(1-x^2) \ (\geq 0 \text{ for } |x| \leq 1).$$

Hence

$$\begin{aligned} V &= \pi\int_{-1}^{1} 4(1-x^2)^2\,\mathrm{d}x = 8\pi\int_0^1 \big(1-2x^2+x^4\big)\,\mathrm{d}x \\ &= 8\pi\Big(1 - \tfrac{2}{3} + \tfrac{1}{5}\Big) = 8\pi\cdot\tfrac{8}{15} = \frac{64\pi}{15}. \end{aligned}$$

Checks

- At $\theta = 0$ we get $(0,1)$ and at $\theta = \pm\pi/2$ we get $(\pm1,-1)$, which agree with $y = 1-2x^2$.
- The length $\sqrt{17} + \frac{1}{4}\sinh^{-1}4 \approx 4.6468$ should be a little more than the length of the two straight chords from $(\pm1,-1)$ to $(0,1)$, which is $2\sqrt{5} \approx 4.472$. It is. A midpoint-rule evaluation of $\int \sqrt{\cos^2\theta + 4\sin^2 2\theta}\,\mathrm{d}\theta$ over $[-\pi/2, \pi/2]$ also gives 4.64678.

- Pappus's theorem: the region has area $\int_{-1}^{1}(2-2x^2)\,\mathrm{d}x = \frac{8}{3}$, and its centroid is $\frac{4}{5}$ above the axis. So $V = 2\pi \cdot \frac{4}{5} \cdot \frac{8}{3} = \frac{64\pi}{15} \approx 13.404$, which also matches numerical integration.

Common slips

- Rotating about the x-axis by mistake and using radius y instead of $y+1$.
- Forgetting the factor $\frac{1}{4}$ from $\mathrm{d}x = \frac{1}{4}\cosh u\,\mathrm{d}u$, or leaving $\sinh 2\alpha$ unsimplified instead of writing it as $2\sinh\alpha\cosh\alpha = 8\sqrt{17}$.

Beyond STEP

Arc length is a first glimpse of the differential geometry of curves. If we measure position along a curve by arc length s, the speed is always 1, and how fast the direction turns with s is the *curvature*. For the parabola $y = 1 - 2x^2$ this is largest at the vertex. Hyperbolic substitutions like the one above appear all over first-year calculus. Siklos, *Advanced Problems in Mathematics*, has many more questions in this style.

WORKED SOLUTION

STEP III 1993, Question 2

Pure • Inflection and Polar Area

STEP III 1993, Question 2

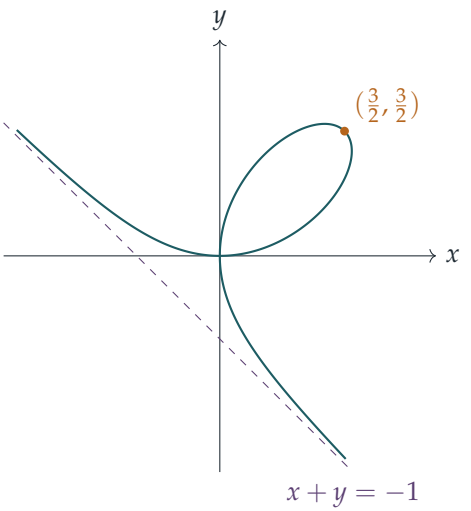
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q2

The key idea

At a point of inflection the second derivative must vanish (and change sign), so the plan for (i) is to find $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ by implicit differentiation and see when it can be zero. The algebra looks heavy, but the equation of the curve itself collapses the numerator to something tiny. For (ii), the cubic terms and the xy term have different degrees, so in polar coordinates a factor r^2 cancels and r comes out as an explicit function of θ.

Solution



The sketch shows the curve (the folium of Descartes): a loop in the first quadrant and two branches in the second and fourth quadrants approaching the asymptote $x + y = -1$.

Part (i)

Differentiating $x^3 + y^3 = 3xy$ implicitly with respect to x,

$$3x^2 + 3y^2\frac{\mathrm{d}y}{\mathrm{d}x} = 3y + 3x\frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \text{so} \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{N}{D}, \quad N = y - x^2,\ D = y^2 - x,$$

wherever $D \neq 0$. Differentiating the quotient, and using $\frac{\mathrm{d}N}{\mathrm{d}x} = y' - 2x$ and $\frac{\mathrm{d}D}{\mathrm{d}x} = 2yy' - 1$ with $y' = N/D$,

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= \frac{(y' - 2x)D - N(2yy' - 1)}{D^2} = \frac{(N - 2xD) - \left(2yN^2/D - N\right)}{D^2} \\ &= \frac{2\left(ND - xD^2 - yN^2\right)}{D^3}.\end{aligned}$$

Now expand the bracket:

$$\begin{aligned}ND &= x^3 + y^3 - x^2y^2 - xy, \\ xD^2 &= xy^4 - 2x^2y^2 + x^3, \\ yN^2 &= y^3 - 2x^2y^2 + x^4y,\end{aligned}$$

so

$$ND - xD^2 - yN^2 = 3x^2y^2 - xy - xy^4 - x^4y = -xy\left(1 + x^3 + y^3 - 3xy\right).$$

On C we have $x^3 + y^3 - 3xy = 0$, so the bracket is just 1 and

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{-2xy}{(y^2 - x)^3}.$$

Ask yourself Where could this be zero, and what about points where the formula breaks down?

The second derivative vanishes only if $x = 0$ or $y = 0$. On C, $x = 0$ forces $y^3 = 0$, and $y = 0$ forces $x^3 = 0$; either way we are at the origin, which we may assume is not a point of inflection. Away from the origin and from points with $y^2 = x$, the second derivative is continuous and non-zero, so it cannot change sign: no inflection there.

That leaves the points with $D = y^2 - x = 0$, where y is not a differentiable function of x. If N were also zero there, we would have $x = y^2$ and $y = x^2$, so $x = x^4$, giving $(0,0)$ or $(1,1)$; but $(1,1)$ is not on C since $1 + 1 \neq 3$. So at every such point other than the origin $N \neq 0$: the tangent is vertical, and we should treat x as a function of y. The equation of C is symmetric in x and y, so swapping the roles of the letters in the work above gives

$$\frac{\mathrm{d}^2x}{\mathrm{d}y^2} = \frac{-2xy}{(x^2 - y)^3},$$

which is defined (because $x^2 - y = -N \neq 0$) and non-zero (because $xy \neq 0$ away from the origin). So the curve does not change its sense of bending at these points either.

Hence C has no point of inflection.

Part (ii)

Put $x = r\cos\theta$, $y = r\sin\theta$. Then

$$r^3(\cos^3\theta + \sin^3\theta) = 3r^2\cos\theta\sin\theta,$$

and for $r \neq 0$ we can divide by r^2:

$$r = \frac{3\cos\theta\sin\theta}{\cos^3\theta + \sin^3\theta}.$$

For $0 \leq \theta \leq \frac{\pi}{2}$ the numerator and denominator are both non-negative (and the denominator is positive), so $r \geq 0$, with $r = 0$ exactly at $\theta = 0$ and $\theta = \frac{\pi}{2}$. So as θ runs from 0 to $\frac{\pi}{2}$ the point goes once round the loop, starting and finishing at the origin. The area is

$$A = \frac{1}{2}\int_0^{\pi/2} r^2\,\mathrm{d}\theta = \frac{9}{2}\int_0^{\pi/2} \frac{\cos^2\theta\sin^2\theta}{(\cos^3\theta + \sin^3\theta)^2}\,\mathrm{d}\theta.$$

Dividing top and bottom by $\cos^6\theta$ (positive for $0 \leq \theta < \frac{\pi}{2}$),

$$A = \frac{9}{2}\int_0^{\pi/2} \frac{\tan^2\theta\sec^2\theta}{(1+\tan^3\theta)^2}\,\mathrm{d}\theta.$$

Substituting $u = \tan\theta$, $\mathrm{d}u = \sec^2\theta\,\mathrm{d}\theta$, with u running from 0 to ∞,

$$A = \frac{9}{2}\int_0^{\infty} \frac{u^2}{(1+u^3)^2}\,\mathrm{d}u = \frac{9}{2}\left[-\frac{1}{3(1+u^3)}\right]_0^{\infty} = \frac{9}{2}\cdot\frac{1}{3} = \frac{3}{2}.$$

The improper integral converges because the antiderivative tends to 0 as $u \to \infty$. So the area of the loop is $\frac{3}{2}$.

Checks

- The loop's tip is where $y = x$: $2x^3 = 3x^2$ gives $(\frac{3}{2}, \frac{3}{2})$, at distance $\frac{3}{\sqrt{2}} \approx 2.12$ from the origin. A loop of that length and width roughly 1 should have area of order 1 to 2, which fits.
- Using the parametrisation $x = \frac{3t}{1+t^3}$, $y = \frac{3t^2}{1+t^3}$ (the line $y = tx$ meets C there), we compared $\frac{-2xy}{(y^2-x)^3}$ with the parametric second derivative at random points, and checked that the curvature never changes sign on any branch.
- Midpoint quadrature of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ and a shoelace sum round the parametrised loop both give 1.500000.

Common slips

- Expanding the second derivative fully and never noticing that $x^3 + y^3 - 3xy$ appears inside it; the simplification is the whole point.
- Forgetting the points with vertical tangents, where $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ is undefined rather than zero.
- Writing $\frac{1}{2}\int r\,\mathrm{d}\theta$ instead of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ for the area.

Beyond STEP

Implicit differentiation is justified by the implicit function theorem: near any point where $\partial F/\partial y \neq 0$, the equation $F(x, y) = 0$ really does define y as a smooth function of x. The origin, where both partial derivatives vanish, is a singular point where two branches cross, which is why it needed special treatment. The folium is also a rational curve, parametrised by lines $y = tx$ through its double point, an idea that leads into algebraic geometry. For the analysis behind all this, see Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP III 1993, Question 3

Pure • LU Matrix Decomposition

STEP III 1993, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q3

The key idea

Writing $\mathbf{M}$ as a lower-triangular matrix times an upper-triangular one (an *LU decomposition*) turns everything into a chain of one-unknown equations. Multiply out $\mathbf{AB}$ and match entries in a sensible order: each new equation then contains only one unknown, which gives both the values and their uniqueness at once. Triangular matrices are also easy to invert, and $\mathbf{Mx} = \mathbf{v}$ splits into two easy triangular systems.

A warning first. The paper prints $e = 1$, but this is a misprint: the correct value is $e = -1$. With $e = 1$ the bottom row of $\mathbf{AB}$ would be $(3, 10, 1)$, not $(3, 8, 7)$. Everything below uses $e = -1$.

Solution

Finding A and B

Multiplying out,

$$\mathbf{AB} = \begin{pmatrix} a & ap & aq \\ b & bp + c & bq + cr \\ d & dp + e & dq + er + f \end{pmatrix}.$$

We set this equal to $\mathbf{M}$ and take the entries in the following order, so that each equation has just one unknown left in it.

- Entry $(1,1)$: $a = 1$.
- Entries $(1,2)$, $(1,3)$: $ap = 3$ and $aq = 2$, so $p = 3$, $q = 2$ (since $a = 1$).
- Entries $(2,1)$, $(3,1)$: $b = 4$, $d = 3$.
- Entry $(2,2)$: $bp + c = 13$, so $c = 13 - 12 = 1$.
- Entry $(2,3)$: $bq + cr = 5$, so $8 + r = 5$ and $r = -3$ (using $c = 1$).
- Entry $(3,2)$: $dp + e = 8$, so $9 + e = 8$ and $e = -1$.

- Entry $(3,3)$: $dq + er + f = 7$, so $6 + 3 + f = 7$ and $f = -2$.

Every unknown was *forced* by an equation in which everything else was already known, and the only divisions were by $a = 1$ and $c = 1$, which are non-zero. So if $\mathbf{M} = \mathbf{AB}$ then the unknowns must take these values. Conversely, these values satisfy all nine equations. So the solution exists and is unique:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 3 & -1 & -2 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix},$$

with $a = 1$, $b = 4$, $c = 1$, $d = 3$, $e = -1$, $f = -2$, $p = 3$, $q = 2$, $r = -3$.

Ask yourself Where does uniqueness come from? We only found one solution.

Uniqueness comes from the *order* of the deductions. No choice was made at any stage. Each equation was linear in a single new unknown with a non-zero coefficient, so it had exactly one solution. Any $\mathbf{A}$, $\mathbf{B}$ of the given forms with $\mathbf{AB} = \mathbf{M}$ must therefore be the pair above. (The 1s on the diagonal of $\mathbf{B}$ matter: without them we could scale $\mathbf{A}$ up and $\mathbf{B}$ down.)

The inverses

To invert $\mathbf{A}$, we solve $\mathbf{Au} = \mathbf{v}$ from the top down:

$$\begin{aligned} u_1 &= v_1, \\ u_2 &= v_2 - 4u_1 = -4v_1 + v_2, \\ -2u_3 &= v_3 - 3u_1 + u_2 = -7v_1 + v_2 + v_3 \implies u_3 = \tfrac{1}{2}(7v_1 - v_2 - v_3). \end{aligned}$$

To invert $\mathbf{B}$, we solve $\mathbf{Bw} = \mathbf{u}$ from the bottom up:

$$\begin{aligned} w_3 &= u_3, \\ w_2 &= u_2 + 3w_3 = u_2 + 3u_3, \\ w_1 &= u_1 - 3w_2 - 2w_3 = u_1 - 3u_2 - 11u_3. \end{aligned}$$

Reading off the coefficients,

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ \frac{7}{2} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix}, \qquad \mathbf{B}^{-1} = \begin{pmatrix} 1 & -3 & -11 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplying out confirms that $\mathbf{AA}^{-1} = \mathbf{I}$ and $\mathbf{BB}^{-1} = \mathbf{I}$. Notice that the inverse of a lower (upper) triangular matrix is again lower (upper) triangular.

The simultaneous equations

The equations say $\mathbf{Mx} = \mathbf{v}$ with $\mathbf{x} = (x, y, z)^{\mathrm{T}}$ and $\mathbf{v} = (7, 18, 25)^{\mathrm{T}}$. Since $\mathbf{M} = \mathbf{AB}$,

$$\mathbf{x} = \mathbf{M}^{-1}\mathbf{v} = \mathbf{B}^{-1}\mathbf{A}^{-1}\mathbf{v}.$$

Note the order: $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$. First,

$$\mathbf{u} = \mathbf{A}^{-1}\mathbf{v} = \begin{pmatrix} 7 \\ -28+18 \\ \frac{1}{2}(49-18-25) \end{pmatrix} = \begin{pmatrix} 7 \\ -10 \\ 3 \end{pmatrix},$$

and then

$$\mathbf{x} = \mathbf{B}^{-1}\mathbf{u} = \begin{pmatrix} 7+30-33 \\ -10+9 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix}.$$

So the unique solution is

$$x = 4, \qquad y = -1, \qquad z = 3.$$

It is unique because $\mathbf{M}$ is invertible: $\det \mathbf{M} = \det \mathbf{A} \det \mathbf{B} = (1 \cdot 1 \cdot (-2))(1) = -2 \neq 0$.

Checks

Substitute back: $4-3+6=7$, $16-13+15=18$, $12-8+21=25$. All three equations hold. We also checked with exact fractions that $\mathbf{AB} = \mathbf{M}$ and that $\mathbf{B}^{-1}\mathbf{A}^{-1}$ really is $\mathbf{M}^{-1}$; its bottom row is $(\frac{7}{2}, -\frac{1}{2}, -\frac{1}{2})$. With the printed value $e = 1$, the product $\mathbf{AB}$ has bottom row $(3, 10, 1)$, which confirms the misprint.

Common slips

- Trusting the printed $e = 1$ and then forcing the $(3,2)$ and $(3,3)$ entries to fit. Always multiply your $\mathbf{A}$ and $\mathbf{B}$ back together.
- Using $\mathbf{A}^{-1}\mathbf{B}^{-1}$ instead of $\mathbf{B}^{-1}\mathbf{A}^{-1}$.
- Showing that the values work without arguing that no other values could.

Beyond STEP

This is the *LU decomposition*, which is how computers actually solve linear systems. Gaussian elimination is really the process of factorising $\mathbf{M} = \mathbf{LU}$. After that, each new right-hand side costs only a forward substitution and a back substitution, with no fresh elimination. In general a row swap may be needed first, which gives $\mathbf{PM} = \mathbf{LU}$. Strang's *Introduction to Linear Algebra* builds a whole chapter around this idea.

WORKED SOLUTION

STEP III 1993, Question 4

Pure • Summing Series

STEP III 1993, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q4

The key idea

Each series is a familiar one in disguise. The usual way to sum a series is to recognise it as a standard expansion (logarithm, geometric series or its derivative) evaluated at a particular point, or to split the general term so that the sum telescopes. So for each part we ask which standard series it could be, and what we would have to multiply, differentiate or rearrange to get it into that form.

Solution

Part (i)

The odd denominators $1, 3, 5, \ldots$ suggest the logarithm series with only odd powers left. For $|x| < 1$,

$$\ln(1+x) = x - \tfrac{x^2}{2} + \tfrac{x^3}{3} - \cdots, \qquad \ln(1-x) = -x - \tfrac{x^2}{2} - \tfrac{x^3}{3} - \cdots,$$

and subtracting removes the even powers:

$$\ln\frac{1+x}{1-x} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right) = 2\sum_{n=0}^{\infty}\frac{x^{2n+1}}{2n+1}.$$

Our series is $\sum x^{2n}/(2n+1)$ with $x = \frac{1}{2}$. The powers are even, so we divide by x (allowed, since $x \neq 0$):

$$\sum_{n=0}^{\infty}\frac{x^{2n}}{2n+1} = \frac{1}{2x}\ln\frac{1+x}{1-x}.$$

At $x = \frac{1}{2}$, which is inside the interval of convergence, $\frac{1+x}{1-x} = \frac{3/2}{1/2} = 3$ and $\frac{1}{2x} = 1$, so

$$1 + \frac{1}{3}\left(\frac{1}{2}\right)^2 + \frac{1}{5}\left(\frac{1}{2}\right)^4 + \cdots = \ln 3.$$

Part (ii)

The coefficients repeat with period 4: they are $2, -1, 0, -1$ for $x^{4k}, x^{4k+1}, x^{4k+2}, x^{4k+3}$. So we group each block of four terms:

$$\sum_{k=0}^{\infty} x^{4k}(2 - x - x^3) = \frac{2 - x - x^3}{1 - x^4},$$

a geometric series with ratio x^4, which converges since $|x^4| < 1$. The grouping is legitimate because the original series converges absolutely for $|x| < 1$: its terms are bounded in size by $2|x|^m$.

Now simplify. The numerator vanishes at $x = 1$, so $1 - x$ is a factor:

$$2 - x - x^3 = (1 - x)(x^2 + x + 2), \qquad 1 - x^4 = (1 - x)(1 + x)(1 + x^2).$$

Cancelling $1 - x$ (non-zero, as $|x| < 1$),

$$\frac{x^2 + x + 2}{(1 + x)(1 + x^2)} = \frac{A}{1 + x} + \frac{Bx + C}{1 + x^2}.$$

Putting $x = -1$ gives $2 = 2A$, so $A = 1$. Then $x^2 + x + 2 - (1 + x^2) = x + 1 = (Bx + C)(1 + x)$, so $B = 0$ and $C = 1$. Therefore

$$2 - x - x^3 + 2x^4 - \cdots = \frac{1}{1 + x} + \frac{1}{1 + x^2}.$$

This is easy to confirm: $1 - x + x^2 - x^3 + x^4 - \cdots$ plus $1 - x^2 + x^4 - \cdots$ gives coefficients $2, -1, 0, -1, 2, \ldots$, as required.

Part (iii)

Write the general term as

$$\frac{r\,2^{r-2}}{3^{r-1}} = \frac{1}{2}\, r \left(\frac{2}{3}\right)^{r-1}.$$

Ask yourself Where have we seen $\sum r y^{r-1}$ before?

It is the derivative of the geometric series. For $|y| < 1$, differentiating $\sum_{r=0}^{\infty} y^r = \frac{1}{1-y}$ term by term (valid inside the interval of convergence) gives

$$\sum_{r=1}^{\infty} r y^{r-1} = \frac{1}{(1 - y)^2}.$$

With $y = \frac{2}{3}$ this is $\frac{1}{(1/3)^2} = 9$. Our sum starts at $r = 2$, so we remove the $r = 1$ term, which is 1:

$$\sum_{r=2}^{\infty} \frac{r\,2^{r-2}}{3^{r-1}} = \frac{1}{2}(9 - 1) = 4.$$

Part (iv)

Since $r(r^2-1) = (r-1)r(r+1)$ is a product of three consecutive integers, we look for a difference of two consecutive "pair" fractions:

$$\frac{1}{(r-1)r} - \frac{1}{r(r+1)} = \frac{(r+1)-(r-1)}{(r-1)r(r+1)} = \frac{2}{r(r^2-1)}.$$

So the sum telescopes. Summing from $r=2$ to N, every term cancels except the first and last:

$$\sum_{r=2}^{N} \frac{2}{r(r^2-1)} = \frac{1}{1\cdot 2} - \frac{1}{N(N+1)}.$$

As $N \to \infty$ the last term tends to 0, so

$$\sum_{r=2}^{\infty} \frac{2}{r(r^2-1)} = \frac{1}{2}.$$

Checks

- (i): the first four terms give $1 + 0.0833 + 0.0125 + 0.0022 \approx 1.098$, already close to $\ln 3 \approx 1.0986$.
- (ii): at $x = 0$ both sides equal 2. At $x = \frac{1}{2}$ the closed form gives $\frac{2}{3} + \frac{4}{5} = \frac{22}{15}$, which matches the partial sums numerically.
- (iii): the first terms are $\frac{2}{3}, \frac{2}{3}, \frac{16}{27}, \ldots$; partial sums climb slowly towards 4.
- (iv): the partial sum to $N = 4$ is $\frac{1}{3} + \frac{1}{12} + \frac{1}{30} = \frac{9}{20} = \frac{1}{2} - \frac{1}{20}$, matching the formula.

Common slips

- In (i), forgetting to divide by x: $\sum x^{2n+1}/(2n+1)$ at $x = \frac{1}{2}$ is $\frac{1}{2}\ln 3$, not $\ln 3$.
- In (iii), summing from $r = 1$ and forgetting to subtract the missing first term, which gives $\frac{9}{2}$.
- In (iv), telescoping with $\frac{1}{r-1} - \frac{1}{r+1}$ instead: that splits $\frac{2}{r^2-1}$, not $\frac{2}{r(r^2-1)}$.

Beyond STEP

The steps we took without fuss (regrouping terms in (ii) and differentiating a power series term by term in (iii)) are exactly what a first analysis course justifies. Inside its radius of convergence a power series converges absolutely and can be differentiated term by term, but rearranging a series that converges only conditionally can change its sum (Riemann's rearrangement theorem). Abbott's *Understanding Analysis* covers both carefully.

WORKED SOLUTION

STEP III 1993, Question 5

Pure • Groups and Quaternions

STEP III 1993, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 6.7, *The Quaternion Group*: https://beyondstep.keithmcnulty.org/ch06/quaternions.html

The key idea

The operation looks arbitrary, but the four products in the first part show its structure: pairs of the form $(z,0)$ multiply like ordinary complex numbers, and the second slot behaves like a new symbol j with $j^2=-1$, which conjugates anything it moves past. Once you have those four products, the eight elements of S_1 are just $(u,0)$ and $(0,u)$ with $u\in\{1,i,-1,-i\}$. So closure, inverses and orders come from a few general lines, with no need for a 64-entry table.

Solution

Associativity and commutativity

Write $z=(z_1,z_2)$, $w=(w_1,w_2)$, $u=(u_1,u_2)$. We use $(ab)^*=a^*b^*$, $(a\pm b)^*=a^*\pm b^*$ and $a^{**}=a$ throughout. First,

$$z\circ w=(z_1w_1-z_2w_2^*,\ z_1w_2+z_2w_1^*),$$

so the two components of $(z\circ w)\circ u$ are

$$\begin{aligned}(z_1w_1-z_2w_2^*)u_1-(z_1w_2+z_2w_1^*)u_2^* &= z_1w_1u_1-z_1w_2u_2^*-z_2w_1^*u_2^*-z_2w_2^*u_1,\\ (z_1w_1-z_2w_2^*)u_2+(z_1w_2+z_2w_1^*)u_1^* &= z_1w_1u_2+z_1w_2u_1^*+z_2w_1^*u_1^*-z_2w_2^*u_2.\end{aligned}$$

On the other side, $w\circ u=(w_1u_1-w_2u_2^*,\ w_1u_2+w_2u_1^*)$, so the two components of $z\circ(w\circ u)$ are

$$\begin{aligned}z_1(w_1u_1-w_2u_2^*)-z_2(w_1u_2+w_2u_1^*)^* &= z_1w_1u_1-z_1w_2u_2^*-z_2w_1^*u_2^*-z_2w_2^*u_1,\\ z_1(w_1u_2+w_2u_1^*)+z_2(w_1u_1-w_2u_2^*)^* &= z_1w_1u_2+z_1w_2u_1^*+z_2w_1^*u_1^*-z_2w_2^*u_2.\end{aligned}$$

The right-hand sides agree term by term, so $\circ$ is associative.

It is not commutative. For example,

$$(0,1)\circ(i,0) = (0-0,\ 0+1\cdot i^*) = (0,-i), \qquad (i,0)\circ(0,1) = (0-0,\ i\cdot 1+0) = (0,i).$$

The four products

Putting zeros into the definition,

$$\begin{aligned} (z,0)\circ(w,0) &= (zw,\ 0), & (z,0)\circ(0,w) &= (0,\ zw),\\ (0,z)\circ(w,0) &= (0,\ zw^*), & (0,z)\circ(0,w) &= (-zw^*,\ 0). \end{aligned}$$

The set S_1

Let $U = \{1, i, -1, -i\}$. Then

$$S_1 = \{(u,0) : u \in U\} \cup \{(0,u) : u \in U\},$$

with A, C, E, G of the first kind and B, D, F, H of the second.

Closure. The set U is closed under multiplication, under complex conjugation ($i^* = -i$) and under negation. By the four products, the product of two elements of S_1 is $(zw,0)$, $(0,zw)$, $(0,zw^*)$ or $(-zw^*,0)$ with $z, w \in U$. Each of zw and $\pm zw^*$ lies in U, so every product lies in S_1.

Identity. For any $(w_1,w_2) \in S$,

$$(1,0)\circ(w_1,w_2) = (w_1,\ w_2), \qquad (w_1,w_2)\circ(1,0) = (w_1,\ w_2\cdot 1^*) = (w_1,w_2),$$

so $A = (1,0)$ is a two-sided identity, and it lies in S_1.

Inverses and orders. For elements of the first kind, $(z,0)\circ(w,0) = (zw,0)$, so they behave exactly like the complex numbers in U. Hence A has order 1 and is its own inverse, $E = (-1,0)$ has order 2 and is its own inverse, and $C = (i,0)$ and $G = (-i,0)$ each have order 4 and are inverses of each other.

> **Ask yourself** What is the square of an element of the second kind?

By the last of the four products, for $u \in U$,

$$(0,u)\circ(0,u) = (-uu^*,\ 0) = (-|u|^2,\ 0) = (-1,0) = E.$$

So each of B, D, F, H squares to E. Since E has order 2, each of these has order 4. For the inverse, $(0,u)\circ(0,w) = (-uw^*,0)$ equals $(1,0)$ exactly when $w^* = -1/u = -u^*$ (using $uu^* = 1$), that is, when $w = -u$. Checking the other order, $(0,-u)\circ(0,u) = (uu^*,0) = (1,0)$ too. So the inverse of $(0,u)$ is $(0,-u)$.

In summary:

element	A	B	C	D	E	F	G	H
inverse	A	F	G	H	E	B	C	D
order	1	4	4	4	2	4	4	4

Group. We now have all four axioms. S_1 is closed under $\circ$. The operation is associative on all of S, so in particular on S_1. $A \in S_1$ is an identity, and every element has an inverse in S_1. So S_1 is a group under $\circ$, of order 8. It is non-abelian: $B \circ C = H$ but $C \circ B = D$.

The subgroup and the matrix

Using the products above, the powers of $B = (0,1)$ are

$$B^2 = E, \qquad B^3 = E \circ B = (-1,0) \circ (0,1) = (0,-1) = F, \qquad B^4 = E \circ E = A.$$

So $\{A, B, E, F\} = \{B^0, B^1, B^2, B^3\}$ is the cyclic subgroup generated by B. It is closed, because $B^m \circ B^n = B^{m+n}$ and exponents can be reduced mod 4, and it contains A and the inverse B^{4-k} of each B^k. So it is a subgroup of order 4.

Let $M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then

$$M^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I, \qquad M^3 = -M, \qquad M^4 = I,$$

so the group generated by M is $\{I, M, -I, -M\}$, also cyclic of order 4. Any two cyclic groups of the same order are isomorphic. Here the map

$$A \mapsto I, \quad B \mapsto M, \quad E \mapsto -I, \quad F \mapsto -M,$$

that is, $B^k \mapsto M^k$, is a bijection and satisfies $B^m \circ B^n = B^{m+n} \mapsto M^{m+n} = M^m M^n$. So the two groups *are* isomorphic.

Checks

- Order of an element divides the order of the group: all the orders $1, 2, 4$ divide 8, and 2 and 4 divide 4 for the subgroup.
- The complex-number view: $(z,0) \circ (w,0) = (zw,0)$, so $C \circ C = (i^2,0) = E$, exactly as $i^2 = -1$.
- A short computer check confirmed associativity on 10 000 random triples of complex pairs, built the full 8×8 table to confirm closure, the identity, the inverses and orders above and $B \circ C = H \neq D = C \circ B$, and confirmed that $B^k \mapsto M^k$ respects multiplication.

Common slips

- Forgetting the conjugates: $(0,z) \circ (w,0)$ is $(0, zw^*)$, not $(0, zw)$. This is where non-commutativity comes from.
- Showing associativity with one numerical example. An example can show that a law *fails*, but proving it needs the general expansion.

- Concluding "not isomorphic" because one group is made of pairs of complex numbers and the other of matrices. Isomorphism is about the structure of the multiplication, not what the elements are.

Beyond STEP

The operation is Hamilton's multiplication of *quaternions*. The pair (z_1, z_2) stands for $z_1 + z_2 j$, where $j^2 = -1$ and $jz = z^* j$. The group S_1 is the *quaternion group* $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$. It is the smallest non-abelian group in which every subgroup is normal, and it differs from the other non-abelian group of order 8 (the symmetries of a square) by having only one element of order 2. Allenby's *Rings, Fields and Groups* covers groups of small order and the quaternions.

WORKED SOLUTION

STEP III 1993, Question 6

Pure • Loci in the Argand Diagram

STEP III 1993, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q6

The key idea

For a circle, $|z-k|=r$ is the natural description, and squaring it with $|u|^2=uu^*$ turns it into an equation in z and z^* with no square roots. Once a locus is written that way, a Möbius map such as $w=z/(z-1)$ is easy to handle: invert it to get z in terms of w, substitute, clear denominators, and read off the new centre and radius by comparing with the general form.

Solution

The equation of a circle

P lies on the circle exactly when its distance from K is r, that is $|z-k|=r$. Squaring, and using $|u|^2=uu^*$ with $(z-k)^*=z^*-k^*$,

$$r^2=(z-k)(z^*-k^*)=zz^*-kz^*-k^*z+kk^*,$$

so $zz^*-kz^*-k^*z+kk^*-r^2=0$, as required.

Conversely, any equation $ww^*-cw^*-c^*w+cc^*-s^2=0$ with $s>0$ says $|w-c|^2=s^2$, so it is the circle with centre c and radius s. We use this to recognise circles below.

The locus of P

The circle on OA as diameter has centre i (the midpoint of 0 and $2i$) and radius 1. With $k=i$, $k^*=-i$, $kk^*=1$ and $r=1$, the first part gives

$$zz^*+iz-iz^*=0. \qquad (*)$$

The locus L of Q_1

Note that $z = 1$ is not on the circle (its distance from i is $\sqrt{2}$), so $w_1 = z/(z-1)$ is defined for every P. Write w for w_1. Solving $w(z-1) = z$ for z gives

$$z = \frac{w}{w-1}, \qquad z^* = \frac{w^*}{w^*-1},$$

and $w \neq 1$, because $w = 1$ would need $z = z - 1$. Substituting into $(*)$:

$$\frac{ww^*}{(w-1)(w^*-1)} + \frac{iw}{w-1} - \frac{iw^*}{w^*-1} = 0.$$

Multiplying through by $(w-1)(w^*-1)$, which is non-zero,

$$\begin{aligned} 0 &= ww^* + iw(w^*-1) - iw^*(w-1) \\ &= ww^* + iww^* - iw - iww^* + iw^* \\ &= ww^* - iw + iw^*. \end{aligned}$$

So L has equation

$$w_1 w_1^* - iw_1 + iw_1^* = 0.$$

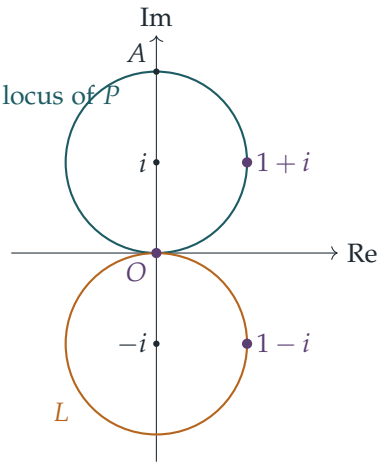
To identify it, compare with the general form: we need $-k^* = -i$ and $-k = i$, so $k = -i$; then $kk^* = 1$, and the constant term $kk^* - r^2$ is 0, so $r = 1$. Hence L is the circle with centre $-i$ and radius 1, i.e. $|w_1 + i| = 1$. It is the reflection of P's circle in the real axis, and it also passes through the origin.

Ask yourself Is every point of this circle actually reached, or only part of it?

Every step above can be reversed. If $w \neq 1$ lies on $|w + i| = 1$, then $z = w/(w-1)$ satisfies $(*)$, so it is a point P on the circle with $w_1 = w$. The only point that could be missed is $w = 1$, and that is not on L, since $|1 + i| = \sqrt{2}$. So the locus is the whole circle.

The locus of Q_2

Taking the conjugate of $|z - i| = 1$ gives $|z^* + i| = 1$, since $|u^*| = |u|$ and $(z-i)^* = z^* + i$. So $w_2 = z^*$ satisfies $|w_2 + i| = 1$. As z runs over the whole circle centred at i, z^* runs over the whole circle centred at $-i$. Equivalently, conjugating $(*)$ gives $z^*z - iz^* + iz = 0$, which is the equation of L with $w = z^*$. So the locus of Q_2 is also L.



When Q_1 and Q_2 coincide

We need $\dfrac{z}{z-1} = z^*$, that is

$$z = zz^* - z^*.$$

On the circle, $(*)$ gives $zz^* = iz^* - iz$. Substituting,

$$z = iz^* - iz - z^* \quad \Longrightarrow \quad (1+i)z = (i-1)z^*.$$

One solution is $z = 0$. Otherwise $z^* \neq 0$ and

$$\frac{z}{z^*} = \frac{i-1}{1+i} = \frac{(i-1)(1-i)}{(1+i)(1-i)} = \frac{2i}{2} = i.$$

Writing $z = \rho e^{i\theta}$ with $\rho > 0$, this says $e^{2i\theta} = i$, so $\theta = \frac{\pi}{4}$ or $\frac{5\pi}{4}$: P lies on the line $y = x$. That line meets the circle $x^2 + (y-1)^2 = 1$, i.e. $x^2 + y^2 = 2y$, where $2x^2 = 2x$, so at $x = 0$ or $x = 1$. The non-zero point is $z = 1 + i$.

Checking both: $z = 0$ gives $w_1 = 0 = z^*$; and $z = 1 + i$ gives

$$w_1 = \frac{1+i}{i} = 1 - i = z^*.$$

So Q_1 and Q_2 coincide exactly when P is the origin or the point $1 + i$. These are the two points where $y = x$ meets the circle, and the common positions of $Q_1 = Q_2$ are 0 and $1 - i$.

Checks

- A sample point: $P = 2i$ (the point A) gives $w_1 = \frac{2i}{2i-1} = \frac{4-2i}{5}$, and $|w_1 + i| = \left|\frac{4+3i}{5}\right| = 1$, so it lies on L.
- Numerically, we took 10^4 random points on $|z - i| = 1$ and confirmed that both $z/(z-1)$ and z^* satisfy $|w + i| = 1$ to within 10^{-9}. Scanning 2×10^5 points round the circle for $\left|z/(z-1) - z^*\right| < 10^{-3}$ found only $z = 0$ and $z = 1 + i$.

Common slips

- Dividing by z^* without first setting aside $z = 0$, and so losing the solution $P = O$.
- Concluding $z/z^* = i$ and stopping there: this fixes only the direction of P, and you still have to intersect that line with the circle.
- Getting the sign of k wrong when matching $ww^* - iw + iw^* = 0$ to the general form. The coefficient of w^* is $-k$, so $k = -i$, not i.

Beyond STEP

The map $z \mapsto z/(z-1)$ is a Möbius transformation, and such maps always send circles and lines to circles and lines. Here the image circle happens to be the mirror image of the original, which is why conjugation produces the same locus. Conjugation itself reverses orientation, so it is not Möbius, but composing it with a Möbius map gives the "anti-Möbius" maps, which include reflection in a circle. Needham's *Visual Complex Analysis* gives a beautiful geometric treatment of Möbius maps, inversion and the circle-preserving property.

WORKED SOLUTION

STEP III 1993, Question 7

Pure • Hyperbolic Simultaneous Equations

STEP III 1993, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q7

The key idea

Expand both double-angle sinhs. The second equation then gives $\cosh x$ directly in terms of y, and dividing the first by it gives $\sinh x$ in terms of y too. The identity $\cosh^2 x - \sinh^2 x = 1$ eliminates x completely, leaving an equation in $\sinh^2 y$ alone. After that it is a matter of locating the right root of a cubic and pinning it down numerically.

Solution

Eliminating x

Using $\sinh 2u = 2\sinh u \cosh u$, the two equations become

$$2\sinh x \cosh x = \cosh y, \qquad 2\sinh y \cosh y = 2\cosh x.$$

The second gives

$$\cosh x = \sinh y \cosh y.$$

Since $\cosh x \geq 1 > 0$, we may divide the first equation by $2\cosh x$:

$$\sinh x = \frac{\cosh y}{2\sinh y \cosh y} = \frac{1}{2\sinh y},$$

where dividing by $\cosh y$ is fine because $\cosh y > 0$, and $\sinh y \neq 0$ because $\sinh y \cosh y = \cosh x \neq 0$.

Now substitute both into $\cosh^2 x - \sinh^2 x = 1$. Write $t = \sinh^2 y$, so that $\cosh^2 y = 1 + t$:

$$t(1+t) - \frac{1}{4t} = 1.$$

Multiplying by $4t$ (which is non-zero) gives $4t^2 + 4t^3 - 1 = 4t$, that is,

$$4t^3 + 4t^2 - 4t - 1 = 0,$$

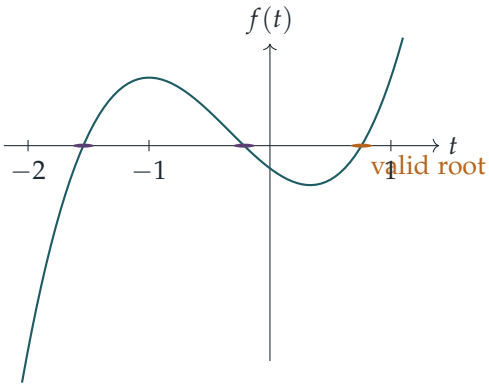
so $t = \sinh^2 y$ is a root of the given cubic, as required.

At most one valid solution

Let $f(t) = 4t^3 + 4t^2 - 4t - 1$. Since $t = \sinh^2 y$ with $\sinh y \neq 0$, a valid root must satisfy $t > 0$. Now

$$f(-2) = -9, \quad f(-1) = 3, \quad f(0) = -1, \quad f(1) = 3,$$

so by the intermediate value theorem the cubic has a root in each of $(-2, -1)$, $(-1, 0)$ and $(0, 1)$. A cubic has at most three roots, so these are all of them, and exactly one root is positive.



Ask yourself A positive t gives two values $\sinh y = \pm\sqrt{t}$. Why does only one survive?

From $\cosh x = \sinh y \cosh y$ and $\cosh x > 0$, $\cosh y > 0$, we need $\sinh y > 0$. So $\sinh y = +\sqrt{t}$, and since sinh is strictly increasing this gives exactly one y, namely $y > 0$. Hence the cubic gives at most one valid y (and then $\sinh x = 1/(2\sinh y)$ fixes x uniquely, as sinh is one-to-one).

Locating and computing t

We have

$$f(0.7) = 1.372 + 1.96 - 2.8 - 1 = -0.468 < 0,$$
$$f(0.8) = 2.048 + 2.56 - 3.2 - 1 = 0.408 > 0,$$

so by the intermediate value theorem the positive root lies between 0.7 and 0.8.

We use the Newton–Raphson iteration

$$t_{n+1} = t_n - \frac{f(t_n)}{f'(t_n)}, \qquad f'(t) = 12t^2 + 8t - 4,$$

starting from $t_0 = 0.75$, the midpoint. Here $f(0.75) = -0.0625$ and $f'(0.75) = 8.75$, so

$$\begin{aligned} t_1 &= 0.75 + \frac{0.0625}{8.75} = 0.7571429, \\ t_2 &= 0.7570685, \\ t_3 &= 0.75706846, \end{aligned}$$

and t_4 agrees with t_3 to all the figures shown. To confirm six decimal places, we check the sign change

$$f(0.7570680) < 0 < f(0.7570690),$$

so the root lies in that interval and

$$t = 0.757068 \quad \text{(6 d.p.)}.$$

(A slower but perfectly good alternative: rearrange as $t^2(4t+4) = 4t+1$, i.e. $t = \sqrt{(4t+1)/(4t+4)}$. Near the root the derivative of the right-hand side is about 0.16, so this iteration converges, gaining a little under one decimal place per step.)

Finding y and x

Taking the positive square root, $\sinh y = \sqrt{0.7570685} = 0.870097$, so

$$y = \operatorname{arsinh}(0.870097) = \ln\bigl(0.870097 + \sqrt{1.757068}\bigr) = 0.786474,$$

and

$$\sinh x = \frac{1}{2\sinh y} = 0.574649, \qquad x = \ln\bigl(0.574649 + \sqrt{1.330221}\bigr) = 0.546965.$$

Checking in the original equations:

$$\begin{aligned} \sinh 2x &= 1.325545 = \cosh y, \\ \sinh 2y &= 2.306704 = 2\cosh x, \end{aligned}$$

so both equations hold. To four decimal places,

$$x = 0.5470, \qquad y = 0.7865.$$

Checks

- The other two roots of the cubic are about -0.2140 and -1.5431: both negative, as the argument requires. The product of all three roots is $\frac{1}{4}$, which matches $0.757068 \times 0.2140 \times 1.5431 \approx 0.250$.
- A direct numerical scan over $y \in [-5, 5]$, taking $x = \pm\operatorname{arcosh}(\frac{1}{2}\sinh 2y)$ from the second equation and testing the first, finds a single solution, at $y \approx 0.7865$ with $x > 0$.
- The negative choice $\sinh y = -\sqrt{t}$ would make $\sinh y \cosh y < 0$, which can never equal $\cosh x$: a quick confirmation that it really is invalid.

Common slips

- Squaring early (for instance squaring both original equations) and then forgetting to discard the sign that doesn't work.
- Stopping the iteration as soon as two iterates agree to six places without checking the sign change across the rounding interval.
- Rounding t to six places before taking square roots and logarithms. Carry extra figures and round only the final x and y.

Beyond STEP

The two iterations here converge at very different rates. A fixed-point iteration $t \mapsto g(t)$ converges linearly, with error shrinking by about $|g'(\text{root})|$ each step, while Newton–Raphson converges quadratically, roughly doubling the number of correct digits each step. Making this precise uses the mean value theorem and Taylor's theorem, and it is the starting point of numerical analysis. For the analysis behind it, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1993, Question 8

Pure • Planes and Lines in 3D

STEP III 1993, Question 8

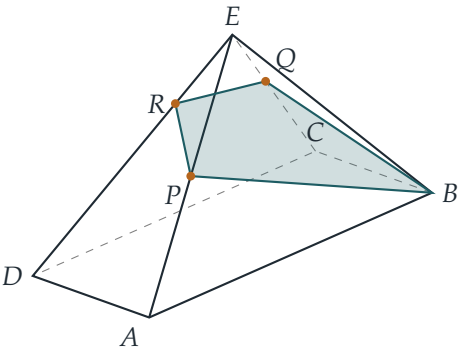
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q8

The key idea

Every point on a slant edge has a simple parametric form, because each edge joins a base vertex to E. So P, Q and R can all be written down at once, and the plane through B, P and Q can be found as a linear equation whose coefficients come from substituting those points. For the second part, the cleanest test for a parallelogram is that its diagonals bisect each other; one coordinate is enough to show this fails.

Solution



Finding γ

The edge AE consists of the points $(ta, 0, (1-t)a)$ for $0 \le t \le 1$, since $x + z = a$ along it. Similarly CE is $\{(-ta, 0, (1-t)a)\}$ and DE is $\{(0, -ta, (1-t)a)\}$. So

$$P = \big(\lambda a,\, 0,\, (1-\lambda)a\big), \qquad Q = \big(-\mu a,\, 0,\, (1-\mu)a\big), \qquad R = \big(0,\, -\gamma a,\, (1-\gamma)a\big).$$

We look for the plane BPQ in the form

$$px + qy + rz = a.$$

The points B, P, Q are not collinear, so they determine a unique plane. If we can find p, q, r satisfying all three conditions, that equation is the plane. Substituting each point:

$$\begin{aligned} B: &\quad qa = a, && \text{so } q = 1,\\ P: &\quad p\lambda + r(1-\lambda) = 1,\\ Q: &\quad -p\mu + r(1-\mu) = 1. \end{aligned}$$

To eliminate p, multiply the P equation by μ, the Q equation by λ, and add:

$$r\big(\mu - \lambda\mu + \lambda - \lambda\mu\big) = \lambda + \mu, \qquad \text{so} \qquad r = \frac{\lambda+\mu}{\lambda+\mu-2\lambda\mu}.$$

The denominator is $\lambda(1-\mu) + \mu(1-\lambda) > 0$, so this is fine. (We won't need it, but $p = (\lambda-\mu)/(\lambda+\mu-2\lambda\mu)$.)

Now R lies in the plane, so

$$-\gamma + r(1-\gamma) = 1 \quad\Longrightarrow\quad \gamma(1+r) = r-1 \quad\Longrightarrow\quad \gamma = \frac{r-1}{r+1}.$$

Putting everything over the common denominator $\lambda+\mu-2\lambda\mu$:

$$r-1 = \frac{2\lambda\mu}{\lambda+\mu-2\lambda\mu}, \qquad r+1 = \frac{2(\lambda+\mu-\lambda\mu)}{\lambda+\mu-2\lambda\mu},$$

and hence

$$\gamma = \frac{\lambda\mu}{\lambda+\mu-\lambda\mu},$$

as required.

It is worth noting that $0 < \gamma < 1$, so R really does lie on the edge DE strictly between D and E. Positivity is clear. For $\gamma < 1$, the denominator exceeds the numerator by $\lambda+\mu-2\lambda\mu = \lambda(1-\mu) + \mu(1-\lambda) > 0$.

$BPRQ$ is not a parallelogram

Ask yourself What is the quickest test for whether four points, taken in a given order, form a parallelogram?

A quadrilateral $BPRQ$ is a parallelogram exactly when its diagonals BR and PQ bisect each other, that is, when

$$\overrightarrow{OB} + \overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{OQ}.$$

(This is the same as $\overrightarrow{BP} = \overrightarrow{QR}$, i.e. one pair of opposite sides is equal and parallel.) Compare the y-coordinates of the two sides:

$$\text{left: } a - \gamma a = (1-\gamma)a, \qquad \text{right: } 0 + 0 = 0.$$

These would have to be equal, forcing $\gamma = 1$. But we showed $\gamma < 1$; indeed $\gamma = 1$ would put R at E. So the diagonals never bisect each other, and $BPRQ$ cannot be a parallelogram. ■

For completeness: the vertices really do occur in the order B, P, R, Q round the quadrilateral, because the plane cuts the four slant edges in the cyclic order EB, EA, ED, EC. Even pairing the points differently does not help. $B + P = Q + R$ and $B + Q = P + R$ would both need $a = -\gamma a$ in the y-coordinate, which is impossible since $\gamma > 0$.

A geometric way to see the same thing: BP lies in face ABE and QR lies in face CDE. These two faces meet in the line through E parallel to AB. Two parallel lines lying in these two planes would have to be parallel to that common line. So $BP \parallel QR$ would force $BP \parallel AB$, which is impossible since $P \neq A$.

Checks

- Symmetric case $\lambda = \mu$: the formula gives $\gamma = \lambda/(2 - \lambda)$. With $\lambda = \mu = \frac{1}{2}$ this is $\frac{1}{3}$. Directly: $p = 0$, $r = 2$ and the plane is $y + 2z = a$, which meets DE at $(0, -\frac{1}{3}a, \frac{2}{3}a)$, as predicted.
- Limit $\lambda, \mu \to 1$: $P \to A$, $Q \to C$, the plane becomes the base, and $\gamma \to 1$, so $R \to D$, as it should.
- A neat rewriting is $\dfrac{1-\gamma}{\gamma} = \dfrac{1-\lambda}{\lambda} + \dfrac{1-\mu}{\mu}$, which is easy to test on numbers. For example, $\lambda = \frac{1}{2}$, $\mu = \frac{2}{5}$ gives $\gamma = \frac{2}{7} \approx 0.2857$.
- We confirmed the formula, $0 < \gamma < 1$ and the parallelogram claim exactly (with fractions) on 2000 random choices of a, λ, μ. We found R independently, via a cross-product normal to the plane.

Common slips

- Parametrising AE wrongly: on AE the coordinates satisfy $x + z = a$ and $y = 0$, so P's z-coordinate is $(1 - \lambda)a$, not λa.
- Testing only that $BP \parallel QR$ by slopes in a 2D picture. The quadrilateral is in a tilted plane, so work with 3D vectors.
- Forgetting to show $\gamma < 1$, which is exactly what the parallelogram argument needs.

Beyond STEP

Finding where a plane cuts a line is the simplest case of solving a linear system. Since $EP : PA = (1 - \lambda) : \lambda$, and similarly on the other edges, the result says

$$\frac{ER}{RD} = \frac{EP}{PA} + \frac{EQ}{QC},$$

a coordinate-free statement about any plane through B. Because it uses only ratios along lines, it holds for every pyramid on a parallelogram base: an affine map takes any such pyramid to this one. Strang's *Introduction to Linear Algebra* develops planes, normals and linear systems properly.

WORKED SOLUTION

STEP III 1993, Question 9

Pure • Sums of Squares Inequalities

STEP III 1993, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q9

The key idea

Every inequality here is a disguised statement that a sum of squares is non-negative. Part (i) is the simplest case, $(a_1 - a_2)^2 \geq 0$, and the pattern for the later parts is to add up $(a_i - a_j)^2$ over *all* pairs and count how often each square and each product appears. Once the general version (iv) is in hand, (v) follows by expanding $(\sum a_i)^2$ into squares plus cross terms.

Throughout we write

$$S = \sum_{i=1}^{n} a_i^2, \qquad P = \sum_{i<j} a_i a_j,$$

where the second sum runs over the $\binom{n}{2}$ pairs with $1 \leq i < j \leq n$.

Solution

Part (i)

Since the square of a real number is non-negative,

$$0 \leq (a_1 - a_2)^2 = a_1^2 + a_2^2 - 2a_1a_2,$$

so $a_1^2 + a_2^2 \geq 2a_1a_2$, with equality exactly when $a_1 = a_2$.

Part (ii)

Apply (i) to each of the three pairs and add:

$$\begin{aligned} a_2^2 + a_3^2 &\geq 2a_2a_3, \\ a_3^2 + a_1^2 &\geq 2a_3a_1, \\ a_1^2 + a_2^2 &\geq 2a_1a_2. \end{aligned}$$

Each square appears in exactly two of these lines, so adding gives

$$2(a_1^2 + a_2^2 + a_3^2) \geq 2(a_2a_3 + a_3a_1 + a_1a_2),$$

and dividing by 2 gives the result. Equivalently, $a_1^2 + a_2^2 + a_3^2 - (a_2a_3 + a_3a_1 + a_1a_2) = \frac{1}{2}\left[(a_1 - a_2)^2 + (a_2 - a_3)^2 + (a_3 - a_1)^2\right] \geq 0$, with equality exactly when $a_1 = a_2 = a_3$.

Part (iii)

Now there are six pairs (i, j) with $1 \leq i < j \leq 4$. Apply (i) to each:

$$\sum_{1 \leq i < j \leq 4} (a_i^2 + a_j^2) \;\geq\; 2 \sum_{1 \leq i < j \leq 4} a_i a_j.$$

On the left, how often does a_1^2 appear? Once for every pair containing the index 1, that is, with each of the three other indices. The same holds for every a_k^2, so the left-hand side is $3(a_1^2 + a_2^2 + a_3^2 + a_4^2)$, and we have

$$3(a_1^2 + a_2^2 + a_3^2 + a_4^2) \geq 2(a_1a_2 + a_1a_3 + a_1a_4 + a_2a_3 + a_2a_4 + a_3a_4),$$

as required.

Part (iv)

The pattern in (i)–(iii) is that the coefficient on the left is one less than the number of variables: 1, then 2 (part (ii) multiplied by 2), then 3.

Claim. For any $n \geq 2$ and real numbers $a_1, \ldots, a_n$,

$$(n-1) \sum_{i=1}^{n} a_i^2 \;\geq\; 2 \sum_{1 \leq i < j \leq n} a_i a_j, \qquad \text{that is,} \qquad (n-1)S \geq 2P,$$

with equality if and only if $a_1 = a_2 = \cdots = a_n$.

> **Ask yourself** When we add up $a_i^2 + a_j^2$ over every pair $i < j$, how many times does a particular a_k^2 turn up?

Proof. Consider the sum of the squared differences over all pairs:

$$D = \sum_{1 \leq i < j \leq n} (a_i - a_j)^2 = \sum_{1 \leq i < j \leq n} \left(a_i^2 + a_j^2\right) - 2 \sum_{1 \leq i < j \leq n} a_i a_j.$$

The index k belongs to exactly $n - 1$ pairs (it is paired with each of the other $n - 1$ indices), so a_k^2 appears $n - 1$ times in the first sum. Hence

$$D = (n-1)S - 2P.$$

Each term of D is a real square, so $D \geq 0$, which gives $(n-1)S \geq 2P$. Equality holds if and only if every $(a_i - a_j)^2$ is zero, that is, if and only if all the a_i are equal. ■

Putting $n = 2, 3, 4$ recovers (i), twice (ii), and (iii).

Part (v)

Here we need $n \geq 2$ so that $n-1>0$. Expanding the square of the sum, every square appears once and every product $a_i a_j$ with $i \neq j$ appears twice (as $a_i a_j$ and $a_j a_i$), so

$$\left(\sum_{i=1}^{n} a_i\right)^2 = S + 2P.$$

Ask yourself How can (iv), which bounds S, help with a bound that only mentions P?

Since $n-1>0$, part (iv) can be divided by $n-1$ without reversing the inequality, giving $S \geq \dfrac{2P}{n-1}$. Substituting this lower bound for S:

$$\begin{aligned}\left(\sum_{i=1}^{n} a_i\right)^2 &= S + 2P \\ &\geq \frac{2P}{n-1} + 2P \\ &= \frac{2P + 2(n-1)P}{n-1} \\ &= \frac{2n}{n-1} \sum_{1 \leq i < j \leq n} a_i a_j,\end{aligned}$$

which is the required inequality. Equality holds exactly when it holds in (iv), that is, when all the a_i are equal.

Note that we never needed P to be positive: the argument only adds $2P$ to both sides of $S \geq 2P/(n-1)$, which is valid whatever the sign of P.

Checks

- All equal: with every $a_i = c$, we get $S = nc^2$ and $P = \binom{n}{2}c^2 = \frac{1}{2}n(n-1)c^2$, so $(n-1)S = n(n-1)c^2 = 2P$, and in (v) both sides equal n^2c^2. Both inequalities are tight, as they should be.
- A numerical test of (iii): $(a_1, a_2, a_3, a_4) = (1, 2, -3, 5)$ gives $S = 39$ and $P = -7$, so $3S = 117 \geq -14 = 2P$; and in (v), $(\sum a_i)^2 = 25 \geq \frac{8}{3}(-7) = -\frac{56}{3}$. (Also $S + 2P = 39 - 14 = 25$, as it should be.)
- We also tested the identity $D = (n-1)S - 2P$ and both inequalities exactly, with rational arithmetic, on 20 000 random lists of length 2 to 9, plus lists clustered very close to equality. No counterexamples appeared.

Common slips

- Miscounting: each a_k^2 appears $n-1$ times in $\sum_{i<j}(a_i^2+a_j^2)$, not n times and not $\binom{n}{2}$ times.
- Writing $\left(\sum a_i\right)^2 = S+P$: each cross term appears twice in the expansion.
- Dividing (iv) by $n-1$ without noting that $n-1>0$, or forgetting to state the equality case.

Beyond STEP

The quantity $D=(n-1)S-2P$ is n times the sum of squared deviations from the mean, $n\sum(a_i-\bar{a})^2$, so (iv) is really the statement that a variance is non-negative. Part (v) can also be read as the Cauchy–Schwarz inequality $\left(\sum a_i \cdot 1\right)^2 \le n\sum a_i^2$ in disguise. In linear algebra this is a statement that the quadratic form $(n-1)S-2P$ is positive semi-definite, with the all-ones vector spanning its kernel. Strang's *Introduction to Linear Algebra* covers quadratic forms and positive definiteness, and Siklos's *Advanced Problems in Mathematics* has more STEP-style inequality practice.

WORKED SOLUTION

STEP III 1993, Question 10

Pure • Matrix Transformations

STEP III 1993, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

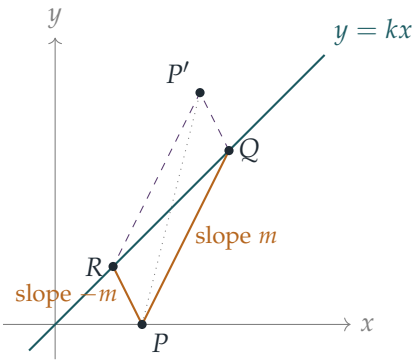
step.maths.org/questions/93-s3-q10

The key idea

Everything here is the intersection of two straight lines, so the coordinates come from solving one linear equation at a time. The parallelogram gives P' with no further work: in a parallelogram $PQP'R$ the diagonals bisect each other, so $P + P' = Q + R$ as position vectors. Once P' is written as a matrix times P, the area question is just a determinant.

Solution

Throughout we need $m^2 \neq k^2$: if $m = \pm k$, one of the two lines through P is parallel to $y = kx$ and never meets it (or lies along it), so the construction fails.



(The picture uses $m = 2$, $k = 1$ and $P = (1, 0)$.)

The point Q

The line through $P(x_1, y_1)$ parallel to $y = mx$ is $y - y_1 = m(x - x_1)$. At Q we also have $y = kx$, so

$$kx - y_1 = mx - mx_1 \quad \Longrightarrow \quad (m-k)x = mx_1 - y_1 \quad \Longrightarrow \quad x = \frac{mx_1 - y_1}{m-k},$$

dividing by $m-k \neq 0$. Then $y = kx$ gives

$$Q = \left(\frac{mx_1 - y_1}{m-k},\ \frac{k(mx_1 - y_1)}{m-k} \right),$$

as required.

The point R

The line through P parallel to $y = -mx$ is the same calculation with m replaced by $-m$:

$$x = \frac{-mx_1 - y_1}{-m-k} = \frac{mx_1 + y_1}{m+k}, \qquad R = \left(\frac{mx_1 + y_1}{m+k},\ \frac{k(mx_1 + y_1)}{m+k} \right).$$

The point P'

> **Ask yourself** How do we find the fourth vertex of a parallelogram without any more intersecting lines?

The vertices go round in the order P, Q, P', R, so PP' and QR are the diagonals. The diagonals of a parallelogram bisect each other, so they have the same midpoint:

$$\tfrac{1}{2}(P + P') = \tfrac{1}{2}(Q + R) \quad \Longrightarrow \quad P' = Q + R - P.$$

(Equivalently, $\overrightarrow{QP'} = \overrightarrow{PR}$.) Both Q and R lie on $y = kx$, so it is easiest to add their x-coordinates first. Over the common denominator $m^2 - k^2$,

$$\begin{aligned} x_Q + x_R &= \frac{(mx_1 - y_1)(m+k) + (mx_1 + y_1)(m-k)}{m^2 - k^2} \\ &= \frac{(m^2x_1 + mkx_1 - my_1 - ky_1) + (m^2x_1 - mkx_1 + my_1 - ky_1)}{m^2 - k^2} \\ &= \frac{2(m^2x_1 - ky_1)}{m^2 - k^2}, \end{aligned}$$

and $y_Q + y_R = k(x_Q + x_R)$. Therefore, writing $P' = (x', y')$,

$$\begin{aligned} x' &= \frac{2(m^2x_1 - ky_1)}{m^2 - k^2} - x_1 = \frac{(m^2 + k^2)x_1 - 2ky_1}{m^2 - k^2}, \\ y' &= \frac{2k(m^2x_1 - ky_1)}{m^2 - k^2} - y_1 = \frac{2km^2x_1 - (m^2 + k^2)y_1}{m^2 - k^2}. \end{aligned}$$

T as a matrix

Both coordinates are linear in x_1, y_1 with no constant term, so

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \mathbf{M} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \qquad \mathbf{M} = \frac{1}{m^2 - k^2} \begin{pmatrix} m^2 + k^2 & -2k \\ 2km^2 & -(m^2 + k^2) \end{pmatrix}.$$

Areas

A linear map with matrix $\mathbf{M}$ multiplies every area by $|\det \mathbf{M}|$. Taking the factor $\frac{1}{m^2-k^2}$ out of each row,

$$\det \mathbf{M} = \frac{-(m^2+k^2)^2 + 4k^2m^2}{(m^2-k^2)^2} = \frac{-(m^4 - 2m^2k^2 + k^4)}{(m^2-k^2)^2} = \frac{-(m^2-k^2)^2}{(m^2-k^2)^2} = -1.$$

So $|\det \mathbf{M}| = 1$ and every region is mapped to a region of the same area, as required. The minus sign tells us that T reverses orientation, as a reflection does: a figure traced anticlockwise has an image traced clockwise.

? **Ask yourself** Is there a picture of what T is doing?

Yes. The midpoint of PP' equals the midpoint of QR, which lies on $y = kx$, and points of $y = kx$ are fixed (there $Q = R = P$). Solving $\mathbf{Mv} = -\mathbf{v}$ gives the row $2m^2x - 2ky = 0$, so the vectors reversed by T are the multiples of (k, m^2). Thus T is a *skew reflection*: it reflects in the line $y = kx$, but along the direction (k, m^2) rather than perpendicular to the line. Like any reflection it satisfies $\mathbf{M}^2 = \mathbf{I}$, which also fits the construction: starting from P' gives the same Q and R, and so leads back to P.

Checks

- With $m = 2$, $k = 1$, $P = (1,0)$: $Q = (2,2)$, $R = (\frac{2}{3}, \frac{2}{3})$, so $P' = (\frac{5}{3}, \frac{8}{3})$. The matrix is $\frac{1}{3}\begin{pmatrix} 5 & -2 \\ 8 & -5 \end{pmatrix}$, which sends $(1,0)$ to $(\frac{5}{3}, \frac{8}{3})$, and its determinant is $\frac{1}{9}(-25+16) = -1$.
- A point (t, kt) on the line should be fixed: $x' = \frac{(m^2+k^2)t - 2k^2t}{m^2-k^2} = t$. ✓
- The trace of $\mathbf{M}$ is 0 and its determinant is -1, so its eigenvalues are ± 1, as expected for a reflection.
- We checked all of these formulas, together with the preservation of triangle areas, exactly in rational arithmetic on about 3000 random choices of m, k and P.

Common slips

- Taking $P' = Q + R$, or $P' = P + Q + R$, instead of $Q + R - P$. The order of the letters $PQP'R$ tells you which pairs of vertices are opposite.
- Concluding from $\det \mathbf{M} = -1$ that areas become negative. The area scale factor is $|\det \mathbf{M}|$; the sign only records orientation.

Beyond STEP

A linear map that fixes a line pointwise and reverses one other direction is an *involution*, $T^2 = I$. Ordinary orthogonal reflections are the special case in which the reversed

direction is perpendicular to the line; here that happens exactly when (k, m^2) is perpendicular to $(1, k)$, that is, only when $k = 0$ (the lines $y = \pm mx$ are symmetric about the x-axis, not about $y = kx$). At university this sits inside the study of eigenvalues and diagonalisation, where T is seen as $\mathrm{diag}(1, -1)$ in the basis $(1, k)$, (k, m^2), and of the determinant as a signed volume scale factor. Strang's *Introduction to Linear Algebra* covers both ideas clearly.

WORKED SOLUTION

STEP III 1993, Question 11

Mechanics • Couples and Moments of Inertia

STEP III 1993, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q11

The key idea

Put coordinates on the cube and write every force as a vector. The six forces add up to zero, so all they do is apply a *couple*, and a couple has the same moment about every point. Part (i) is then about the direction of that moment. Part (ii) uses only its component along OO', together with a moment of inertia that we build up rod by rod using the distance from a point to a line.

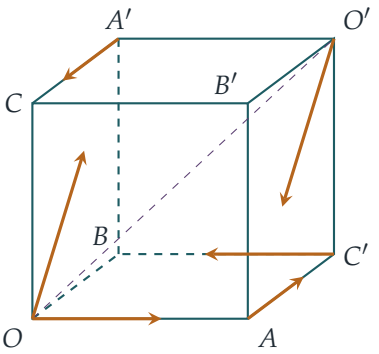
Solution

We model the frame as rigid, made of thin uniform rods, with no gravity. Take O as origin and unit vectors along OA, OB, OC, so that

$$A = (a,0,0), \quad B = (0,a,0), \quad C = (0,0,a),$$

and the opposite vertices are

$$O' = (a,a,a), \quad A' = (0,a,a), \quad B' = (a,0,a), \quad C' = (a,a,0).$$



The amber arrows show the six lines of action, and the diagonal OO' is dashed in plum.

Part (i)

Each force is its magnitude times the unit vector along its line. For example, $O'A$ is the vector $(0, -a, -a)$, of length $\sqrt{2}\,a$, so the force $\sqrt{2}\,mg$ along it is $mg(0, -1, -1)$. Doing the same for all six:

line	point on line	force
OA	O	$mg(2, 0, 0)$
AC'	A	$mg(0, 1, 0)$
$O'A$	A	$mg(0, -1, -1)$
OA'	O	$mg(0, 1, 1)$
$C'B$	B	$mg(-2, 0, 0)$
$A'C$	C	$mg(0, -1, 0)$

The forces sum to $\mathbf{0}$. A system with zero resultant is equivalent to a couple, whose moment is the same about every point. Taking moments about O (the forces along OA and OA' pass through O and contribute nothing):

$$\begin{aligned}\mathbf{G} &= A \times mg(0,1,0) + A \times mg(0,-1,-1) + B \times mg(-2,0,0) + C \times mg(0,-1,0)\\ &= amg\big[(0,0,1) + (0,1,-1) + (0,0,2) + (1,0,0)\big]\\ &= amg\,(1,1,2).\end{aligned}$$

The pivot at O can push on the frame, but its force acts through O and so has no moment about O. Starting from rest, the frame therefore begins to turn about the line through O along the axis of the couple, which has direction

$$\begin{pmatrix}1\\1\\2\end{pmatrix},$$

as required.

A note for the curious. This is the answer the question intends, and it's what you should write. Strictly, though, for a body free to turn in three dimensions the angular acceleration is $\mathbf{I}^{-1}\mathbf{G}$, where $\mathbf{I}$ is the inertia *tensor* about O. This points along $\mathbf{G}$ only when $\mathbf{G}$ lies along a principal axis. For this frame, about O, the moment of inertia is $14ma^2/3$ about OO' (from part (ii)) and $41ma^2/3$ about every axis through O perpendicular to OO'. This gives an initial axis in direction $(25, 25, 32)$, about 12.6° away from $(1, 1, 2)$.

Part (ii)

Let $\mathbf{u} = \frac{1}{\sqrt{3}}(1,1,1)$ be the unit vector along OO'. A point $\mathbf{p}$ is at squared distance

$$d^2 = |\mathbf{p}|^2 - (\mathbf{p}\cdot\mathbf{u})^2$$

from the line OO', by Pythagoras, since $\mathbf{p}\cdot\mathbf{u}$ is the component of $\mathbf{p}$ along the line.

Rod OA. The point at distance x from O is $(x, 0, 0)$, so $d^2 = x^2 - \frac{x^2}{3} = \frac{2}{3}x^2$. An element of length $\mathrm{d}x$ has mass $\frac{m}{a}\,\mathrm{d}x$, so

$$I_{OA} = \int_0^a \frac{m}{a}\cdot\frac{2x^2}{3}\,\mathrm{d}x = \frac{2ma^2}{9}.$$

The midpoint $(\frac{a}{2}, 0, 0)$ is at squared distance $\frac{2}{3} \cdot \frac{a^2}{4} = \frac{a^2}{6}$ from OO'. By the parallel axis theorem, about the parallel axis through the midpoint,

$$I_{\text{mid}} = \frac{2ma^2}{9} - m \cdot \frac{a^2}{6} = \frac{ma^2}{18}.$$

Rod $B'C$. This rod runs from $(a, 0, a)$ to $(0, 0, a)$, so it is parallel to OA. It is also a rod of the same length, placed at the same angle to the axis. So about the parallel axis through its own midpoint its moment of inertia is again $ma^2/18$. Its midpoint is $\mathbf{p} = (\frac{a}{2}, 0, a)$, with

$$d^2 = \frac{a^2}{4} + a^2 - \frac{1}{3}\left(\frac{3a}{2}\right)^2 = \frac{5a^2}{4} - \frac{3a^2}{4} = \frac{a^2}{2}.$$

By the parallel axis theorem,

$$I_{B'C} = \frac{ma^2}{18} + \frac{ma^2}{2} = \frac{5ma^2}{9}.$$

Ask yourself Do I really have to do this for all twelve rods?

No: symmetry does it for us. A rotation of 120° about OO' (cycling $x \to y \to z \to x$) and the point reflection in the centre of the cube, $\mathbf{p} \mapsto (a, a, a) - \mathbf{p}$, both map the frame and the line OO' to themselves. So they don't change any moment of inertia about OO'. The rotation carries OA to OB and OC, and the reflection carries these to $O'A'$, $O'B'$ and $O'C'$. These are the six rods with an end at O or O', and each has moment of inertia $2ma^2/9$. In the same way, $B'C$ is carried to the other five rods of the "zig-zag" AB', $B'C$, CA', $A'B$, BC', $C'A$ round the middle of the cube, and each of these has moment of inertia $5ma^2/9$. Hence

$$I_{\text{frame}} = 6 \cdot \frac{2ma^2}{9} + 6 \cdot \frac{5ma^2}{9} = \frac{42ma^2}{9} = \frac{14ma^2}{3},$$

as required.

Angular acceleration. Now the frame can only turn about the fixed line OO'. The bearings supply whatever couple is needed perpendicular to that line, so only the component of the applied couple along OO' turns the frame. The forces still form the couple $\mathbf{G} = amg(1, 1, 2)$, and its moment about any line is its component along that line:

$$\mathbf{G} \cdot \mathbf{u} = \frac{amg(1 + 1 + 2)}{\sqrt{3}} = \frac{4amg}{\sqrt{3}}.$$

The equation $I\ddot{\theta} = \mathbf{G} \cdot \mathbf{u}$ for rotation about a fixed axis gives

$$\ddot{\theta} = \frac{4amg/\sqrt{3}}{14ma^2/3} = \frac{12g}{14\sqrt{3}\,a} = \frac{2\sqrt{3}\,g}{7a}.$$

Its sense is anticlockwise when viewed looking from O' towards O, that is, a right-handed rotation about the direction $\overrightarrow{OO'}$.

Checks

- Each of the six forces has a "partner": OA with $C'B$, AC' with $A'C$, and $O'A$ with OA'. Each pair is equal and opposite on parallel lines, which is why the resultant is zero.
- The frame has cubic symmetry about its centre, so its moment of inertia is the same about every axis through the centre. About the axis through the centre parallel to OC, the four vertical rods contribute $4m(a/\sqrt{2})^2 = 2ma^2$. The eight horizontal rods each contribute $\frac{ma^2}{12} + \frac{ma^2}{4} = \frac{ma^2}{3}$. The total is $\frac{14ma^2}{3}$, agreeing with the answer above, since OO' also passes through the centre.
- Numerically, with $a = m = g = 1$, midpoint sums over the twelve rods give $I = 4.66667$ and $\ddot{\theta} = 0.494872$, matching $2\sqrt{3}/7 \approx 0.494872$.

Common slips

- Forgetting the factor $1/\sqrt{2}$ when converting the face-diagonal forces $\sqrt{2}\,mg$ into components: each has components of size mg, not $\sqrt{2}\,mg$.
- Using the full couple $|\mathbf{G}| = \sqrt{6}\,amg$ in part (ii) instead of its component along the fixed axis.
- Applying the parallel axis theorem from an axis that doesn't pass through the centre of mass: it must always go *via* the centre of mass, as in the step from $2ma^2/9$ to $ma^2/18$.

Beyond STEP

For rotation in three dimensions, the moment of inertia becomes a symmetric 3×3 matrix, the inertia tensor, with angular momentum $\mathbf{L} = \mathbf{I}\boldsymbol{\omega}$. Its eigenvectors are the principal axes. As the note in part (i) shows, a body does not in general start to turn about the axis of the applied couple. The cube frame is a nice example: it has the same moment of inertia about every axis through its centre, but this is not true about a corner. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid body motion in three dimensions, including the inertia tensor.

WORKED SOLUTION

STEP III 1993, Question 12

Mechanics • Elastic Strings and Energy

STEP III 1993, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

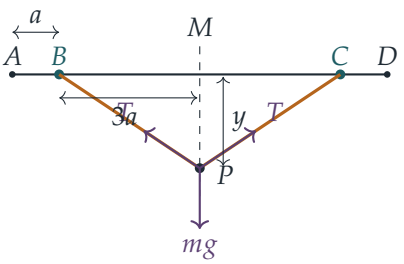
step.maths.org/questions/93-s3-q12

The key idea

Because the pegs are smooth and the particle sits at the midpoint, the string splits into two identical halves, each of natural length a, with the same tension all along each half. Each half runs from a fixed end, over a peg, down (or up) to P, so its stretched length depends only on the distance BP. Once we write the extension in terms of the height of P, equilibrium gives k and conservation of energy links Q to R.

Solution

We model P as a particle, the string as light and elastic, and the pegs as smooth. By symmetry P moves on the vertical line through the midpoint M of BC. Let y be the depth of P below M; it is negative when P is above BC.



Finding k

The half $A \to B \to P$ has length $a + BP = a + \sqrt{9a^2 + y^2}$ and natural length a, so its extension is

$$x = \sqrt{9a^2 + y^2}.$$

By Hooke's law the tension in each half is

$$T = \frac{kmg\,x}{a}.$$

If θ is the angle BP makes with the horizontal, then $\sin\theta = |y|/x$. Resolving vertically at P gives

$$2T\sin\theta = mg \quad\Longrightarrow\quad 2\cdot\frac{kmg\,x}{a}\cdot\frac{y}{x} = mg \quad\Longrightarrow\quad \frac{2ky}{a} = 1.$$

The square root has cancelled, so the ugly value $BP = \frac{\sqrt{145}}{4}a$ never matters. With $y = a/4$ we get $k/2 = 1$, so

$$k = 2.$$

Energy from Q to R

Each half stores elastic energy $\dfrac{kmg\,x^2}{2a}$, so with $k = 2$ the two halves together store

$$E(y) = 2\cdot\frac{2mg\,x^2}{2a} = \frac{2mg\,(9a^2+y^2)}{a}.$$

Ask yourself Why does P stop at R, and why is it fine that P ends up above the pegs?

P is released from rest at Q, and R is the highest point it rises to, so it is momentarily at rest there too. Above BC the halves are still stretched (each still has $x \geq 3a > 0$), so the same formula for E applies. The string never goes slack.

At Q, $y = pa$; at R, $y = -3a$. The particle rises $(p+3)a$. Since the pegs are smooth and there is no other loss, the kinetic energy is zero at both ends, so

$$\begin{aligned} E(pa) &= E(-3a) + mg(p+3)a \\ 2mga(9+p^2) &= 2mga(9+9) + mga(p+3) \\ 18 + 2p^2 &= 36 + p + 3, \end{aligned}$$

which gives

$$2p^2 - p - 21 = 0, \qquad \text{that is,} \qquad (2p+6)\left(p-\tfrac{7}{2}\right) = 0.$$

Since $p > 0$, we get $p = \frac{7}{2}$.

This does *not* agree with the printed $2p^2 - p - 17 = 0$. We have checked the model carefully (see Checks), including a direct numerical simulation, and -21 is what the stated data give. The clean root $p = \frac{7}{2}$ suggests that 21 was the intended constant and that 17 is a misprint. The final part does not depend on which constant is right.

To confirm that P really does stop for the first time at R, write $y = ta$. The kinetic energy at depth ta is

$$E(pa) - E(ta) - mg(p-t)a = mga\,(p-t)(2p+2t-1).$$

With $p = \frac{7}{2}$ this is $mga\,(\frac{7}{2}-t)(2t+6)$. It is positive for $-3 < t < \frac{7}{2}$ and first vanishes at $t = -3$, which is R.

Comparing the tensions

The tension is $T = 2mg\,x/a$, which increases with x, so we need to show that x is smaller at R than at Q. The energy equation does this without using the value of p:

$$E(Q) - E(R) = mg(p+3)a > 0.$$

The elastic energy is $\dfrac{2mg\,x^2}{a}$, so $x_R^2 < x_Q^2$, giving $x_R < x_Q$ and hence $T_R < T_Q$. In words, the particle gains potential energy on the way up, and that energy can only come out of the string, so the string must be less stretched at R.

Explicitly, with $p = \frac{7}{2}$, $x_Q = \frac{\sqrt{85}}{2}a$ and $x_R = 3\sqrt{2}\,a$, so

$$T_Q = \sqrt{85}\,mg \approx 9.22\,mg, \qquad T_R = 6\sqrt{2}\,mg \approx 8.49\,mg.$$

Checks

- *SHM check.* The upward force on P is $2T \cdot \dfrac{y}{x} = \dfrac{4mg\,y}{a}$, which is linear in y, above BC as well as below. So P moves in simple harmonic motion about $y = a/4$ (the equilibrium found above). Released from rest at $y = \frac{7}{2}a$, its amplitude is $\frac{13}{4}a$, so its highest point is $y = \frac{1}{4}a - \frac{13}{4}a = -3a$. This is exactly R, which confirms $p = \frac{7}{2}$ independently.
- A numerical integration of the equation of motion with $a = 1$, $k = 2$, $p = 3.5$ reaches a highest point of $y = -3.0000$.
- The printed equation would give $p = \frac{1+\sqrt{137}}{4} \approx 3.18$, and the energy balance then fails by $4mga$.

Common slips

- Leaving out the segment AB when computing the length of each half. The half is $AB + BP = a + BP$ long, and its extension is BP only because AB equals its natural length a.
- Treating the whole string as one spring with tension $kmg \cdot (\text{total extension})/(2a)$. This does give the same answer, but only if the extension is the *total* one, $2x$.
- Forgetting the elastic energy is still there when P is above BC. The string is still stretched there.

Beyond STEP

The vertical restoring force came out exactly proportional to displacement, even though the string geometry is nonlinear. This is because the vertical component of a tension proportional to length is proportional to the vertical separation, the same reason a

zero-natural-length spring makes an ideal suspension (the principle behind the LaCoste long-period seismometer). Potential energy that is quadratic in the coordinates leads to linear equations of motion, and this is the starting point of the theory of small oscillations and normal modes. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats springs, energy and oscillations in this spirit.

WORKED SOLUTION

STEP III 1993, Question 13

Mechanics • Variable Mass Chain

STEP III 1993, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q13

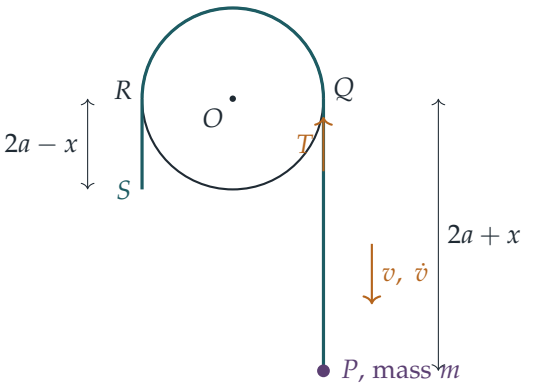
The key idea

Because the chain cannot slip, the disc, the whole chain and the particle all move at the same speed: there is a single unknown, v. The kinetic energy is then $\frac{1}{2}(\text{effective mass})v^2$. In the potential energy only the two hanging pieces change, since the arc over the top just keeps being replaced by identical chain. For the tension, look at the moving piece PQ together with the particle. Then think hard about how fast the chain that joins it at Q is already moving.

Solution

Model

The disc is a uniform lamina turning about a smooth fixed axis through O, with moment of inertia $\frac{1}{2}(4m)a^2 = 2ma^2$. The chain is uniform with mass m/a per unit length, is inextensible, and does not slip on the rim. The particle is a point mass. When P has descended x, the right-hand piece QP has length $2a + x$, the left-hand piece RS has length $2a - x$, and the arc RQ is a semicircle of length πa. Every point of the chain moves at speed v, and since there is no slipping the rim does too, so the disc's angular speed is $\omega = v/a$.



The speed

Kinetic energy. The disc contributes $\frac{1}{2}(2ma^2)(v/a)^2 = mv^2$, the chain $\frac{1}{2}(4+\pi)mv^2$, and the particle $\frac{1}{2}mv^2$. So

$$\text{KE} = \tfrac{1}{2}mv^2\big(2+4+\pi+1\big) = \tfrac{1}{2}(\pi+7)mv^2.$$

Potential energy. The arc RQ is always the same semicircle of chain, so its potential energy never changes. Take the level of O as zero. A hanging piece of length ℓ below that level has mass $m\ell/a$ and its centre of mass at depth $\frac{1}{2}\ell$, so its potential energy is $-\dfrac{mg\ell^2}{2a}$. The loss of potential energy of the two hanging pieces is therefore

$$\frac{mg}{2a}\Big[(2a+x)^2+(2a-x)^2-(2a)^2-(2a)^2\Big] = \frac{mg}{2a}\cdot 2x^2 = \frac{mgx^2}{a}.$$

The particle loses mgx.

Energy. The axis is smooth and the rough contact does no work, because there is no slipping. So the energy is conserved, and starting from rest

$$\tfrac{1}{2}(\pi+7)mv^2 = \frac{mgx^2}{a} + mgx \quad\Longrightarrow\quad (\pi+7)av^2 = 2g(x^2+ax),$$

as required.

The acceleration

Differentiating with respect to time and using $\dot{x} = v$:

$$2(\pi+7)av\dot{v} = 2g(2x+a)v \quad\Longrightarrow\quad \dot{v} = \frac{g(2x+a)}{(\pi+7)a} \qquad (v \neq 0).$$

S reaches R when $x = 2a$, and then

$$\dot{v} = \frac{5g}{\pi+7}, \qquad v^2 = \frac{2g(4a^2+2a^2)}{(\pi+7)a} = \frac{12ga}{\pi+7}.$$

The tension at Q

Let T be the tension at Q. Our system is the particle together with the chain hanging below Q. Its mass is

$$M = \frac{m}{a}(2a + x) + m,$$

which increases at the rate $\dot{M} = \dfrac{m}{a}v$ as chain comes off the disc. The external forces on it are its weight Mg downwards and T upwards. All of it moves downwards at speed v.

Ask yourself How fast is the chain that is joining PQ moving just before it joins?

At Q the rim is vertical, so the chain arriving there is already moving vertically downwards at speed v. It brings its momentum $v\,\delta M$ with it. The variable-mass equation (rate of change of momentum = force + momentum brought in per unit time) is then

$$\frac{\mathrm{d}}{\mathrm{d}t}(Mv) = Mg - T + v\dot{M}, \qquad \text{that is,} \qquad M\dot{v} = Mg - T.$$

When S reaches R, $M = 5m$, so

$$T = 5m\Big(g - \frac{5g}{\pi + 7}\Big) = \frac{5(\pi + 2)}{\pi + 7}\,mg.$$

A cross-check by moments. Take the disc and the arc RQ of chain together. The axle force acts through O, the arc's weight has zero total moment about O (the arc is symmetric about the vertical through O), and the friction between chain and disc is internal. Every particle of the arc moves at speed v at distance a from O. So the angular momentum is $(2ma^2 + \pi ma^2)(v/a)$, and

$$(T - T_R)a = (2 + \pi)ma\,\dot{v}.$$

When S reaches R the left-hand piece has shrunk to nothing, so $T_R = 0$. This gives $T = (2 + \pi)m \cdot \dfrac{5g}{\pi + 7} = \dfrac{5(\pi + 2)}{\pi + 7}mg$ again.

The printed answer. The paper's value $\dfrac{5\pi - 2}{\pi + 7}mg$ is exactly what we get by writing $\dfrac{\mathrm{d}}{\mathrm{d}t}(Mv) = Mg - T$, which treats the chain joining PQ as if it arrived at rest:

$$T = M(g - \dot{v}) - \frac{m}{a}v^2 = \frac{5(\pi + 2)}{\pi + 7}mg - \frac{12}{\pi + 7}mg = \frac{5\pi - 2}{\pi + 7}mg.$$

That is the right equation for a raindrop picking up stationary mist. It is not right here, because the chain coming round the rim is already moving with PQ. Both of our methods agree on $\dfrac{5(\pi + 2)}{\pi + 7}mg$, so we believe the printed answer contains this error. If you need the printed value, use the "joins at rest" equation, but know that it undercounts T by mv^2/a.

Checks

- At $x = 0$ the formula gives $v = 0$ and $\dot{v} = g/(\pi + 7)$. The out-of-balance force is just the particle's weight mg, acting on effective mass $(\pi + 7)m$.
- Stepping the equation of motion $(\pi + 7)m\dot{v} = mg(2x + a)/a$ numerically from rest reproduces $(\pi + 7)av^2 = 2g(x^2 + ax)$ to within 10^{-11}. With $m = a = 1$ and $g = 9.81$, the momentum method and the moments method both give $T/mg \approx 2.5349 = 5(\pi + 2)/(\pi + 7)$. The "joins at rest" version gives $1.3517 = (5\pi - 2)/(\pi + 7)$.
- $T < 5mg$, as it must be, since the hanging mass $5m$ is accelerating downwards.

Common slips

- Forgetting the disc's kinetic energy, or using ma^2 instead of $\frac{1}{2}(4m)a^2$ for its moment of inertia.
- Including the arc of chain in the change of potential energy. Its shape and position never change.
- Using $\frac{\mathrm{d}}{\mathrm{d}t}(Mv) = F$ without asking what velocity the incoming mass already had.

Beyond STEP

Variable-mass problems are really about choosing a system and keeping track of the momentum that crosses its boundary. That is the idea of a control volume, which underlies rocket motion and much of fluid mechanics. The common error here, assuming $F = \frac{\mathrm{d}}{\mathrm{d}t}(mv)$ holds for any changing mass, is a famous trap: in general that equation isn't even independent of the frame of reference. Kleppner and Kolenkow, *An Introduction to Mechanics*, treat systems of variable mass carefully.

WORKED SOLUTION

STEP III 1993, Question 14

Mechanics • Vertical Circular Motion

STEP III 1993, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

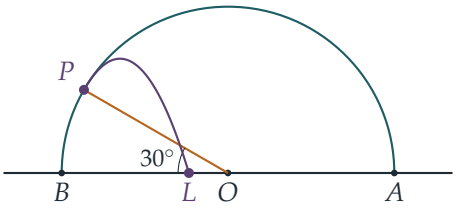
step.maths.org/questions/93-s3-q14

The key idea

Because the string is horizontal and the particle is projected vertically, the velocity starts perpendicular to the string, so the particle moves on a vertical circle of radius c about O, and it lands at the point B diametrically opposite A. Each impact is with the velocity vertical, so it simply halves the speed. We need to find the first bounce at which the particle can no longer get over the top of the circle, and then use energy and the radial equation to find where the tension vanishes. After that it is an ordinary projectile.

Solution

We model the particle as a particle of mass m, the string as light and inextensible, and the table as smooth and horizontal. Take O as origin, with $A = (c, 0)$ and $B = (-c, 0)$. Let the particle be at angle θ above the horizontal radius on whichever side it is rising.



Motion between impacts

Suppose the particle leaves the table at A or B moving vertically upwards with speed V. Its velocity is perpendicular to the taut string, so while the string stays taut it moves on the circle of radius c about O. By conservation of energy, at angle θ its speed v satisfies

$$v^2 = V^2 - 2gc\sin\theta.$$

Resolving towards O, with tension T,

$$T + mg\sin\theta = \frac{mv^2}{c} \quad \Longrightarrow \quad \frac{T}{m} = \frac{V^2}{c} - 3g\sin\theta.$$

For $0 \le \theta \le 90^\circ$ the tension is least at the top, so the string stays taut over the whole semicircle exactly when $V^2 \ge 3gc$. (Past the top, $\sin\theta$ decreases again, so T only grows.) In that case the particle comes down to the table at the opposite end of the diameter, moving vertically downwards with speed V again, by energy.

At that impact the velocity is normal to the table, so the rebound is vertical with speed $\frac{1}{2}V$, and the string is still horizontal and taut. So the whole process repeats with V halved.

Where the string goes slack

The particle starts with $V_0 = 64\sqrt{6gc}$, so $V_0^2 = 24576\,gc$. After k impacts the speed is $V_k = V_0/2^k$, so

$$V_k^2 = \frac{24576}{4^k}\,gc: \qquad V_6^2 = 6gc, \qquad V_7^2 = \tfrac{3}{2}gc.$$

So the semicircles starting with speeds $V_0, \ldots, V_6$ are all completed (each has $V_k^2 \ge 3gc$), giving impacts at B, A, B, A, B, A, B. The seventh impact is at B, and the particle rises from B with $V_7^2 = \frac{3}{2}gc < 3gc$.

> **Ask yourself** Can the particle fall back before the string goes slack, rather than the string going slack first?

No. The speed would vanish where $V_7^2 = 2gc\sin\theta$, that is $\sin\theta = \frac{3}{4}$, whereas the tension vanishes earlier, where

$$\frac{V_7^2}{c} = 3g\sin\theta \quad \Longrightarrow \quad \tfrac{3}{2}gc = 3gc\sin\theta \quad \Longrightarrow \quad \sin\theta = \tfrac{1}{2}.$$

So the string becomes slack at $\theta = 30^\circ$, at the point P on the B side, at height $c\sin\theta = c/2$ above the table, as required.

The first rebound between A and B

At P we have $\theta = 30^\circ$, so $P = \left(-\frac{\sqrt{3}}{2}c,\ \frac{1}{2}c\right)$ and

$$v^2 = \tfrac{3}{2}gc - 2gc\cdot\tfrac{1}{2} = \tfrac{1}{2}gc.$$

The velocity is along the tangent, perpendicular to OP, pointing up and towards O; it makes 60° with the horizontal. So its components are

$$v_x = \tfrac{1}{2}v, \qquad v_y = \tfrac{\sqrt{3}}{2}v, \qquad \text{with } v_x^2 = \tfrac{1}{8}gc, \quad v_y^2 = \tfrac{3}{8}gc.$$

From now on the particle is a projectile. It reaches the table when $\frac{1}{2}c + v_y t - \frac{1}{2}gt^2 = 0$; taking the positive root,

$$t = \frac{v_y + \sqrt{v_y^2 + gc}}{g} = \frac{\sqrt{3}+\sqrt{11}}{\sqrt{8}}\sqrt{\frac{c}{g}}.$$

Its horizontal displacement in this time is

$$v_x t = \sqrt{\frac{gc}{8}} \cdot \frac{\sqrt{3}+\sqrt{11}}{\sqrt{8}}\sqrt{\frac{c}{g}} = \frac{\sqrt{3}+\sqrt{11}}{8}\,c,$$

so it lands at the point L with

$$x_L = -\frac{\sqrt{3}}{2}c + \frac{\sqrt{3}+\sqrt{11}}{8}c = \frac{\sqrt{11}-3\sqrt{3}}{8}\,c \approx -0.235\,c.$$

(Throughout this flight the particle stays within distance c of O, so the string does not become taut again before L; in particular $|x_L| < c$.)

Since $\sqrt{11} < \sqrt{27} = 3\sqrt{3}$, we have $x_L < 0$: the landing point lies on the same side of O as B. As O is the midpoint of AB, the first rebound between A and B is **nearer to** B, at distance about $0.77c$ from B and $1.23c$ from A.

Checks

- The test $V_k^2 \geq 3gc$ picks out $k = 7$ as the first incomplete semicircle, and $V_7^2 = \frac{3}{2}gc$; the slack condition gives $\sin\theta = \frac{1}{2}$.
- At P the radial component of gravity, $g\sin 30^\circ = \frac{1}{2}g$, equals $v^2/c = \frac{1}{2}g$, just as zero tension requires.
- A numerical step-by-step simulation (with $g = 9.8$, $c = 1.3$) found the string going slack at height $0.5000c$, a landing point at $x \approx -0.2349c$, and a distance from O never exceeding c during the flight.

Common slips

- Using $V^2 \geq 4gc$ (just reaching the top) as the condition for completing the semicircle. A string, unlike a rod, needs $V^2 \geq 3gc$.
- Losing track of which end the seventh impact is at: odd impacts are at B.
- Giving the velocity at P the wrong direction: it is perpendicular to OP, at 60° to the horizontal, not 30°.

Beyond STEP

The moment when the string goes slack is a constraint switching off: the tension is the Lagrange multiplier that enforces $|OP| = c$, and when it would have to become

negative the constraint stops acting. Problems with one-sided constraints like this, and with impacts, are treated systematically in undergraduate mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see energy and radial equations used together in this way.

WORKED SOLUTION

STEP III 1993, Question 15

Probability and Statistics • Waiting Time Distribution

STEP III 1993, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q15

The key idea

Focus on the *last* throw. The experiment stops at throw n exactly when that throw supplies the second of whichever face was still short, so the first $n-1$ throws must contain exactly one of that face and all the rest of the other. That makes the distribution a sum of two simple terms. The expectation is then a pair of series of the form $\sum n(n-1)x^{n-2}$, which we get by differentiating a geometric series twice.

Solution

We assume $0 < p < 1$ (otherwise the experiment never ends) and that throws are independent.

The distribution of N

We need at least two heads and two tails, so $N \geq 4$. Fix $n \geq 4$. On the event $N = n$, the nth throw is either a head or a tail, and these two cases are disjoint.

The nth throw is a head. Before it, the target was not reached, and throw n reached it by adding a head. So the first $n-1$ throws already contained at least two tails but only one head: exactly one head and $n-2$ tails (and $n-2 \geq 2$ because $n \geq 4$). Conversely, any such sequence followed by a head stops at exactly n. The single head can be in any of $n-1$ positions, so this case has probability

$$(n-1)\,p\,q^{n-2} \cdot p = (n-1)p^2q^{n-2}.$$

The nth throw is a tail. By the same argument with heads and tails swapped, the probability is $(n-1)q^2p^{n-2}$.

Adding,

$$\boxed{\mathrm{P}(N=n) = (n-1)\left(p^2q^{n-2} + q^2p^{n-2}\right), \qquad n = 4,5,6,\ldots}$$

and $\mathrm{P}(N=n) = 0$ for $n \leq 3$.

Ask yourself Why can't the first $n-1$ throws contain two heads when the last throw is a head?

If they did, they would contain two heads and (as the stop needs) two tails already, so the experiment would have ended before throw n. This is what pins the count at exactly one.

The expectation

For $|x|<1$, differentiating $\sum_{n\geq 0} x^n = (1-x)^{-1}$ twice gives

$$\sum_{n=2}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3}.$$

Our sums start at $n=4$, so we remove the $n=2$ and $n=3$ terms, which are 2 and $6x$:

$$\sum_{n=4}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3} - 2 - 6x.$$

Now

$$\mathrm{E}(N) = \sum_{n=4}^{\infty} n\,\mathrm{P}(N=n) = p^2 \sum_{n=4}^{\infty} n(n-1)q^{n-2} + q^2 \sum_{n=4}^{\infty} n(n-1)p^{n-2}.$$

Using the series with $x=q$ (so $1-x=p$) and then with $x=p$ (so $1-x=q$), both valid since $0<p,q<1$:

$$\begin{aligned}\mathrm{E}(N) &= p^2\Big(\frac{2}{p^3} - 2 - 6q\Big) + q^2\Big(\frac{2}{q^3} - 2 - 6p\Big)\\ &= \frac{2}{p} + \frac{2}{q} - 2(p^2+q^2) - 6pq(p+q).\end{aligned}$$

Since $p+q=1$ we have $\frac{1}{p}+\frac{1}{q} = \frac{p+q}{pq} = \frac{1}{pq}$ and $p^2+q^2 = (p+q)^2 - 2pq = 1-2pq$. Hence

$$\begin{aligned}\mathrm{E}(N) &= \frac{2}{pq} - 2(1-2pq) - 6pq\\ &= \frac{2}{pq} - 2 - 2pq = 2\Big(\frac{1}{pq} - 1 - pq\Big),\end{aligned}$$

as required.

Checks

- The probabilities sum to 1: the same series with one fewer factor gives $\sum_{n\geq 4}(n-1)p^2q^{n-2} = 1 - p^2 - 2p^2q$, and adding the swapped version leaves $1-(p^2+q^2)-2pq(p+q) = 1-1+2pq-2pq = 1$, using $p^2+q^2 = 1-2pq$.

- For a fair coin, $pq = \frac{1}{4}$ and $\mathrm{E}(N) = 2(4 - 1 - \frac{1}{4}) = \frac{11}{2}$. Directly: $\mathrm{P}(N = 4) = \frac{3}{8}$, $\mathrm{P}(N = 5) = \frac{1}{4}$, …, and the partial sums of $n\,\mathrm{P}(N = n)$ approach 5.5.
- As $p \to 0$, $\mathrm{E}(N) \approx 2/p$: we wait for two heads, each taking about $1/p$ throws.
- Enumerating all sequences up to length 10 reproduces the distribution exactly, and a simulation with $p = 0.3$ gave a mean of 7.10, against 7.1038 from the formula.

Common slips

- Writing $\binom{n}{2}$ instead of $n - 1$: the last throw's position is fixed, so only the lone head (or tail) among the first $n - 1$ throws is free.
- Forgetting to subtract the $n = 2$ and $n = 3$ terms when using the standard series.

Beyond STEP

N is a mixture of two shifted *negative binomial* waiting times, and the trick of differentiating a geometric series is a hand-made version of a *probability generating function*. With $G(s) = \mathrm{E}(s^N)$ you would get the mean as $G'(1)$ and the variance from $G''(1)$ just as quickly. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions and waiting-time problems like this one.

WORKED SOLUTION

STEP III 1993, Question 16

Probability and Statistics • Truncated Normal Distribution

STEP III 1993, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q16

The key idea

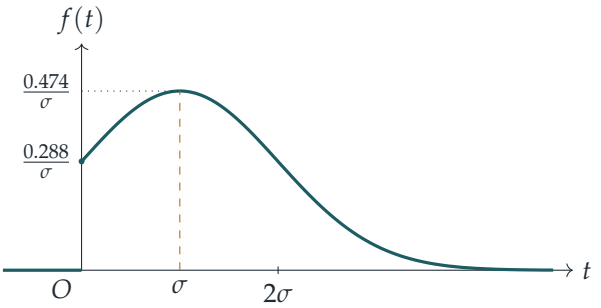
Apart from the factor $1/k$, the density is exactly that of a normal distribution with mean σ and standard deviation σ, cut off at $t = 0$. So every probability about T is a normal probability divided by the probability that the untruncated variable is non-negative. Standardising with $z = (t - \sigma)/\sigma$ turns everything into statements about $Z \sim \mathrm{N}(0,1)$ and the table of Φ.

Solution

Throughout, $Z \sim \mathrm{N}(0,1)$ with density $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and distribution function Φ.

The sketch

For $t \geq 0$ the graph is the normal bell curve centred at $t = \sigma$, scaled up by $1/k$. It has its maximum at $t = \sigma$, where $f(\sigma) = \frac{1}{k\sigma\sqrt{2\pi}} \approx \frac{0.474}{\sigma}$, and a point of inflexion at $t = 2\sigma$. At the cut-off the curve starts at a positive height, $f(0) = \frac{e^{-1/2}}{k\sigma\sqrt{2\pi}} \approx \frac{0.288}{\sigma}$, and it jumps to 0 for $t < 0$. As $t \to \infty$, $f(t) \to 0$.



The value of k

The total probability must be 1:

$$1 = \int_0^\infty \frac{1}{k\sigma\sqrt{2\pi}} \exp\left(-\tfrac{1}{2}\left(\tfrac{t-\sigma}{\sigma}\right)^2\right) \mathrm{d}t.$$

Substituting $z = (t-\sigma)/\sigma$, so $\mathrm{d}t = \sigma\,\mathrm{d}z$ and $t = 0$ becomes $z = -1$,

$$1 = \frac{1}{k}\int_{-1}^\infty \phi(z)\,\mathrm{d}z = \frac{1-\Phi(-1)}{k} = \frac{\Phi(1)}{k}.$$

Hence $k = \Phi(1) \approx 0.8413$, so $k \approx 0.841$, as required.

The mean

With the same substitution, $t = \sigma(1+z)$, and

$$\mathrm{E}(T) = \frac{1}{k}\int_{-1}^\infty \sigma(1+z)\,\phi(z)\,\mathrm{d}z = \sigma\left[\frac{1}{k}\int_{-1}^\infty \phi(z)\,\mathrm{d}z + \frac{1}{k}\int_{-1}^\infty z\,\phi(z)\,\mathrm{d}z\right].$$

The first integral is k, as we have just shown.

Ask yourself How do we integrate $z\,\phi(z)$ when we can't integrate $\phi(z)$ itself?

We use the fact that $\phi'(z) = -z\,\phi(z)$, so $z\,\phi(z)$ has antiderivative $-\phi(z)$:

$$\int_{-1}^\infty z\,\phi(z)\,\mathrm{d}z = \Big[-\phi(z)\Big]_{-1}^\infty = \phi(-1) = \frac{e^{-1/2}}{\sqrt{2\pi}} \approx 0.24197.$$

Therefore

$$\mathrm{E}(T) = \sigma\left(1 + \frac{e^{-1/2}}{k\sqrt{2\pi}}\right) \approx \sigma\left(1 + \frac{0.24197}{0.84134}\right) \approx 1.288\,\sigma.$$

This is larger than σ, as it should be: removing the left tail pushes the mean to the right.

Finding σ

The mean setting time is 3 hours, so

$$\sigma = \frac{3}{1.2876} \approx 2.330 \text{ hours}.$$

For any $a \geq 0$, the same substitution gives

$$\mathrm{P}(T > a) = \frac{1}{k}\int_{(a-\sigma)/\sigma}^\infty \phi(z)\,\mathrm{d}z = \frac{1-\Phi\left(\frac{a}{\sigma}-1\right)}{\Phi(1)}. \qquad (*)$$

Part (i)

With $a = 4$: $\frac{4}{\sigma} - 1 \approx \frac{4}{2.330} - 1 \approx 0.717$, and $\Phi(0.717) \approx 0.7633$. By $(*)$,

$$\mathrm{P}(T > 4) \approx \frac{1-0.7633}{0.8413} \approx 0.2814, \qquad \mathrm{P}(T \leq 4) \approx 0.7186.$$

We now assume that the setting times of the 16 questions are independent, each with the distribution of T (the only reasonable model available). Then

$$\mathrm{P}(\text{no question takes more than 4 hours}) = \mathrm{P}(T \le 4)^{16} \approx 0.7186^{16} \approx 0.0051.$$

So it is very unlikely (about one chance in 200) that every question is set within 4 hours.

Part (ii)

Ask yourself Which event are we conditioning on?

"Unsatisfactory after 2 hours" means the question needs more than 2 hours in total, so it is the event $T > 2$. "Still unacceptable after a further 2 hours" is $T > 4$. Since $\{T > 4\} \subseteq \{T > 2\}$,

$$\mathrm{P}(T > 4 \mid T > 2) = \frac{\mathrm{P}(T > 4)}{\mathrm{P}(T > 2)}.$$

For $a = 2$: $\frac{2}{\sigma} - 1 \approx -0.142$, so $1 - \Phi(-0.142) = \Phi(0.142) \approx 0.5563$. In the ratio of the two expressions $(*)$ the factor $\Phi(1)$ cancels:

$$\mathrm{P}(T > 4 \mid T > 2) \approx \frac{1 - 0.7633}{0.5563} = \frac{0.2367}{0.5563} \approx 0.426.$$

Checks

- A direct numerical integration of the density (Simpson's rule, $\sigma = 1$) gives total area 0.84134 before dividing by k, and mean 1.2876σ.
- A simulation of 4×10^5 normal values with mean and standard deviation $\sigma = 2.330$, discarding the negative ones, gives sample mean 3.004, $\mathrm{P}(T > 4) \approx 0.282$ and $\mathrm{P}(T > 4 \mid T > 2) \approx 0.426$.
- Part (i) is very sensitive to rounding: $0.719^{16} \approx 0.0051$ but $0.718^{16} \approx 0.0050$. Quote it as about 0.005.
- Without the truncation, $\mathrm{P}(T > 4 \mid T > 2)$ would be the same, since the factor $1/k$ cancels. Only part (i) and the mean feel the effect of k.

Common slips

- Forgetting to divide by k in $\mathrm{P}(T > 4)$: this gives 0.237 instead of 0.281, and gives $0.7633^{16} \approx 0.013$ in (i), more than double the correct value.
- Taking the mean of T to be σ (the mean before truncation), which gives $\sigma = 3$.
- In (ii), finding $\mathrm{P}(2 < T \le 4)$ or $\mathrm{P}(T > 2)$ instead of the conditional probability.

Beyond STEP

The quantity $\phi(1)/\Phi(1)$ in the mean is an example of the *inverse Mills ratio*: in general a normal $\mathrm{N}(\mu, \sigma^2)$ truncated below at a has mean $\mu + \sigma\,\phi(\alpha)/(1-\Phi(\alpha))$ where $\alpha = (a-\mu)/\sigma$. Truncated distributions appear wherever data can only be observed above a threshold, for example in survival analysis and in econometric models of censored data. Part (ii) is about the *survival function* $\mathrm{P}(T > t)$; the exponential distribution is the one case where $\mathrm{P}(T > s+t \mid T > s)$ does not depend on s. Grimmett and Welsh, *Probability: An Introduction*, covers continuous distributions and conditioning.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

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